

Report Number: R120-17

MISO DPP 2016 February West Area Phase 2 Study

Prepared for

MISO

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03/05/2018

Siemens PTI Project Number 62OT-001494

Revision History

Date	Rev.	Description
01/18/2018	A	Draft Report
02/01/2018	B	Draft Report
03/05/2018	C	Final Report

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Executive Summary

This report presents results of a System Impact Study (SIS) performed to evaluate the interconnection of the DPP 2016 February Phase 2 West Area Group (DPP West Area) generating facilities. The results for 2022 scenario are summarized below.

1.1 Project List

The DPP West Area study group has twenty five (25) generation projects with a combined nameplate rating of 4871 MW. The DPP West Area generating facilities are listed in Table ES-1. The modeling details and projects' slider diagrams are shown in Appendix B.

Table ES-1: Generating Facilities in DPP 2016 February West Area Group

MISO Project #	Service Type	TO	County, State	Point Of Interconnection	Fuel Type	Max Output	SH MW	SPK MW	Stability MW
J432	NRIS	XEL	Brookings, SD	Brookings 345 kV	Wind	98	98	15.29	98
J460	NRIS	GRE	Lincoln, MN	Brookings-H081 345 kV line	Wind	200	200	31.20	200
J475	NRIS	MEC	Poweshiek, IA	Montezuma 345 kV	Wind	200	200	31.20	200
J485	NRIS	RPU	Olmsted, MN	West Side 161 kV	CT	46.85	0	46.85	46.85
J488	ERIS	OTP	Grant, Roberts, SD	Big Stone-Ellendale 345 kV	Wind	151.8	151.8	23.68	151.8
J489	ERIS	OTP	Grant, Roberts, SD	Big Stone-Ellendale 345 kV	Wind	151.8	151.8	23.68	151.8
J493	NRIS	OTP	Yellow Medicine, MN	Big Stone-Brookings 345 kV	Wind	150	150	23.40	150
J495	NRIS	ITCM	Winnebago, IA	Ledyard-Colby 345 kV line	Wind	200	200	31.20	200
J498	NRIS	MEC	Boone, Greene, IA	Grimes-Lehigh 345 kV	Wind	340	340	53.04	340
J499	NRIS	MEC	Adair, Madison, IA	Fallow-Grimes 345 kV	Wind	340	340	53.04	340
J500	NRIS	MEC	Adair, IA	New substation at intersection of Boone-Atchison 345 kV and Rolling Hills-Madison Co 345 kV	Wind	500	500	78.00	500
J504	NRIS	ITCM	Linn, IA	Bertram-Duane Arnold 161 kV	Solar	50	25	50	50
J506	NRIS	MEC	Plymouth, IA	Raun-Lakefield Jct 345 kV	Wind	200	200	31.20	200
J510	NRIS	OTP	Deuel, SD	Scandinavia Township	CT	266 sum / 284.5 win	0	284.5	284.5
J514	NRIS	ITCM	Marshall, IA	Marshalltown substation	CC	30 sum / 65 win	32.5	65	65

MISO Project #	Service Type	TO	County, State	Point Of Interconnection	Fuel Type	Max Output	SH MW	SPK MW	Stability MW
J523	NRIS	ITCM	Mower, MN	Adams 161 kV	Solar	50	25	50	50
J524	NRIS	MEC	Webster, IA	Webster 161 kV	Solar	100	50	100	100
J525	NRIS	GRE	Murray, MN	Lake Wilson–Hadley 69 kV	Solar	33	16.5	33	33
J526	NRIS	OTP	Deuel, SD	Brooking Co-Big Stone South 345 kV	Wind	300	300	46.80	300
J527	NRIS	MEC	Adair, IA	Booneville-Cooper 345 kV	Wind	250	250	39.00	250
J528	NRIS	MEC	Adair, IA	Rolling Hills-Madison 345 kV	Wind	200	200	31.20	200
J529	NRIS	MEC	Palo Alto, IA	Obrien-Kossuth 345 kV	Wind	250	250	39.00	250
J530	NRIS	MEC	Iowa, IA	Montezuma-Hills 345 kV	Wind	250	250	39.00	250
J534	NRIS	MEC	Humboldt, Kossuth, IA	Kossuth–Webster 345 kV	Wind	250	250	39.00	250
J535	NRIS	MEC	Sac, IA	J411–Lehigh 345 kV	Wind	210	210	32.76	210

1.2 Reactive Power Requirements for Non-Synchronous Generation (FERC Order 827)

For non-synchronous generation projects in the DPP 2016 February West Area study group, if they do not have signed Generator Interconnection Agreement (GIA) or Provisional GIA (PGIA) by September 21, 2016, they are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

Except for generation projects J498, J499, J500, and J529, all other non-synchronous generation projects in this study group are required to meet the reactive power requirements per FERC Order 827.

The reactive power requirement analysis results are summarized as following:

- J498, J499, J500, and J529 are exempted from the FERC Order 827 reactive power requirements.
- All other non-synchronous generation projects can meet the reactive power requirements per FERC Order 827.

1.3 Total Network Upgrades for all Projects

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection Service as of the System Impact Study report date. The total cost of network upgrades in the interconnection plan required for each generation project is listed in Table ES-2. The costs for Network Upgrades are planning level estimates and subject to be revised in the facility studies.

Table ES-2: Total Cost of Network Upgrades for DPP 2016 February West Area Generation Projects

Project Num	ERIS Network Upgrades (\$)								NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		SNU (\$)	Total Cost (Exclude TOIF) (\$)	M2 Received (\$)	M3 Received (\$)	M4 (\$)
	MISO Thermal & Voltage	Voltage Collapse & Stability	Short-circuit	OTP LPC	GRE LPC	CIPCO Affected System	PJM Affected System	SPP Affected System		TO Network Upgrades	TO - Owned Direct assigned					
J432	\$361,568	\$632,506	\$0	\$0	\$0	\$0	\$0	\$8,683,175	\$0	\$2,000,000	NA	\$0	\$2,994,074	\$691,615	\$4,952,490	\$0
J460	\$3,574,464	\$1,287,905	\$0	\$0	\$0	\$0	\$0	\$13,737,136	\$0	\$10,000,000	NA	\$0	\$14,862,369	\$1,281,664	\$11,343,766	\$0
J475	\$1,157,376	\$0	\$0	\$0	\$0	\$0	\$0	\$7,797,263	\$3,180	\$2,622,000	NA	\$0	\$3,782,555	\$729,210	\$4,345,872	\$0
J485	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$2,015,047	\$0	\$2,046,000	\$340,600	\$0	\$2,046,000	\$190,507	\$14,093	\$204,600
J488	\$8,842,142	\$28,067,796	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$500,000	NA	\$0	\$37,409,938	\$979,992	\$9,857,744	\$0
J489	\$8,842,142	\$28,067,796	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$500,000	NA	\$0	\$37,409,938	\$979,992	\$9,857,744	\$0
J493	\$7,155,321	\$5,494,230	\$0	\$0	\$0	\$0	\$0	\$14,343,824	\$0	\$3,774,376	\$1,073,340	\$0	\$16,423,928	\$1,005,589	\$9,508,992	\$0
J495	\$17,867,238	\$0	\$0	\$0	\$0	\$0	\$0	\$10,583,796	\$0	\$11,000,000	NA	\$0	\$28,867,238	\$1,279,452	\$5,750,598	\$0
J498	\$9,239,130	\$0	\$0	\$0	\$0	\$0	\$2,452,991	\$23,562,689	\$34,899	\$10,799,800	\$634,600	\$0	\$20,073,829	\$1,585,468	\$15,019,709	\$0
J499	\$2,098,350	\$0	\$0	\$0	\$0	\$0	\$2,420,165	\$27,402,574	\$10,614	\$11,750,200	\$634,600	\$0	\$13,859,164	\$1,572,468	\$9,749,503	\$0
J500	\$1,954,352	\$0	\$0	\$0	\$0	\$0	\$2,748,000	\$48,316,644	\$129,105	\$23,714,000	\$634,600	\$16,284	\$25,813,741	\$2,141,629	\$10,095,182	\$0
J504	\$4,285	\$0	\$0	\$0	\$0	\$78,072	\$0	\$1,513,429	\$0	\$4,500,000	NA	\$0	\$4,504,285	\$242,863	\$299,743	\$358,251
J506	\$703,949	\$0	\$0	\$0	\$0	\$0	\$0	\$13,637,357	\$0	\$11,500,000	NA	\$0	\$12,203,949	\$1,166,452	\$5,868,139	\$0
J510	\$251,240	\$1,760,965	\$0	\$0	\$0	\$0	\$0	\$27,964,394	\$0	\$8,225,624	\$663,687	\$0	\$10,237,828	\$1,145,081	\$0	\$902,485
J514	\$5,456,063	\$0	\$0	\$0	\$0	\$56,176	\$0	\$3,319,256	\$0	\$15,000	NA	\$50,369	\$5,521,432	\$650,000	\$4,826,530	\$0
J523	\$71,517	\$0	\$0	\$0	\$0	\$0	\$0	\$2,071,050	\$0	\$1,200,000	NA	\$0	\$1,271,517	\$177,863	\$0	\$76,440
J524	\$6,475,617	\$0	\$0	\$0	\$0	\$0	\$0	\$6,187,232	\$0	\$1,000,000	NA	\$0	\$7,475,617	\$355,726	\$0	\$1,139,397
J525	\$52,036	\$0	\$0	\$0	\$0	\$0	\$0	\$1,870,361	\$0	\$3,000,000	NA	\$0	\$3,052,036	\$340,363	\$2,649,637	\$0
J526	\$17,403,193	\$10,988,802	\$0	\$0	\$0	\$0	\$2,221,418	\$30,237,002	\$0	\$11,250,000	NA	\$0	\$39,641,996	\$1,487,177	\$20,997,853	\$0
J527	\$985,215	\$0	\$0	\$0	\$0	\$0	\$1,716,000	\$23,628,190	\$66,750	\$11,500,000	NA	\$0	\$12,551,964	\$1,160,315	\$4,213,132	\$0
J528	\$788,170	\$0	\$0	\$0	\$0	\$0	\$34,426	\$18,930,826	\$53,748	\$11,500,000	NA	\$0	\$12,341,918	\$969,452	\$3,538,358	\$0
J529	\$1,014,572	\$0	\$0	\$0	\$0	\$0	\$1,786,000	\$15,122,513	\$0	\$11,328,700	\$634,600	\$0	\$12,343,272	\$1,440,315	\$7,377,204	\$0
J530	\$361,198	\$0	\$0	\$0	\$0	\$465,752	\$1,828,000	\$7,783,127	\$0	\$11,500,000	NA	\$0	\$11,861,198	\$959,315	\$5,846,477	\$0
J534	\$1,150,449	\$0	\$0	\$0	\$0	\$0	\$1,786,000	\$15,153,928	\$0	\$11,500,000	NA	\$0	\$12,650,449	\$1,890,315	\$7,105,121	\$0
J535	\$945,414	\$0	\$0	\$0	\$0	\$0	\$0	\$14,639,189	\$26,704	\$11,500,000	NA	\$0	\$12,472,119	\$1,522,177	\$6,293,685	\$0
Total (\$)	\$96,755,000	\$76,300,000	\$0	\$0	\$0	\$600,000	\$16,993,000	\$338,500,002	\$325,000	\$188,225,700	\$4,616,027	\$66,653	\$361,672,353	\$25,945,000	\$159,511,570	\$2,681,174

Executive Summary

The study was performed under the direction of MISO by Siemens PTI and an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Ameren, American Transmission Company, Basin Electric Power, Cedar Falls Utilities, Central Iowa Power Cooperative, City of Springfield (IL) Water Light & Power, Columbia (MO) Water and Light, Commonwealth Edison, Corn Belt Power Cooperative, Dairyland Power, Great River Energy, ITC Midwest, Lincoln Electric System, Manitoba Hydro, MidAmerican Energy Company, MISO, Minnesota Power, Minnkota Power, Missouri River Energy Services, Montana-Dakota Utilities Co., Muscatine Power & Water, Nebraska Public Power District, Northwestern Public Service, Omaha Public Power District, Otter Tail Power, PJM, Southern Illinois Power Cooperative, Southern Minnesota Municipal Power Agency, SPP, Western Area Power Administration, and Xcel Energy.

1.4 Per Project Summary

This section provides estimated cost of Network Upgrades on a per project basis for the 2022 scenario. The shared cost of Network Upgrades for all the generation projects are listed below.

The Interconnection Customers are required to mitigate the constraints observed from the 2022 scenario.

1.4.1 J432 Summary

Network Upgrade	Cost	J432	NUs Type
Hankinson-Wahpeton 230 kV	\$2,300,000	\$74,552	SH MISO and/or NRIS
Hankinson-Wahpeton 230 kV (2nd J510-Brookings)	\$19,513,283	\$632,506	Thermal mitigated by 2nd J510-Brookings
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$36,401	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$29,375	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$58,137	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$163,103	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$5,444,670	SPP AFS
Required to mitigate APPLIEDORN 4 - WATERTOWN 230 KV CKT 1 NRIS thermal constraints for contingency conditions.	\$1,000,000	\$113,087	SPP AFS
Required to mitigate BLAIR - GRANITE FALLS 230KV CKT 1 NRIS thermal constraints for contingency conditions.	\$33,000,000	\$2,849,076	SPP AFS

Network Upgrade	Cost	J432	NUs Type
Required to mitigate Granite Falls – MN Valley Tap 230kV (WAPA) for contingency conditions	\$2,500,000	\$176,165	SPP AFS
Required to mitigate SPLIT ROCK - WHITE 345KV CKT 1 NRIS thermal constraint for contingency conditions to at least 1210MVA.	\$1,000,000	\$100,177	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$9,677,249	

1.4.2 J460 Summary

Network Upgrade	Cost	J460	NUs Type
Hazel Creek 345-230-13.8 kV xfmr #6	\$4,900,000	\$2,843,511	SH MISO
Hankinson-Wahpeton 230 kV	\$2,300,000	\$151,803	SH MISO and/or NRIS
Hankinson-Wahpeton 230 kV (2nd J510-Brookings)	\$19,513,283	\$1,287,905	Thermal mitigated by 2nd J510-Brookings
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$72,376	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$55,306	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$114,948	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$336,521	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$10,875,957	SPP AFS
Required to mitigate APPLIEDORN 4 - WATERTOWN 230 KV CKT 1 NRIS thermal constraints for contingency conditions.	\$1,000,000	\$80,038	SPP AFS
Required to mitigate BLAIR - GRANITE FALLS 230KV CKT 1 NRIS thermal constraints for contingency conditions.	\$33,000,000	\$2,307,267	SPP AFS
Required to mitigate Granite Falls – MN Valley Tap 230kV (WAPA) for contingency conditions	\$2,500,000	\$299,920	SPP AFS
Required to mitigate SPLIT ROCK - WHITE 345KV CKT 1 NRIS thermal constraint for contingency conditions to at least 1210MVA.	\$1,000,000	\$173,954	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$18,599,505	

1.4.3 J475 Summary

Network Upgrade	Cost	J475	NUs Type
J530 POI-Hills 345 kV	\$300,000	\$115,097	SH MISO
Lehigh-Webster 345 kV	\$100,000	\$3,180	NRIS
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$52,116	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$22,532	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$967,631	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$7,797,263	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$8,957,818	

1.4.4 J485 Summary

Network Upgrade	Cost	J485	NUs Type
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$2,009,749	SPP AFS
Required to mitigate SPLIT ROCK - WHITE 345KV CKT 1 NRIS thermal constraint for contingency conditions to at least 1210MVA.	\$1,000,000	\$5,298	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$2,015,047	

1.4.5 J488 Summary

Network Upgrade	Cost	J488	NUs Type
Johnson Jct-Ortonville 115 kV	\$16,850,000	\$3,263,129	SH MISO
Johnson Jct-Morris 115 kV	\$8,175,000	\$1,582,585	SH MISO
Big Stone 230-115-13.8 kV xfmr	\$3,250,000	\$632,318	SH MISO
Big Stone-Big Stone South 230 kV #1	\$1,300,000	\$249,611	SH MISO
Big Stone-Browns Valley 230 kV	\$350,000	\$55,925	SH MISO and/or NRIS
Big Stone-Blair 230 kV	\$80,000	\$15,566	SH MISO
Hankinson-Wahpeton 230 kV	\$2,300,000	\$390,253	SH MISO and/or NRIS

Network Upgrade	Cost	J488	NUs Type
Oakes-Forman 230 kV	\$1,600,000	\$800,000	SH MISO
Oakes-Ellendale 230 kV	\$2,850,000	\$1,425,000	SH MISO
Hankinson-Forman 230 kV (2nd J488&489-BSS)	\$2,700,000	\$1,350,000	Thermal mitigated by 2nd J488&489-BSS
Oakes-Ellendale 230 kV (2nd J488&489-BSS)	\$17,650,000	\$8,825,000	Thermal mitigated by 2nd J488&489-BSS
Voltage Collapse (2nd J488&489-BSS)	\$24,550,000	\$12,275,000	Voltage collapse
Big Stone-Highway 12 115 kV (2nd J510-Brookings)	\$3,078,431	\$598,119	Thermal mitigated by 2nd J510-Brookings
Highway 12-Ortonville 115 kV (2nd J510-Brookings)	\$6,851,992	\$1,331,296	Thermal mitigated by 2nd J510-Brookings
Big Stone-Big Stone South 230 kV #2 (2nd J510-Brookings)	\$1,241,303	\$238,340	Thermal mitigated by 2nd J510-Brookings
Big Stone-Blair 230 kV (2nd J510-Brookings)	\$714,991	\$139,124	Thermal mitigated by 2nd J510-Brookings
Hankinson-Wahpeton 230 kV (2nd J510-Brookings)	\$19,513,283	\$3,310,917	Thermal mitigated by 2nd J510-Brookings
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$54,353	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$42,237	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$86,365	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$244,800	Reactive Power NU
Total Cost Per Project for ERIIS for each Project		\$36,909,938	

1.4.6 J489 Summary

Network Upgrade	Cost	J489	NUs Type
Johnson Jct-Ortonville 115 kV	\$16,850,000	\$3,263,129	SH MISO
Johnson Jct-Morris 115 kV	\$8,175,000	\$1,582,585	SH MISO
Big Stone 230-115-13.8 kV xfmr	\$3,250,000	\$632,318	SH MISO
Big Stone-Big Stone South 230 kV #1	\$1,300,000	\$249,611	SH MISO
Big Stone-Browns Valley 230 kV	\$350,000	\$55,925	SH MISO and/or NRIS
Big Stone-Blair 230 kV	\$80,000	\$15,566	SH MISO
Hankinson-Wahpeton 230 kV	\$2,300,000	\$390,253	SH MISO and/or NRIS
Oakes-Forman 230 kV	\$1,600,000	\$800,000	SH MISO

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Network Upgrade	Cost	J489	NUs Type
Oakes-Ellendale 230 kV	\$2,850,000	\$1,425,000	SH MISO
Hankinson-Forman 230 kV (2nd J488&489-BSS)	\$2,700,000	\$1,350,000	Thermal mitigated by 2nd J488&489-BSS
Oakes-Ellendale 230 kV (2nd J488&489-BSS)	\$17,650,000	\$8,825,000	Thermal mitigated by 2nd J488&489-BSS
Voltage Collapse (2nd J488&489-BSS)	\$24,550,000	\$12,275,000	Voltage collapse
Big Stone-Highway 12 115 kV (2nd J510-Brookings)	\$3,078,431	\$598,119	Thermal mitigated by 2nd J510-Brookings
Highway 12-Ortonville 115 kV (2nd J510-Brookings)	\$6,851,992	\$1,331,296	Thermal mitigated by 2nd J510-Brookings
Big Stone-Big Stone South 230 kV #2 (2nd J510-Brookings)	\$1,241,303	\$238,340	Thermal mitigated by 2nd J510-Brookings
Big Stone-Blair 230 kV (2nd J510-Brookings)	\$714,991	\$139,124	Thermal mitigated by 2nd J510-Brookings
Hankinson-Wahpeton 230 kV (2nd J510-Brookings)	\$19,513,283	\$3,310,917	Thermal mitigated by 2nd J510-Brookings
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$54,353	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$42,237	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$86,365	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$244,800	Reactive Power NU
Total Cost Per Project for ERIIS for each Project		\$36,909,938	

1.4.7 J493 Summary

Network Upgrade	Cost	J493	NUs Type
Johnson Jct-Ortonville 115 kV	\$16,850,000	\$3,433,101	SH MISO
Johnson Jct-Morris 115 kV	\$8,175,000	\$1,665,990	SH MISO
Big Stone 230-115-13.8 kV xfmr	\$3,250,000	\$661,755	SH MISO
Big Stone-Big Stone South 230 kV #1	\$1,300,000	\$266,921	SH MISO
Big Stone-Browns Valley 230 kV	\$350,000	\$64,824	SH MISO and/or NRIS
Big Stone-Blair 230 kV	\$80,000	\$16,288	SH MISO
Hankinson-Wahpeton 230 kV	\$2,300,000	\$361,847	SH MISO and/or NRIS
Wahpeton-Fergus Falls 230 kV	\$750,000	\$249,158	SH MISO
Big Stone-Highway 12 115 kV (2nd J510-Brookings)	\$3,078,431	\$627,398	Thermal mitigated by 2nd J510-Brookings

Network Upgrade	Cost	J493	NUs Type
Highway 12-Ortonville 115 kV (2nd J510-Brookings)	\$6,851,992	\$1,396,467	Thermal mitigated by 2nd J510-Brookings
Big Stone-Big Stone South 230 kV #2 (2nd J510-Brookings)	\$1,241,303	\$254,869	Thermal mitigated by 2nd J510-Brookings
Big Stone-Blair 230 kV (2nd J510-Brookings)	\$714,991	\$145,576	Thermal mitigated by 2nd J510-Brookings
Hankinson-Wahpeton 230 kV (2nd J510-Brookings)	\$19,513,283	\$3,069,920	Thermal mitigated by 2nd J510-Brookings
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$55,238	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$44,143	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$88,115	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$247,941	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$8,303,049	SPP AFS
Required to mitigate APPLEDOORN 4 - WATERTOWN 230 KV CKT 1 NRIS thermal constraints for contingency conditions.	\$1,000,000	\$165,451	SPP AFS
Required to mitigate BLAIR - GRANITE FALLS 230KV CKT 1 NRIS thermal constraints for contingency conditions.	\$33,000,000	\$5,349,660	SPP AFS
Required to mitigate Granite Falls – MN Valley Tap 230kV (WAPA) for contingency conditions	\$2,500,000	\$377,303	SPP AFS
Required to mitigate SPLIT ROCK - WHITE 345KV CKT 1 NRIS thermal constraint for contingency conditions to at least 1210MVA.	\$1,000,000	\$148,361	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$26,993,375	

1.4.8 J495 Summary

Network Upgrade	Cost	J495	NUs Type
Hazleton-Washburn 161 kV	\$16,700,000	\$16,700,000	SH MISO
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$69,955	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$35,710	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$115,736	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$945,837	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$10,583,796	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$28,451,034	

1.4.9 J498 Summary

Network Upgrade	Cost	J498	NUs Type
Boone Jct-Sub T Fort Dodge 161 kV	\$7,000,000	\$7,000,000	SH MISO
Grimes-Sycamore 345 kV #2	\$1,200,000	\$519,779	SH MISO
Grimes-Beaver Creek 345 kV	\$600,000	\$318,015	SH MISO
Granger Tap-108th & 54th 161 kV	\$100,000	\$44,481	SH MISO
Twin Branch–Argenta 345 kV	\$208,000	\$59,991	PJM AFS
Benton–T-094 345 kV (N5106)	\$19,000,000	\$2,250,000	PJM AFS
Benton–T-094 345 kV (N5312.3)	\$1,000,000	\$143,000	PJM AFS
Lehigh-Webster 345 kV	\$100,000	\$34,899	NRIS
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$170,045	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$284,732	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$438,493	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$463,585	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$23,562,689	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$35,289,709	

1.4.10 J499 Summary

Network Upgrade	Cost	J499	NUs Type
Grimes-Sycamore 345 kV #2	\$1,200,000	\$680,221	SH MISO
Granger Tap-108th & 54th 161 kV	\$100,000	\$55,519	SH MISO
Bondurant-Montezuma 345 kV	\$200,000	\$54,695	SH MISO
Twin Branch–Argenta 345 kV	\$208,000	\$59,165	PJM AFS
Benton–T-094 345 kV (N5106)	\$19,000,000	\$2,220,000	PJM AFS
Benton–T-094 345 kV (N5312.3)	\$1,000,000	\$141,000	PJM AFS
Lehigh-Webster 345 kV	\$100,000	\$10,614	NRIS
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$201,022	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$366,448	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$548,237	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$192,208	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$27,402,574	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$31,931,703	

1.4.11 J500 Summary

Network Upgrade	Cost	J500	NUs Type
Bondurant-Montezuma 345 kV	\$200,000	\$75,810	SH MISO
Benton–T-094 345 kV (N5106)	\$19,000,000	\$2,584,000	PJM AFS
Benton–T-094 345 kV (N5312.3)	\$1,000,000	\$164,000	PJM AFS
J274 POI-Creston 161 kV	\$175,000	\$16,284	SNU
Lehigh-Webster 345 kV	\$100,000	\$12,770	NRIS
J527 POI-Booneville 345 kV	\$100,000	\$51,591	NRIS
J528 POI-Madison 345 kV	\$125,000	\$64,744	NRIS
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$329,705	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$525,410	Reactive Power NU

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Network Upgrade	Cost	J500	NUs Type
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$772,903	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$250,524	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$48,316,644	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$53,164,385	

1.4.12 J504 Summary

Network Upgrade	Cost	J504	NUs Type
Mqoketa-Wyoming 161 kV	\$600,000	\$78,072	CIPCO AFS
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$4,285	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$1,513,429	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,595,785	

1.4.13 J506 Summary

Network Upgrade	Cost	J506	NUs Type
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$94,165	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$98,800	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$158,247	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$352,736	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$13,637,357	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$14,341,306	

1.4.14 J510 Summary

Network Upgrade	Cost	J510	NUs Type
Big Stone-Browns Valley 230 kV	\$350,000	\$43,678	SH MISO and/or NRIS
Hankinson-Wahpeton 230 kV	\$2,300,000	\$207,562	SH MISO and/or NRIS
Hankinson-Wahpeton 230 kV (2nd J510-Brookings)	\$19,513,283	\$1,760,965	Thermal mitigated by 2nd J510-Brookings
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$16,288,843	SPP AFS
Required to mitigate APPLIEDORN 4 - WATERTOWN 230 kV CKT 1 NRIS thermal constraints for contingency conditions.	\$1,000,000	\$321,789	SPP AFS
Required to mitigate BLAIR - GRANITE FALLS 230KV CKT 1 NRIS thermal constraints for contingency conditions.	\$33,000,000	\$10,338,226	SPP AFS
Required to mitigate Granite Falls – MN Valley Tap 230kV (WAPA) for contingency conditions	\$2,500,000	\$733,395	SPP AFS
Required to mitigate SPLIT ROCK - WHITE 345KV CKT 1 NRIS thermal constraint for contingency conditions to at least 1210MVA.	\$1,000,000	\$282,141	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$29,976,598	

1.4.15 J514 Summary

Network Upgrade	Cost	J514	NUs Type
Marshalltown-Blairstown 115 kV	\$2,700,000	\$2,700,000	SH MISO
Blairstown-Prairie Crk 115 kV	\$2,700,000	\$2,700,000	SH MISO
Mqoketa-Wyoming 161 kV	\$600,000	\$56,176	CIPCO AFS
Jasper-Laurel 161 kV	\$600,000	\$50,369	SNU
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$11,533	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$7,966	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$16,367	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$20,196	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$3,319,256	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$8,881,864	

1.4.16 J523 Summary

Network Upgrade	Cost	J523	NUs Type
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$6,419	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$8,171	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$56,927	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$2,071,050	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$2,142,567	

1.4.17 J524 Summary

Network Upgrade	Cost	J524	NUs Type
Grimes-Beaver Creek 345 kV	\$600,000	\$21,171	SH MISO
Webster-Sub T Ft Dodge 161 kV	\$6,250,000	\$6,250,000	SH MISO
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$22,171	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$27,738	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$45,773	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$108,762	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$6,187,232	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$12,662,849	

1.4.18 J525 Summary

Network Upgrade	Cost	J525	NUs Type
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$6,286	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$5,143	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$10,193	Reactive Power

Network Upgrade	Cost	J525	NUs Type
			NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$30,414	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$1,869,694	SPP AFS
Required to mitigate SPLIT ROCK - WHITE 345KV CKT 1 NRIS thermal constraint for contingency conditions to at least 1210MVA.	\$1,000,000	\$667	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,922,397	

1.4.19 J526 Summary

Network Upgrade	Cost	J526	NUs Type
Hazel Creek 345-230-13.8 kV xfmr #6	\$4,900,000	\$2,056,489	SH MISO
Panther-McLeod 230 kV	\$1,000,000	\$1,000,000	SH MISO
Johnson Jct-Ortonville 115 kV	\$16,850,000	\$6,890,640	SH MISO
Johnson Jct-Morris 115 kV	\$8,175,000	\$3,343,840	SH MISO
Big Stone 230-115-13.8 kV xfmr	\$3,250,000	\$1,323,609	SH MISO
Big Stone-Big Stone South 230 kV #1	\$1,300,000	\$533,858	SH MISO
Big Stone-Browns Valley 230 kV	\$350,000	\$129,648	SH MISO and/or NRIS
Big Stone-Blair 230 kV	\$80,000	\$32,579	SH MISO
Hankinson-Wahpeton 230 kV	\$2,300,000	\$723,730	SH MISO and/or NRIS
Wahpeton-Fergus Falls 230 kV	\$750,000	\$500,842	SH MISO
Twin Branch–Argenta 345 kV	\$208,000	\$54,418	PJM AFS
Benton–T-094 345 kV (N5106)	\$19,000,000	\$2,038,000	PJM AFS
Benton–T-094 345 kV (N5312.3)	\$1,000,000	\$129,000	PJM AFS
Big Stone-Highway 12 115 kV (2nd J510-Brookings)	\$3,078,431	\$1,254,796	Thermal mitigated by 2nd J510-Brookings
Highway 12-Ortonville 115 kV (2nd J510-Brookings)	\$6,851,992	\$2,792,933	Thermal mitigated by 2nd J510-Brookings
Big Stone-Big Stone South 230 kV #2 (2nd J510-Brookings)	\$1,241,303	\$509,753	Thermal mitigated by 2nd J510-Brookings
Big Stone-Blair 230 kV (2nd J510-Brookings)	\$714,991	\$291,167	Thermal mitigated by 2nd J510-Brookings
Hankinson-Wahpeton 230 kV (2nd J510-Brookings)	\$19,513,283	\$6,140,153	Thermal mitigated by 2nd J510-Brookings

Executive Summary

Network Upgrade	Cost	J526	NUs Type
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$110,093	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$87,671	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$175,574	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$494,622	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$16,558,975	SPP AFS
Required to mitigate APPLIEDORN 4 - WATERTOWN 230 KV CKT 1 NRIS thermal constraints for contingency conditions.	\$1,000,000	\$319,635	SPP AFS
Required to mitigate BLAIR - GRANITE FALLS 230KV CKT 1 NRIS thermal constraints for contingency conditions.	\$33,000,000	\$12,155,771	SPP AFS
Required to mitigate Granite Falls – MN Valley Tap 230kV (WAPA) for contingency conditions	\$2,500,000	\$913,218	SPP AFS
Required to mitigate SPLIT ROCK - WHITE 345KV CKT 1 NRIS thermal constraint for contingency conditions to at least 1210MVA.	\$1,000,000	\$289,403	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$60,850,416	

1.4.20 J527 Summary

Network Upgrade	Cost	J527	NUs Type
Bondurant-Montezuma 345 kV	\$200,000	\$38,594	SH MISO
Benton–T-094 345 kV (N5106)	\$19,000,000	\$1,614,000	PJM AFS
Benton–T-094 345 kV (N5312.3)	\$1,000,000	\$102,000	PJM AFS
Lehigh-Webster 345 kV	\$100,000	\$6,605	NRIS
J527 POI-Booneville 345 kV	\$100,000	\$27,773	NRIS
J528 POI-Madison 345 kV	\$125,000	\$32,372	NRIS
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$162,782	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$266,716	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$392,417	Reactive Power

Executive Summary

Network Upgrade	Cost	J527	NUs Type
			NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$124,706	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$23,628,190	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$26,396,154	

1.4.21 J528 Summary

Network Upgrade	Cost	J528	NUs Type
Bondurant-Montezuma 345 kV	\$200,000	\$30,901	SH MISO
Twin Branch–Argenta 345 kV	\$208,000	\$34,426	PJM AFS
Lehigh-Webster 345 kV	\$100,000	\$5,228	NRIS
J527 POI-Booneville 345 kV	\$100,000	\$20,636	NRIS
J528 POI-Madison 345 kV	\$125,000	\$27,884	NRIS
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$130,480	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$213,510	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$314,305	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$98,974	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$18,930,826	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$19,807,170	

1.4.22 J529 Summary

Network Upgrade	Cost	J529	NUs Type
Benton–T-094 345 kV (N5106)	\$19,000,000	\$1,680,000	PJM AFS
Benton–T-094 345 kV (N5312.3)	\$1,000,000	\$106,000	PJM AFS
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$102,575	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$97,896	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$186,864	Reactive Power

Executive Summary

Network Upgrade	Cost	J529	NUs Type
			NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$627,238	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$15,122,513	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$17,923,085	

1.4.23 J530 Summary

Network Upgrade	Cost	J530	NUs Type
J530 POI-Hills 345 kV	\$300,000	\$184,903	SH MISO
Parnell-J438 POI 161 kV	\$100,000	\$100,000	SH MISO
Mqoketa-Wyoming 161 kV	\$600,000	\$465,752	CIPCO AFS
Benton–T-094 345 kV (N5106)	\$19,000,000	\$1,719,000	PJM AFS
Benton–T-094 345 kV (N5312.3)	\$1,000,000	\$109,000	PJM AFS
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$46,509	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$29,786	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$7,783,127	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$10,438,077	

1.4.24 J534 Summary

Network Upgrade	Cost	J534	NUs Type
Grimes-Beaver Creek 345 kV	\$600,000	\$101,951	SH MISO
Benton–T-094 345 kV (N5106)	\$19,000,000	\$1,680,000	PJM AFS
Benton–T-094 345 kV (N5312.3)	\$1,000,000	\$106,000	PJM AFS
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$103,849	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$108,564	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$202,297	Reactive Power NU

Network Upgrade	Cost	J534	NUs Type
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$633,788	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$15,153,928	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$18,090,377	

1.4.25 J535 Summary

Network Upgrade	Cost	J535	NUs Type
Grimes-Beaver Creek 345 kV	\$600,000	\$158,863	SH MISO
Lehigh-Webster 345 kV	\$100,000	\$26,704	NRIS
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$103,289	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$137,866	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$212,865	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$332,531	Reactive Power NU
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	\$301,000,000	\$14,639,189	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$15,611,308	

1.5 Study Compliance with NERC FAC-002-2 Standard

This DPP 2016 February West Area Phase 2 study was completed in compliance with NERC FAC-002-2:

R1.1: The reliability impact of the new interconnection, or materially modified existing interconnection, on affected system(s).

Section 2 covers steady-state analysis results which include thermal and voltage constraints impacted by the DPP West Area generating facilities. Thermal and voltage upgrades required to interconnect the new generating facilities are also identified.

Executive Summary

Section 3.1 covers reliability impact of the new generating facilities in the CIPCO and MPC affected systems. Network Upgrades required to interconnect the new generating facilities are also identified.

Section 3.2 covers reliability impact of the J525 generating facility per GRE Local Planning Criteria (LPC). Network Upgrades required to interconnect the new generating facilities are also identified.

Section 3.3 covers reliability impact of the new generating facilities per OTP LPC. Network Upgrades required to interconnect the new generating facilities are also identified.

Section 3.4 covers reliability impact of the new generating facilities in the PJM affected systems. Network Upgrades required to interconnect the new generating facilities are also identified.

Section 3.5 covers reliability impact of the new generating facilities in the SPP affected systems. Network Upgrades required to interconnect the new generating facilities are also identified.

Section 4 covers transient stability analysis results.

Section 5 covers voltage stability (PV) analysis on the MWEX System Operating Limit (SOL). No additional Network Upgrades are required for transient stability in the 2022 scenario.

Section 6 covers short circuit reliability impact of the new generating facilities.

Section 7 covers Deliverability reliability impact of the new NRIS generating facilities.

R1.2: Adherence to applicable NERC Reliability Standards; regional and Transmission Owner planning criteria; and Facility interconnection requirements.

Sections 2.3-2.5, Section 3, and Section 4 all cover NERC Reliability Standard TPL-001-4.

Section 3.1 covers CIPCO and MPC's system planning criteria.

Section 3.2 covers GRE Local planning Criteria (LPC).

Section 3.3 covers OTP LPC.

Section 3.4 covers PJM system planning criteria.

Section 3.5 covers SPP system planning criteria.

Section 5 (voltage stability analysis) covers individual system planning criteria (ATC).

Section 7 covers MISO system planning criteria.

R1.3: Steady-state, short-circuit, and dynamics studies, as necessary, to evaluate system performance under both normal and contingency conditions.

Section 2 covers MISO steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 3.1 covers CIPCO and MPC's affected system assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 3.2 covers GRE's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 3.3 covers OTP's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 3.4 covers PJM affected system assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 3.5 covers SPP affected system assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 4 covers transient stability studies under NERC category P0 to P7 contingencies (TPL-001-4).

Section 6 covers short-circuit assessment.

Section 7 covers MISO deliverability study (steady-state assessment) including NERC category P0 to P7 contingencies (TPL-001-4).

R1.4: Study assumptions, system performance, alternatives considered, and coordinated recommendations. While these studies may be performed independently, the results shall be evaluated and coordinated by the entities involved.

Section 2.2, Section 2.3, Section 2.4, Section 2.5, Section 4.2, Section 4.3, and Section 4.4 cover study assumptions and system performance criteria.

Jointly coordinated recommendations can be found in Sections 3.1 (MISO, CIPCO, and MPC), Section 3.2 (MISO and GRE), Section 3.3 (MISO and OTP), Section 3.4 (MISO and PJM), Section 3.5 (MISO and SPP), and Section 5 (MISO and ATC). Results in Section 2, 3, 4, 6, and 7 have also been reviewed by PJM, SPP, CIPCO, and MPC.

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Introduction

Twenty five (25) generation projects, listed in Appendix A.1, Table A-1, have requested to interconnect to the MISO transmission network in the West Area and have advanced to the Definitive Planning Phase (DPP) 2016 February Phase 2 study (DPP West Area). Generating facilities J488 and J489 have requested Energy Resource Interconnection Service (ERIS), and other generating facilities have requested both ERIS and Network Resource Interconnection Service (NRIS).

This report presents the study results of a System Impact Study (SIS) performed to evaluate the interconnection of the generating facilities in the DPP West Area Phase 2 study.

The study was performed under the direction of MISO by Siemens PTI and an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Ameren, American Transmission Company, Basin Electric Power, Cedar Falls Utilities, Central Iowa Power Cooperative, City of Springfield (IL) Water Light & Power, Columbia (MO) Water and Light, Commonwealth Edison, Corn Belt Power Cooperative, Dairyland Power, Great River Energy, ITC Midwest, Lincoln Electric System, Manitoba Hydro, MidAmerican Energy Company, MISO, Minnesota Power, Minnkota Power, Missouri River Energy Services, Montana-Dakota Utilities Co., Muscatine Power & Water, Nebraska Public Power District, Northwestern Public Service, Omaha Public Power District, Otter Tail Power, PJM, Southern Illinois Power Cooperative, Southern Minnesota Municipal Power Agency, SPP, Western Area Power Administration, and Xcel Energy.

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Steady-State Analysis

Steady-state analysis was performed to identify thermal and voltage upgrades required to interconnect the generating facilities in the DPP 2016 February Phase 2 West Area group (DPP West Area) to the transmission system.

2.1 Study Procedure

2.1.1 Computer Programs

Steady-state analyses were performed using PSS®E version 33.10 and PSS®MUST version 12.0.

2.1.2 Study Methodology

Study cases representing 2022 summer shoulder (SH) and summer peak (SPK) system conditions with generating facilities in the DPP 2016 February Phase 2 West Area group (DPP West Area) were developed from the MTEP17 steady state models with LBA dispatch. Benchmark cases were created by removing the DPP West Area generating facilities from the study cases. Nonlinear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP West Area generating facilities was evaluated by comparing the steady-state performance of the transmission system in the benchmark and study cases. Network upgrades were identified to mitigate any constraints.

2.2 Model Development

2.2.1 Study Cases

Power flow study cases representing 2022 summer shoulder and summer peak conditions were developed from the MTEP17 steady state models with LBA dispatch.

The study cases for DPP 2016 February Phase 2 West Area study were created as follows:

- MISO Prior queued generation projects and their associated Network Upgrades (NU) were modeled.
- DPP 2016 February generation projects in the West Area (DPP West Area, Table A-1) were modeled and dispatched.
- DPP 2016 February generation projects in the Central Area (Table A-4), Michigan Area (Table A-5), and Wisconsin Area (Table A-6) were modeled and dispatched.
- For MISO generation projects, their output was sunk to the MISO Classic (Appendix A.3, Table A-8), where generation was scaled uniformly;
- PJM generation projects were modeled and dispatched. The generation output was sunk to the PJM market (Appendix A.4, Table A-9), where generation was scaled uniformly.

- SPP generation projects were modeled and dispatched. The generation output was sunk to the SPP market (Appendix A.5, Table A-10), where generation was scaled uniformly.
- Network Upgrades based on DPP 2015 August West J298 withdrawn restudy were modeled.
- The Hickory Creek–Cardinal 345 kV project (MVP project 3127 with in-service date of 12/31/2023) was not modeled.
- Models were further reviewed by the Ad Hoc study members (transmission owners and customers). Model corrections and changes were made based on the comments and feedback. These modeling changes are listed in Appendix A.2.
- Adjusted Square Butte DC to match the total output of the Bison (Bison 1 to 6) and Oliver County (Oliver County 1 and 2) wind farms.
- Adjusted CU DC to match the total output of Coal Creek generation units #1 and #2.
- Set MHEX interface transfer level at 492 MW in summer shoulder and 1406 MW in summer peak cases.

The study power flow cases were solved with transformer tap adjustment enabled, area interchange disabled, phase shifter adjustment enabled and switched shunt adjustment enabled.

The interface transfer levels in the 2022 study cases are summarized in Table 2-1.

Table 2-1: Interface Transfer Levels in 2022 Steady State Study Cases

Interface	2022 SH Case (MW)	2022 SPK Case (MW)
MHEX	493	1406
MWEX	1226	640
Arrowhead – Stone Lake 345 kV	464	238

2.2.2 Benchmark Cases

Benchmark cases are used to benchmark system performance without the DPP West Restudy generating facilities, and were created by removing the DPP West Area generating facilities (Table A-1) from the study cases. The MISO Classic was used for power balance, where generation was scaled uniformly.

2.3 Contingency Criteria

A variety of contingencies were considered for steady-state analysis:

- NERC Category P0 with system intact (no contingencies)
- NERC Category P1 contingencies
 - NERC Category P1 contingencies, at buses with a nominal voltage of 69 kV and above, in the following areas: CWLD (area 333), AMMO (area 356), AMIL (area 357), CWLP (area 360), SIPC (area 361), WEC (area 295), WEC MI (area 296), XCEL (area 600), MP (area 608), SMMPA (area 613), GRE

- (area 615), OTP (area 620), ITCM (area 627), MPW (area 633), MEC (area 635), MDU (area 661), BEPC-MISO (area 663), MHEB (area 667), DPC (area 680), ALTE (area 694), WPS (area 696), MGE (area 697), UPPC (area 698), CE (area 222), NPPD (area 640), OPPD (area 645), LES (area 650), WAPA (area 652), BEPC-SPP (area 659), AECI (area 330), MIPU (area 540), KCPL (area 541), KACY (area 542), INDN (area 545).
- o Multiple-element NERC Category P1 contingencies, in Dakotas, Illinois, Iowa, Minnesota, Missouri, and Wisconsin. These specified Category P1 contingencies are listed in Appendix A.6.
- NERC Category P2-P7 contingencies
 - o Selected NERC Category P2-P7 contingencies provided by the Ad Hoc Study Group, in the study region of Dakotas, Illinois, Iowa, Minnesota, Missouri, and Wisconsin. These specified Category P2-P7 contingencies are listed in Appendix A.6.

For all contingency and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

2.4 Monitored Elements

The study area is defined in Table 2-2. Facilities in the study area were monitored for system intact and contingency conditions. Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal (PSS[®] E rate A) rating. Under NERC category P1-P7 conditions, branches were monitored for loading as shown in the column labeled "Post-Disturbance Thermal Limits".

Table 2-2: Monitored Elements

Owner / Area	Monitored Facilities	Thermal Limits ¹		Voltage Limits ²	
		Pre-Disturbance	Post-Disturbance	Pre-Disturbance	Post-Disturbance
AECI	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
AMIL	69 kV and above	100% of Rate A	100% of Rate B	1.05/1.00	1.05/0.95
AMMO	69 kV and above	100% of Rate A	100% of Rate B	1.05/1.00	1.05/0.95
ATCLLC	69 kV and above	95% of Rate A	95% of Rate B	1.05/0.95	1.10/0.90
BEPC-MISO	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
BEPC-SPP	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
CWLD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
CWLP	69 kV and above	100% of Rate A	100% of Rate B	1.05/1.00	1.10/0.95
CE	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
DPC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
GMO	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
GRE	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.92
INDN	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90

Owner / Area	Monitored Facilities	Thermal Limits ¹		Voltage Limits ²	
		Pre-Disturbance	Post-Disturbance	Pre-Disturbance	Post-Disturbance
ITCM	69 kV and above	100% of Rate A	100% of Rate B	1.07/1.05/0.95	1.10/0.93
KACY	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
KCPL	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
LES	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
MDU	57 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
MEC	69 kV and above	100% of Rate A	100% of Rate B	1.07/1.05/0.96/0.95	1.05/0.96/0.95 ³
MHEB	69 kV and above	100% of Rate A	100% of Rate B	1.12/1.1/1.07/1.05/1.04/0.99/0.97/0.96/0.95	1.15/1.10/0.94/0.90
MP	69 kV and above	100% of Rate A	100% of Rate B	1.05/1.00	1.10/0.95
MPW	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.06/0.92
NPPD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
OPPD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
OTP	40 kV and above	100% of Rate A	100% of Rate B	1.07/1.05/0.97	1.10/0.92
PPI	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.05/0.95
SIPC	69 kV and above	100% of Rate A	100% of Rate B	1.07/0.95	1.09/0.91
SMMPA	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
WAPA	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
XEL	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.05/0.92

Notes

- 1: PSS[®]E Rate A, Rate B or Rate C
- 2: Limits dependent on nominal bus voltage
- 3: For facilities in Cedar Falls Utilities or Ames Municipal Utilities, post-contingency voltage limits are 1.05/0.94 for >200 kV, and 1.05/0.93 for others.

2.5 Performance Criteria

A branch is considered as a thermal injection constraint if the branch is loaded above its applicable normal or emergency rating for the post-change case, and any of the following conditions are met:

- 1) the generator (NR/ER) has a larger than 20% DF on the overloaded facility under post contingent condition or 5% DF under system intact condition, or
- 2) the megawatt impact due to the generator is greater than or equal to 20% of the applicable rating (normal or emergency) of the overloaded facility, or
- 3) the overloaded facility or the overload-causing contingency is at generator's outlet, or
- 4) for any other constrained facility, where none of the study generators meet one of the above criteria in 1), 2), or 3), however, the cumulative megawatt impact of the group of study generators (NR/ER) is greater than 20% of the applicable rating, then only

those study generators whose individual MW impact is greater than 5% of the applicable rating and has DF greater than 5% (OTDF or PTDF) will be responsible for mitigating the cumulative MW impact constraint.

A bus is considered a voltage constraint if both of the following conditions are met. All voltage constraints must be resolved before a project can receive interconnection service.

- 1) the bus voltage is outside of applicable normal or emergency limits for the post-change case, and
- 2) the change in bus voltage is greater than 0.01 per unit.

All DPP 2016 February Phase 2 West Area study generators must mitigate thermal injection constraints and voltage constraints in order to obtain unconditional Interconnection Service.

Further, all generators requesting Network Resource Interconnection Service (NRIS) must mitigate constraints found by using the deliverability algorithm, to meet the system performance criteria for NERC category P0-P1 events, if the constraint demonstrates an incremental flow caused by the generator equal to or greater than 5% of the generator's maximum dispatch level in each case.

2.6 Reactive Power Requirements for Non-Synchronous Generation (FERC Order 827)

For non-synchronous generation projects in the DPP 2016 February West Area study group, if they do not have signed Generator Interconnection Agreement (GIA) or Provisional GIA (PGIA) by September 21, 2016, they are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

Except for generation projects J498, J499, J500, and J529, all other non-synchronous generation projects in this study group are required to meet the reactive power requirements per FERC Order 827.

Collector system and shunt compensation of DPP West projects are modeled, which are listed in Appendix A.1, Table A-3. An analysis was performed to study the reactive power requirements (FERC Order 827) for the non-synchronous generation projects in the DPP 2016 February West study group. The analysis was performed as follows:

Step 1: Verify whether total dynamic reactive power (reactive power from generators and dynamic compensation devices) in the plant can meet the dynamic reactive power range of 0.95 leading to 0.95 lagging at the generator terminal bus. The verification in Step 1 was performed when generator data was submitted and modeled.

Step 2: Verify whether total reactive power (reactive power from generators, dynamic compensation devices, and static compensation devices) in the plant can meet the dynamic reactive power range of 0.95 leading to 0.95 lagging at the high-side of the generator substation. The testing procedure in Step 2 is described in the following:

- Lock the high-side of the generator substation at 1.0 pu voltage by adding a fictitious SVC. This is to ensure that the test result is not affected by the system condition.
- Lock the reactive power output of the generator to the maximum limit (Q_{max}). Make sure all shunt compensation devices within the substation are at the maximum

capacitive output. Adjust transformer tap to ensure bus voltages within the substation are within 0.95 – 1.05 pu range. Measure real power and reactive power from the generator plant to the high-side of the generator substation. Calculate the power factor to verify if it meets the 0.95 lagging requirement.

- Lock the reactive power output of the generator to the minimum limit (Qmin). Make sure all shunt compensation devices within the substation are at the maximum reactive output. Adjust transformer tap to ensure bus voltages within the substation are within 0.95 – 1.05 pu range. Measure real power and reactive power from the generator plant to the high-side of the generator substation. Calculate the power factor to verify if it meets the 0.95 leading requirement.

Appendix C lists reactive power requirement analysis results for the DPP West generation projects. The results are summarized as following:

- J498, J499, J500, and J529 are exempted from the FERC Order 827 reactive power requirements.
- All other non-synchronous generation projects can meet the reactive power requirements per FERC Order 827.

2.7 2022 Contingency Analysis Results

The incremental impact of the proposed interconnection on individual facilities was evaluated by comparing flows and voltages between benchmark case (without DPP West Area projects) and study case (with DPP West Area projects). Analysis was performed in the 2022 summer shoulder and summer peak scenarios using PSS[®]E and PSS[®]MUST.

2.7.1 2022 Summer Shoulder Contingency Analysis Results

2.7.1.1 System Intact Conditions

For NERC category P0 (system intact) conditions, thermal constraints are listed in Table D-1, and voltage constraints are listed in Table D-2.

2.7.1.2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies.

One category P1 contingency (Big Stone South - J488&J489 POI 345 kV) and two category P2-P7 contingencies including the same P1 contingent branch were solved in the benchmark case but not solved in the study case (Table D-7). Voltage collapse was observed in the study case. Mitigation plan is required for J488 and J489 projects.

Short Circuit Ratio (SCR) is a measure of system strength relative to the rating of the renewable plant and is used by manufacturers to screen for weak grid risks. Composite SCR (CSCR) considers the grid strength as seen by all electrically close converters and is used for renewable plants. CSCR is calculated as the ratio of the composite short-circuit MVA at the points of interconnection (POI) of all renewable plants in a given area to their combined MW rating. When Big Stone South - J488&J489 POI 345 kV is out of service, CSCR of the renewable generation (J436, J437, J488, and J489) is around 0.953 (Table 2-3). Further study is recommended to provide evidence that the J488 and J489 wind farms can operate at the CSCR of 0.953.

Table 2-3: Composite SC MVA and SCR of J488 and J489

System Condition	Composite SC MVA	Composite SCR
P0: System Intact	1617	2.679
P1: Big Stone South-POI of J488, J489	575	0.953

Six category P2-P7 contingencies (Table D-7) were not converged in the benchmark or study cases. No mitigation plan is required for the study projects for these non-converged contingencies.

For P1 contingencies in the 2022 summer shoulder scenario, thermal constraints are listed in Table D-3, and voltage constraints are listed in Table D-4.

For P2-P7 contingencies in the 2022 summer shoulder scenario, thermal constraints are listed in Table D-5, and voltage constraints are listed in Table D-6.

For all non-converged contingencies (Table D-7), the dc thermal results are listed in Table D-8.

2.7.1.3 Worst Thermal Constraints in 2022 Summer Shoulder

Table 2-4 lists worst thermal constraints in the 2022 summer shoulder scenario.

2.7.1.4 Worst Voltage Constraints in 2022 Summer Shoulder

Table 2-5 lists worst voltage constraints in the 2022 summer shoulder scenario.

Table 2-4: 2022 Shoulder Thermal Constraints, Maximum Screened Loading

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Table 2-5: 2022 Shoulder Worst Voltage Constraints

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2.7.2 2022 Summer Peak Contingency Analysis Results

2.7.2.1 System Intact Conditions

For NERC category P0 (system intact) conditions, no thermal or voltage constraints were identified (Table D-9, Table D-10).

2.7.2.2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies. All category P1 contingencies were converged.

For P1 contingencies in the 2022 summer peak scenario, no thermal or voltage constraints were identified (Table D-11, Table D-12).

Seven category P2-P7 contingencies (Table D-15) were not converged, and their dc thermal results are listed in Table D-16. The seven contingencies were not converged in the benchmark or study cases. No mitigation plan is required for the study projects for these contingencies.

For P2-P7 contingencies in the 2022 summer peak scenario, no thermal or voltage constraints were identified (Table D-13, Table D-14).

2.8 Network Upgrades Identified in MISO ERIIS Analysis

Based on the MISO steady state analyses, the highest loading of each facility that violates criteria in the 2022 scenario (SH, SPK) is shown in Table 2-6. Potential thermal network upgrades are also included.

Based on the MISO steady state analyses, the worst voltage constraints are shown in Table 2-7. Potential reactive power network upgrades are also included.

To mitigate voltage collapse under contingencies involving Big Stone South - J488&J489 POI 345 kV line, below two potential solutions have been tested and compared. Details of solution comparisons will be discussed in Section 4.5.1.2.

- Solution 1: Install a (-50, +180) Mvar SVC at Ellendale 230 kV bus, and install a breaker for the fixed line reactor (50 Mvar) at Ellendale 345 kV side.
- Solution 2: Build a 2nd Big Stone South - J488&J489 POI 345 kV line.

When Big Stone South - J488&J489 POI 345 kV is out of service, CSCR of the renewable generation (J436, J437, J488, and J489) is around 0.953. Further study is recommended to provide evidence that the J488 and J489 wind farms can operate at the CSCR of 0.953.

To mitigate voltage constraint identified at Lathrop 161 kV bus (Ameren area), 1×25 Mvar switched shunt capacitor bank at 161 kV bus is required to be installed.

To mitigate voltage constraints identified in the areas of Hickory Creek and Salem (ITCM area), the Hickory Creek-Cardinal 345 kV project (MVP project 3127 with in-service date of 12/31/2023) is required. Also transformer tap should be locked at neutral position for the two 345-161 kV transformers at Salem.

To mitigate voltage constraints identified in the areas of Blackhawk, Montezuma, J530 POI, and J438 POI (MEC area), 150 Mvar switched shunt capacitor bank at Blackhawk 345 kV bus, 1×50 Mvar switched shunt capacitor bank at Midport 161 kV bus, 100 Mvar switched shunt capacitor bank at Montezuma 345 kV bus, 1×25 Mvar switched shunt capacitor bank at J438 POI 161 kV bus are required to be installed.

Costs of the thermal network upgrades identified in the MISO steady state analysis for the 2022 scenario are listed in Table 2-8.

Table 2-6: Network Upgrades for Thermal Constraints in 2022 Scenario, Maximum Screened Loading

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Table 2-7: Network Upgrades for Voltage Constraints in 2022 Scenario, Worst Voltage Constraints

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Table 2-8: Thermal Network Upgrades and Cost Identified in MISO Steady State Analysis

Constraint	Owner	Mitigation	Cost (\$)
J530 POI-Hills 345 kV	MEC	MEC: Structure replacements	\$300,000
Split Rock-White 345 kV	XEL WAPA	Line is currently rated 1075 MVA for SN/SE no mitigation required	\$0
Brookings Co-White 345 kV #2	XEL WAPA	XEL: Line is limited by WAPA terminal limits. NSP's equipment able to handle the required loading. \$0 WAPA: NU is not required unless it is identified as constraint in affected system study.	\$0
Hazel Creek 345-230-13.8 kV xfmr #6	XEL	Upgrade Hazel Creek TR9 to 672 MVA	\$4,900,000
Panther-McLeod 230 kV	GRE	Remediate 4 Structures	\$1,000,000
Johnson Jct-Ortonville 115 kV	MRES GRE OTP	OTP: Ortonville terminal equipment requires replacement. OTP's portion of the 0.56 mile line will be required to be fully rebuilt with a larger conductor. \$500,000 GRE: GRE owned switch at Johnson Jct needs to be replaced. \$350K MRES: The MRES owned section (majority of the line) will have to be rebuilt for rating increases. \$16M	\$16,850,000
Johnson Jct-Morris 115 kV	GRE WAPA	GRE: Rebuild 15 miles of 115 kV line from Johnson Junction to Morris. \$8,175,000 WAPA: NU is not required unless it is identified as constraint in affected system study.	\$8,175,000
Big Stone 115-230-13.8 kV xfmr	OTP	Terminal equipment will be required to be replaced at the substation.	Incl. below
Big Stone 230-115-13.8 kV xfmr	OTP	Transformer and Terminal equipment will be required to be replaced at the substation.	\$3,250,000
Big Stone-Big Stone South 230 kV #1	OTP	Line will be required to be fully rebuilt with a larger conductor as well as replacement of terminal equipment at Big Stone.	\$1,300,000
Big Stone-Big Stone South 230 kV #2	OTP	Line will be required to be fully rebuilt with a larger conductor.	\$1,250,000
Big Stone-Browns Valley 230 kV	OTP MDU	OTP: owns Big Stone and Browns Valley terminal equipment which is sufficient to handle the flow seen in the study. \$0 MDU: Conductor clearance mitigation. \$350,000	\$350,000
Big Stone-Blair 230 kV	OTP BEPC	OTP: Some OTP terminal equipment will be required to be replaced at the Big Stone 230 kV sub. OTP's total cost is \$80,000. BEPC: NU is not required unless it is identified as constraint in affected system study.	\$80,000
Hankinson-Wahpeton 230 kV	OTP	Terminal equipment at both Hankinson and Wahpeton substations will be required to be replaced. In addition, conductor clearance mitigations will be required for the line.	\$2,300,000

Steady-State Analysis

Constraint	Owner	Mitigation	Cost (\$)
Hankinson-Forman 230 kV	OTP	Terminal equipment at the Forman substation will be required to be replaced. In addition, conductor clearance mitigations will be required for the line.	\$2,700,000
Wahpeton-Fergus Falls 230 kV	OTP MRES	Terminal equipment at Wahpeton substation will be required to be replaced. In addition, conductor clearance mitigations will be required for the line.	\$750,000
Oakes-Forman 230 kV	OTP	Terminal equipment at Oakes and Forman substations will be required to be replaced. In addition, conductor clearance mitigations will be required for the line.	\$1,600,000
Oakes-Ellendale 230 kV	OTP MDU	OTP: Terminal equipment at Oakes will be required to be replaced. In addition, conductor clearance mitigation will be required for the line. \$2,850,000 MDU: The Ellendale MVP 230 kV bus will be built to 796 MVA. \$0	\$2,850,000
Marshalltown-Blairstown 115 kV	ITCM	Sag limit at 162F. Raise/replace structures and replace bus conductor at Blairstown. New rating of 120 MVA.	\$2,700,000
Blairstown-Prairie Crk 115 kV	CIPCO ITCM	CIPCO: No Mitigation Needed. \$0 ITCM: Clearance limit at 153F. Raise structures to reach a minimum rating of 114 MVA	\$2,700,000
Lime Creek 161 kV bus tie	ITCM	Not a valid constraint. Bus is rated at 448 MVA	\$0
Hazleton-Washburn 161 kV	ITCM MEC	MEC: Reconductor with structure replacements as appropriate and substation terminal equipment upgrades. \$16.7 M ITCM: Terminal limit is 325 MVA.	\$16,700,000
Boone Jct-Sub T Fort Dodge 161 kV	MEC CIPCO	CIPCO: No Mitigation Needed. \$0 MEC: Reconductor with structure replacements as appropriate and substation terminal equipment upgrades. \$7,000,000	\$7,000,000
Traer-Dysart 161 kV	ITCM	New rating based on revising rating methodology is 297/297	\$0
Parnell-J438 POI 161 kV	MEC ITCM	MEC: Structure replacements. \$100,000 ITCM: ITCM Limit is 335 MVA at Parnell	\$100,000
Arnold-Fairfax 161 kV	ITCM	New rating based on revising rating methodology is 335/335.	\$0
Poweshiek-Reasnor 161 kV	ITCM	New rating based on revising rating methodology is 353/353	\$0
Arnold-Morgan Valley 345 kV	ITCM	New rating based on revising rating methodology is 1043/1043	\$0
Morgan Valley-Tiffin 345 kV	ITCM MEC	MEC: No Mitigation Required. Existing MEC rating is 961 MVA. \$0 ITCM: ITCM future facilities at Morgan Valley will be constructed to a rating greater than what is required for the constraint. \$0	\$0
Raun-S3451 345 kV	MEC OPPD	MEC: No mitigation required. Existing MEC emergency rating is 1152 MVA. \$0 OPPD: NU is not required unless it is identified as constraint in affected system study.	\$0

Steady-State Analysis

Constraint	Owner	Mitigation	Cost (\$)
Grimes-Sycamore 345 kV #2	MEC	Sycamore substation terminal equipment upgrades.	\$1,200,000
Grimes-Beaver Creek 345 kV	MEC	Structure replacements.	\$600,000
Granger Tap-108th & 54th 161 kV	MEC	Reconductor short section of line and terminal equipment upgrades.	\$100,000
Bondurant-Montezuma 345 kV	MEC	Structure replacements	\$200,000
Webster-Sub T Ft Dodge 161 kV	MEC	Reconductor with structure replacements as appropriate.	\$6,250,000
Franklin-Wall Lake 161 kV	MEC	No mitigation required. P6 CON does not need to be mitigated for GI.	\$0
Blair-Granite Falls 230 kV	WAPA	NU is not required unless it is identified as constraint in affected system study.	\$0

Affected System Steady-State Analysis

Steady state analyses were performed to identify constraints on affected systems.

3.1 Affected System Analysis for CIPCO and MPC Companies

Per CIPCO and MPC's Affected System Planning Criteria, transmission facilities which meet all of the following three (3) conditions are considered as constraints:

- 1) the branch is loaded above its applicable normal or emergency rating for the post-change case, and
- 2) the generator has a larger than 3% DF on the overloaded facility under post contingent condition or 5% DF under system intact condition, and
- 3) the loading increase of the overloaded facility is greater than 1 MVA compared with that in the pre-change case under system intact or contingency conditions.

AC contingency analysis was performed for this CIPCO and MPC affected system analysis, using the DPP 2016 February West Phase 2 benchmark (pre-change) cases and study (post-change) cases for the 2022 summer peak and summer shoulder scenarios. All NERC category P0-P7 contingencies described in Section 2.3 were simulated. The CIPCO and MPC affected systems were monitored.

The additional thermal constraints in the affected systems identified in this CIPCO and MPC affected system analysis are listed in Appendix E.1. These include the thermal constraints for the 2022 summer peak and shoulder scenarios. Whether these constraints will be mitigated or not will be determined by the affected transmission owning company's planning criteria.

The highest loading of CIPCO and MPC affected system thermal constraints and potential network upgrades are listed in Table 3-1 for the 2022 summer shoulder scenario. No CIPCO or MPC's thermal constraints were identified in the 2022 summer peak scenario.

Table 3-1: Additional Thermal Constraints Identified in CIPCO/MPC Affected System Analysis and their Highest Loadings

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3.2 J525 GRE Local Planning Criteria Analysis

3.2.1 Study Summary

GRE performed the local planning criteria analysis for J525 based on GRE's LPC. The GRE local planning criteria analysis details can be found in Appendix E.2.

The analysis identified several facility constraints which are owned by non-GRE companies. These non-GRE constraints are not required to be mitigated by the J525 project.

Before the J525 can be allowed to interconnect, the developer and/or system manufacturer must provide assurance that the generation can operate while connected to the relatively weak, local transmissions system at the Point-of-Interconnection.

3.2.2 Study Details

3.2.2.1 Transmission Line Upgrades Identified

Constraints	Owner	Length (miles)
Hadley - Lake Wilson 69 kV	Xcel	2.5
Hadley - Slayton 69 kV	Xcel	4.3
Slayton (GRE) - Slayton (Xcel) 69 kV	Xcel	0.5
Lake Sarah Tap - Tracy Switch 69 kV	Xcel	6.0
Total		13.3

The total cost of the transmission line upgrades is approximately \$10 million. This cost is based on rebuilding the existing transmission with new 477 ACSR transmission line.

3.2.2.2 Fenton Transformer Replacement

The overload of the Fenton 115-69 kV transformer can be mitigated by the addition of a second 115-69 kV transformer at Fenton. The planning estimated cost for the transformer addition is \$3.9 million.

3.2.2.3 Low Short Circuit Ratio

Based on the introduction of a 33 MW generator, the short-circuit ratio (SCR) is between 8.72 and 18.3. GRE-performed stability analysis indicates that the inverter control may have some sensitivity to the low SCR. Additional study may be necessary to assure that the inverter control can operate properly at the requested POI.

3.2.2.4 Possible Inverter Control Issues in Low SCR Conditions

GRE has performed a transient stability analysis in the local area around the proposed interconnection of a 33 MW Solar PV project at the Lake Wilson substation. The results indicate that there are possible control issues due to the relatively weak, local transmission system. The inverter tripped on overvoltages at the inverter bus due to the restarting of the PV inverter into a relatively large system impedance.

In summary, before the interconnection can be allowed to interconnect, the developer and/or system manufacturer must provide assurance that the generation can operate while connected to the relatively weak, local transmissions system at the Point-of-Interconnection.

3.3 OTP Local Planning Criteria Analysis

3.3.1 Introduction

The OTP LPC study is performed for the DPP 2016 February West Area Phase 2 study to adhere to OTP Local Planning Criteria. This LPC analysis is required in addition to MISO's standard DPP analysis.

Three additional scenarios are analyzed in this OTP LPC study:

1. LPC Summer Peak
2. LPC Summer Shoulder
3. LPC Light Load No Wind (LLNW)

3.3.2 Model Development

3.3.2.1 OTP LPC Study Cases

The OTP LPC study cases representing 2022 summer shoulder and summer peak conditions were developed from the DPP 2016 February Phase 2 corresponding summer shoulder (SH) and summer peak (SPK) cases, with the following additional changes:

- Turn on Big Stone generator and its station service load. Dispatch and lock its generation at Pmin (163 MW) in summer shoulder case. No change is made on Big Stone generation in summer peak case since it is dispatched.
- Turn on Coyote generator and its station service load. Dispatch and lock its generation at Pmin (217 MW) in summer shoulder case. No change is made on Coyote generation in summer peak case since it is dispatched.
- Dispatch and lock G736 generation at 90% of its Pmax in shoulder case and 15.6% of its Pmax in peak case. G736 has GIA status as completed.
- Dispatch and lock J436, J437, and J442 generation at 100% of its Pmax in shoulder case and 15.6% of its Pmax in peak case.
- Power mismatch was balanced by generation in the MISO Classic.

The OTP LPC study case representing 2022 light load no wind condition was developed from the OTP LPC summer shoulder case, with the following additional changes:

- Light load profile was applied on OTP loads
- Wind generation in Table E-3 was turned off.
- Wind generation in DPP 2016 February Phase 2 study was turned off, but their GSUs, collector systems, shunt compensation devices, substation transformers, and interconnection tie lines are still in service.
- Power mismatch was balanced by generation in the MISO Classic.

3.3.2.2 OTP LPC Benchmark Cases

OTP LPC benchmark cases representing summer shoulder and summer peak conditions were created by removing the DPP West Area generating facilities from the study cases. The MISO Classic was used for power balance, where generation was scaled uniformly.

OTP LPC benchmark case representing light load no wind condition was created by removing the DPP West Area generating facilities (GSUs, collector systems, shunt compensation devices, substation transformers, and interconnection tie lines) from the study case.

3.3.3 OTP LPC Study Areas

The OTP LPC study areas include OTP (620), Xcel (600), MP (608), GRE (615), WAPA (652), BEPC-SPP (659), MDU (661), BEPC-MISO (663), and MH (667). The contingencies in these study areas (Section 2.3) are all simulated, the applicable thermal and voltage performance criteria are applied (Sections 2.4 and 2.5).

3.3.4 OTP LPC Contingency Analysis Results

3.3.4.1 Summer Shoulder Contingency Analysis Results

For NERC category P0 (system intact) conditions, thermal constraints are listed in Table E-4. No voltage constraints were identified (Table E-5).

One category P1 contingency (Big Stone South - J488&J489 POI 345 kV) and two category P2-P7 contingencies including the same P1 contingent branch were solved in the benchmark case but not solved in the study case (Table E-10). Voltage collapse was observed in the study case. Mitigation plan is required.

One category P2-P7 contingencies (Table E-10) were not converged in the benchmark or study cases. No mitigation plan is required for the study projects for these non-converged contingencies.

For NERC category P1 contingencies, thermal constraints are listed in Table E-6, and voltage constraints are listed in Table E-7.

For NERC category P2-P7 contingencies, thermal constraints are listed in Table E-8, and voltage constraints are listed in Table E-9.

For all non-converged contingencies (Table E-10), the dc thermal results are listed in Table E-11.

3.3.4.2 Summer Peak Contingency Analysis Results

For NERC category P0 (system intact) conditions, no thermal or voltage constraints were identified (Table E-12, Table E-13).

For NERC category P1 contingencies, no thermal or voltage constraints were identified (Table E-14, Table E-15).

Two category P2-P7 contingencies (Table E-18) were not converged, and their dc thermal results are listed in Table E-19. The two contingencies were not converged in the benchmark or study cases. No mitigation plan is required for the study projects for these contingencies.

For NERC category P2-P7 contingencies, no thermal or voltage constraints were identified (Table E-16, Table E-17).

3.3.4.3 Light Load No Wind Contingency Analysis Results

The purpose of light load no wind (LLNW) study is to identify whether generating facilities in DPP 2016 February Phase 2 study will aggravate high voltage issue identified in areas of Big Stone and Ellendale.

All bus voltages are below applicable high voltage limits. Under a few NERC Category P1-P7 contingencies, several bus voltages are above 1.05 p.u. but below their applicable high voltage limits (Appendix E.3.4).

3.3.5 Network Upgrades Identified in OTP LPC Analysis

The highest loading of OTP local planning criteria thermal constraints and potential network upgrades are listed in Table 3-2. Besides the thermal constraints and NUs identified in MISO ACCC analysis, additional thermal constraints and NUs identified in the OTP LPC analysis are listed in Table 3-3.

Low voltage constraints identified in the OTP LPC analysis can all be mitigated by building a 2nd Brookings County - J493-J510 POI 345 kV line.

No high voltage constraints were identified in the light load no wind (LLNW) study. Under a few contingencies, several bus voltages are above 1.05 p.u. but below their applicable high voltage limits.

Table 3-2: Thermal Constraints Identified in OTP LPC Analysis and their Highest Loadings

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Table 3-3: Network Upgrades for Constraints Identified Only in OTP LPC Analysis

Constraint	Owner	Mitigation	Cost (\$)	Comments ¹
Big Stone-Highway 12 115 kV	OTP GRE	OTP: All of OTP's equipment is sufficient for the 244.1 MVA flow seen in this study. GRE: Complete reconstruction of the line and new equipment at Highway 12. \$3,100,000	\$3,100,000	New NU. Cost is \$3,100,000 ¹
Highway 12-Ortonville 115 kV	OTP GRE	OTP: All of OTP's equipment is sufficient for the 235.3 MVA flow seen in this study. GRE: Complete reconstruction of the line and new equipment at Highway 12. \$6,900,000	\$6,900,000	New NU. Cost is \$6,900,000 ¹
Big Stone-Blair 230 kV	OTP BEPC	More NUs are required per OTP LPC. OTP: Conductor clearance mitigation and terminal equipment at Big Stone. OTP's portion \$800,000. BEPC: NU is not required unless it is identified as constraint in affected system study.	\$800,000	NU cost increases by \$720,000 ¹
Hankinson-Wahpeton 230 kV	OTP	More NUs are required per OTP LPC. Complete line rebuild and terminal equipment at Hankinson and Wahpeton.	\$21,950,000	NU cost increases by \$19,650,000 ¹
Oakes-Ellendale 230 kV	OTP MDU	More NUs are required per OTP LPC. OTP: Full line rebuild and terminal equipment at Oakes. OTP's cost \$20,500,000. \$20,500,000 MDU: The Ellendale MVP 230 kV bus will be built to 796 MVA. \$0	\$20,500,000	NU cost increases by \$17,650,000 ²

Note 1: The new NUs identified in OTP LPC study are not required with 2nd J510-Brookings County 345 kV line.

Note 2: The new NUs identified in OTP LPC study are not required with 2nd J488&J489-Big Stone South 345 kV line.

3.4 PJM Affected System AC Contingency Analysis

The PJM affected system analysis details (dated on 01/30/2018) can be found in Appendix E.4.

3.4.1 Steady State Study Results

3.4.1.1 Overload on Twin Branch–Argenta 345 kV line

To relieve the Twin Branch–Argenta 345 kV line overload:

- a. A sag check will be required for the ACSR ~ 954 ~ 45/7 ~ RAIL - Conductor Section 1 to determine if the line section can be operated above its emergency rating of 1409 MVA. The result could prove that no additional upgrades are necessary, that some upgrades on the circuit are necessary, or that the entire 52.0 mile section of line would need to be rebuilt. PJM Network Upgrade N5240.
- b. MISO end rating (1691 MVA) is sufficient through 2016 February MISO DPP.

Cost Estimate: Estimated Cost for the Sag Study: \$208,000. If deemed necessary to rebuild section of line, Estimated Cost: \$104.0 Million.

The cost allocation is as follows:

Queue	MW contribution	% of Cost	Cost (\$208,000)
J498	17.44	28.84	59,991
J499	17.2	28.44	59,165
J526	15.82	26.16	54,418
J528	10.008	16.55	34,426

3.4.1.2 Overload on Benton–T-094 345 kV line

To relieve the Benton–T-094 345 kV overload:

- a) Reconductor or rebuild (depending on the existing structures) the portions of 345 kV lines between the Benton Harbor and Sagreto 345 kV substations. PJM Network Upgrade N5106.

Estimated Cost for rebuilding the line: \$19,000,000

Estimated Cost for reconductoring the line: \$13,500,000

The cost allocation is as follows:

Queue	MW contribution	% of Cost	Cost (\$19 M)
J412	8.44	4.88	0.927
J515	20.836	12.05	2.289

Queue	MW contribution	% of Cost	Cost (\$19 M)
J498	20.48	11.84	2.250
J499	20.21	11.69	2.220
J500	23.52	13.60	2.584
J526	18.55	10.73	2.038
J527	14.69	8.49	1.614
J529	15.29	8.84	1.680
J530	15.65	9.05	1.719
J534	15.29	8.84	1.680

The new expected ratings upon completion of PJM Network upgrade N5106 are no longer sufficient; therefore, additional reinforcements are required in order to resolve this overload.

Additional Reinforcement: Replace the Benton H wavetrap. Replace the Benton H Line Riser and Replace the Segreto Wave trap. PJM Network Upgrades N5312.1, N5312.2, and N5312.3. Cost: \$1,000,000.

The cost allocation for the additional reinforcements is shown below:

Queue	MW contribution	% of Cost	Cost (\$1 M)
J498	20.48	14.25	0.143
J499	20.21	14.07	0.141
J500	23.52	16.37	0.164
J526	18.55	12.91	0.129
J527	14.69	10.22	0.102
J529	15.29	10.64	0.106
J530	15.65	10.89	0.109
J534	15.29	10.64	0.106

3.4.1.3 Overload on Pleasant Prairie–Zion Station R- Zion EC 345 kV line

To relieve the Pleasant Prairie–Zion Station R- Zion EC 345 kV overload:

- The MISO-end of the Pleasant Prairie–Zion Station R 345 kV line SE rating is 1528 MVA and not sufficient. This constraint is expected to be relieved when the Pleasant Prairie generating units 1 and 2 deactivate in April 2018. If the Pleasant Prairie generating units 1 and 2 were to withdrawal their deactivation request and continue to generate, any queue projects who contribute to this constraint may be responsible for mitigating the constraint.

- b) The ComEd SSTE rating of the Zion EC–Zion Station R 345 kV line is 1251 MVA and not sufficient. This constraint is expected to be relieved when the Pleasant Prairie generating units 1 and 2 deactivate in April 2018. If the Pleasant Prairie generating units 1 and 2 were to withdrawal their deactivation request and continue to generate, any queue projects who contribute to this constraint may be responsible for mitigating the constraint.

Per PJM cost allocation rules, the DPP 2016 February West projects presently do not receive any cost allocation for this upgrade.

3.4.1.4 Study Summary

Multiple projects in the MISO DPP 2016 February West Area Phase 2 study contribute loading to the overloads in the PJM system. These projects are responsible for the cost of Network Upgrades per PJM cost allocation rules.

3.4.2 J475 Dynamic Stability Study Results

PJM performed a dynamic simulation analysis of J475 as part of the overall system impact study.

For all fault contingencies tested, the system with J475 included was transiently stable and the J475 generator was able to ride through the faults (except for faults where protective action trips the generator(s)). However, post fault bus voltages at the POI and Ottumwa, Hills, Montezuma, Sub 92, Louisa, Walcott 345 kV buses did not recover to pre-fault values for several stuck breaker faults. Model might need to be tuned to resolve the issue.

No stability or voltage recovery issues were caused due to the model on PJM system. Therefore, no mitigations were found to be required.

3.4.3 J530 Dynamic Stability Study Results

PJM performed a dynamic simulation analysis of J530 as part of the overall system impact study.

For all fault contingencies tested, the system with J530 included was transiently stable and the J530 generator was able to ride through the faults (except for faults where protective action trips the generator(s)). However, post fault J530 terminal voltage does not stabilize – continues ramping up even after 20s. Model might need to be tuned to resolve the issue.

No stability or voltage recovery issues are caused due to the model on PJM system. Therefore, no mitigations were found to be required.

3.5 SPP Affected System AC Contingency Analysis

Southwest Power Pool (SPP) conducted an Affected System Interconnection System Impact Study (ASISIS) to evaluate potential impacts to the SPP Transmission System related to the interconnection of generators on the Mid-Continent Independent System Operation (MISO) Transmission System.

Power flow analysis has indicated that for the power flow cases studied, the SPP Transmission System is impacted by the interconnection of the MISO DPP 2016 February West Phase 2 generation. These impacts are described in Section 8 and Section 9 of Appendix E.5. Steady state thermal and voltage impacts can be mitigated by determinations listed in Table 9A and Table 9B of Appendix E.5. Within the scope of this ASISIS, SPP coordinated with associated SPP Transmission Owners and MISO Transmission Owners for planning level estimates for mitigation of constraints requiring transmission reinforcements and will require Affected System Facilities Study (ASFS) if the MISO Interconnection Request(s) impacting the SPP transmission system proceed forward.

Certain constraints identified can be mitigated with either previously approved SPP or MISO upgrades. These include the Gentleman Generation Station (GGS) – Thedford - Holt 345 kV (R-Plan) transmission line and MISO MVP projects which are scheduled to be placed in service by 2023. Full Interconnection Service for these requests may need to be delayed until that time period.

If the MISO Interconnection Requests proceed, additional analysis will be required through an ASFS. The purpose of the ASFS is to refine the scope, cost estimates, and expected time of construction of Network Upgrades.

Potential mitigations for constraints identified in the SPP affected system are listed in Table 3-4.

MISO Generation Interconnection projects that come into service prior to completion of required Network Upgrades on the SPP transmission system will be subject to SPP's study process for Limited Operation. SPP updates the output limits on all Generator Interconnection Agreements when events occur on the Transmission System that are listed in the Limited Operation Impact Study for that generator to account for changing transmission and generation assumptions. SPP will coordinate project output limitations with MISO on an annual basis or more often as events are needed, and SPP will provide MISO the list of conditions that will be added to MISO Generator Interconnection Agreement.

The Limited Operation Amount determined by SPP impacts are listed in Table 3-5. It should be noted that the amounts listed below may not be the most limited as determined by MISO.

The SPP affected system analysis results (dated on 3/5/2018) for this study are in Appendix E.5.

Table 3-4: SPP Suggested Mitigations for DPP 2016 February West Area Study

Mitigation	Area	Purpose	ERIS and / or NRIS Affecting Requests	Estimated Cost
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	NPPD/KCPL	Required to mitigate COOPER - ST JOE 345KV CKT 1 NRIS thermal constraints for system intact and contingency conditions	NRIS: J432, J460, J475, J485, J493, J495, J498, J499, J500, J504, J506, J510, J514, J523, J524, J525, J526, J527, J528, J529, J530, J534, J535	\$301,000,000
	NPPD/OPPD/KCPL	Required to mitigate COOPER_S flowgate for system intact conditions	NRIS: J432, J460, J475, J485, J493, J495, J498, J499, J500, J504, J506, J510, J514, J523, J524, J525, J526, J527, J528, J529, J530, J534, J535	
	NPPD/OPPD/KCPL	Required to mitigate various steady state voltage violations observed in NPPD, OPPD, KCPL areas after SPP previously allocated, MISO MVP, and MISO current study upgrades are in service. Please refer to Appendix C-V for the incremental voltage constraints.	NRIS: J432, J460, J475, J485, J493, J495, J498, J499, J500, J504, J506, J510, J514, J523, J524, J525, J526, J527, J528, J529, J530, J534, J535	
Replace terminal equipment	WAPA	Required to mitigate APPLIEDORN 4 - WATERTOWN 230 KV CKT 1 NRIS thermal constraints for contingency conditions.	NRIS: J432, J460, J493, J510, J526	\$1,000,000
Rebuild approximately 42 miles of 230kV	WAPA	Required to mitigate BLAIR - GRANITE FALLS 230KV CKT 1 NRIS thermal constraints for contingency conditions.	NRIS: J432, J460, J493, J510, J526	\$33,000,000
Rebuild approximately 3 miles of 230kV	WAPA	Required to mitigate GRANITE FALLS – MN VALLEY TAP 230KV CKT 1 NRIS thermal constraints for contingency conditions.	NRIS: J432, J460, J493, J510, J526	\$2,500,000
Terminal equipment upgrade	WAPA/XEL	Required to mitigate SPLIT ROCK - WHITE 345KV CKT 1 NRIS thermal constraint for contingency conditions to at least 1210MVA.	NRIS: J432, J460, J485, J493, J510, J523, J525, J526	\$1,000,000

Table 3-5: Interconnection Service Availability based on Constraints

MISO Gen #	Requested ERS (MW)	Requested NRIS (MW)	ERIS Availability (MW)	NRIS Availability (MW) for the entire cluster (All Requests must reduce NRIS to this amount or lower to avoid NRIS cost assignment) ¹	NRIS Availability (MW) for individual requests (Each Request must reduce NRIS to this amount or lower to avoid NRIS cost assignment regardless of what other customers do) ²
J432	98	98	98	32	0
J460	200	200	200	65	0
J475	200	200	200	187	0
J485	46.85	46.85	46.85	44	0
J488	151.8	0	151.8	N/A	N/A
J489	151.8	0	151.8	N/A	N/A
J493	150	150	150	17	0
J495	200	200	200	187	0
J498	340	340	340	319	73
J499	340	340	340	319	110
J500	500	500	500	469	312
J504	50	50	50	46	0
J506	200	200	200	187	0
J510	266 sum / 284.5 win	266	266 sum / 284.5 win	32	0
J514	30 sum / 65 win	30	30 sum / 65 win	61	0
J523	50	50	50	46	0
J524	100	100	100	93	0
J525	33	33	33	30	0
J526	300	300	300	34	0
J527	250	250	250	234	57
J528	200	200	200	187	8
J529	250	250	250	234	0
J530	250	250	250	234	0
J534	250	250	250	234	0
J535	210	210	210	197	0

Affected System Steady-State Analysis

Note 1: Amount of NRIS service that is available at no NRIS upgrade cost assignment if all customers reduce their NRIS requests to this amount or lower. Reducing NRIS to this amount may not eliminate the risk of NRIS cost assignment unless all customers reduce NRIS to this amount or lower.

Note 2: Amount of NRIS service that is available at no NRIS upgrade cost assignment if other customers do NOT reduce their NRIS requests. Reducing NRIS to this amount or lower should result in no NRIS upgrade cost assignment regardless of what other customers choose to do. An amount of "0" means that NRIS must be dropped completely in order to avoid the risk of cost assignment.

Stability Analysis

Stability analysis was performed to evaluate the transient stability and impact on the region of the generating facilities in the DPP 2016 February West Area group.

4.1 Procedure

4.1.1 Computer Programs

Stability analysis was performed using PSS®E revision 33.10.

4.1.2 Study Methodology

Study stability package representing 2022 summer shoulder (SH) conditions with generating facilities in the DPP 2016 February Phase 2 West Area group (DPP West Area) was developed from MTEP17 stability package with LBA dispatch. Benchmark case was created by removing the DPP West Area generating facilities from the study case. Disturbances were simulated to evaluate the transient stability and impact on the region of the generating facilities. If a simulation for the study case violates MISO transient stability criteria or local TOs' planning criteria, the simulation was repeated on the benchmark case to assess the impact of the generating facilities on the violation.

4.2 Case Development

4.2.1 Study Case

Study case representing 2022 summer shoulder (SH) condition was developed from the MTEP17 stability package with LBA dispatch.

The stability study case for DPP 2016 February Phase 2 study was created as described in Section 2.2.1. The interface transfer levels are summarized in Table 4-1.

Table 4-1: Interface Transfer Levels in 2022 Stability Study Cases

Interface	2022 SH Case (MW)
MHEX	492
MWEX	1275
Arrowhead – Stone Lake 345 kV	484

4.2.2 Benchmark Case

The DPP West Area generating facilities as described in Table A-1 (Appendix A.1) were removed from the study case. MISO Classic was used for power balance, where generation was scaled uniformly.

4.3 Disturbance Criteria

The stability simulations performed as part of this study considered all the regional and local contingencies listed in Table 4-2. Regional contingencies with the pre-defined switching sequences were selected from the MISO MTEP17 study; switching sequences for local contingencies were developed based on the generic clearing times shown in Table 4-3. The admittance for local single line-to-ground (SLG) faults were estimated by assuming that the Thevenin impedance of the positive, negative and zero sequence networks at the fault point are equal.

Table 4-2: Regional and Local Disturbance Descriptions

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Table 4-3: Generic Clearing Time Assumption

Voltage Level (kV)	Primary Clearing Time (cycle)	Backup Clearing Time (cycle)
345 kV	4	11
230 kV	5	13
161/138 kV	6	18
115 kV	6	20
69 kV	8	24

4.4 Performance Criteria

All generators must mitigate the stability constraints listed below in order to obtain any type of Interconnection Service:

- System instability
- Transient voltage constraint
- Damping violation

4.4.1 MISO Criteria

Stability simulation results are evaluated based on the following MISO criteria:

- All on-line generating units are stable
- No unexpected generator tripping
- Post-fault transient voltage limits: 1.2 per unit maximum, 0.7 per unit minimum.
- Per local TOs' planning criteria, specific transient voltage limits are applied to specific buses, areas or companies that have different requirements.

- All machine rotor angle oscillations must be positively damped with a minimum damping ratio of 0.81633% for disturbances with a fault or 1.6766% for line trips without a fault.

A bus is considered a transient voltage constraint if both of the following conditions are met. All transient voltage constraints must be resolved before a project can receive interconnection service.

1. the bus transient voltage is outside of specified transient voltage limits during transient period, and
2. the bus voltage is at least 0.01 per unit worse than the benchmark case voltage for the same contingency.

4.4.2 Local Planning Criteria

4.4.2.1 ATC Local Planning Criteria

ATC has the following local transient voltage recovery criteria. For facilities in the ATC footprint, transient voltage recovery is evaluated based on ATC's local planning criteria.

- Voltage recovery within 80 percent and 120 percent of nominal for between 2 and 20 seconds following the clearing of a disturbance.

4.4.2.2 ITCM Local Planning Criteria

ITCM has the following local transient voltage and damping criteria. For facilities in the ITCM footprint, transient voltages and dampings are evaluated based on ITCM's local planning criteria.

- Voltages at all busses on the Transmission Systems should not drop below 0.70 per unit after the first swing for more than 5 cycles. The duration for the minimum voltage dip starts after the first swing post clearing of fault.
- Voltage at all Transmission System buses should recover to the applicable post-contingency steady-state voltage level (ITCM post-disturbance limits in Table 2-2), within 1.0 second of the clearing of the fault.
- Rotor angle oscillation damping ratios are not to be less than 0.03.

4.4.2.3 MEC Local Planning Criteria

MEC has the following local transient voltage and damping criteria. For facilities in the MEC footprint, transient voltages and dampings are evaluated based on MEC's local planning criteria.

- Generator bus transient voltage limits shall adhere to the high voltage duration and low voltage duration curve in Attachment 2 of NERC PRC-024, which is:
 - o Generator bus transient over voltage limits (after fault clearing): 1.2 pu voltage from 0.0 to and including 0.2 s; 1.175 pu voltage from 0.2 to and including 0.5 s; 1.15 pu voltage from 0.5 to and including 1.0 s; 1.1 pu voltage for greater than 1.0 s.

- o Generator bus transient low voltage limits (after fault clearing): may be less than 0.45 pu voltage from 0 to 0.15 seconds; Voltage shall remain above 0.45 pu from 0.15 to 0.3 s; Voltage shall remain above 0.65 pu from 0.3 to 2.0 s; Voltage shall remain above 0.75 pu from 2.0 to 3.0 s; Voltage shall recover to 0.9 pu after 3 s.
- Load bus transient voltage limits:
 - o Load bus transient over voltage limits (after fault clearing): 1.6 pu voltage from 0.01 to and including 0.04 s; 1.2 pu voltage from 0.04 to and including 0.5 s; 1.1 pu voltage from 0.5 to and including 5 s; and 1.05 pu voltage for greater than 5 s. These voltage limits also apply to buses without loads or generators.
 - o Load bus transient low voltage limits (after fault clearing): may be less than 0.7 pu voltage from 0 to 2 s; Voltage shall remain above 0.7 pu from 2 to 20 s; Voltage shall recover to 0.9 pu after 20 s.
- Angular transient stability minimum damping ratio (ζ) should not be less than 0.03.

4.5 Stability Results

The contingencies listed in Table 4-2 were simulated using the 2022 summer shoulder study case. If a transient stability criteria violation was identified, the same disturbance was repeated in the benchmark case.

Appendix F.2 contains plots of generator rotor angles, generator power output, generator terminal voltages, bus voltages, and branch flows for each simulation. Simulations were performed with a 2.0 seconds steady-state run followed by the appropriate disturbance. Simulations were run for a 12-second duration.

4.5.1 2022 Stability Analysis Results

Stability study results summary is in Appendix F, Table F-1. The following stability related issues were identified.

4.5.1.1 Voltage Collapse and J510 Angle Instability for Loss of Brookings Co - J493-J510 POI 345 kV Line

Voltage collapse and J510 angle instability were identified with the 3PH or SLG faults resulting in tripping of Brookings County - J493-J510 POI 345 kV line (Table 4-4). In the stability model, J510 (CT) is dispatched at 100% of its Pmax (284.5 MW).

Table 4-4: Disturbances Causing Voltage Collapse and J510 Angle Instability

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Several potential solutions have been investigated and compared for mitigating the voltage collapse and J510 angle instability issue, which are discussed below:

Solution 1: Install a (0, 100) Mvar SVC (\$18,710,000) and 80 Mvar switched capacitor (\$3 M) at Big Stone South 345 kV substation (Estimated cost is \$21,710,000).

With the addition of the (0, 100) Mvar SVC, initial oscillations (angle, power, voltage, etc.) after the fault clearing are very big. But the oscillations can damp out within 20 seconds. There are some severe transient voltage violations in OTP area (Big Stone South 230 kV transient low voltage is 0.53 pu, etc.) due to the initial large oscillations. To mitigate the identified transient voltage violations, at least 80 Mvar switched capacitor is required to be installed at Big Stone South 345 kV bus.

Figure 4-1 shows the oscillations of J510 (angle, P, Q) and voltage at Big Stone 230 kV bus with this potential solution under the 3PH Fault “J493&510-POI_3ph_BRKNGCO_345”.

Solution 2: Build a 2nd Brookings County - J493-J510 POI 345 kV 16-mile line (Estimated cost is \$31,400,000).

With the addition of 2nd Brookings County - J493-J510 POI 345 kV line, initial oscillations are small and no stability violations were identified under these disturbances (Table 4-4).

Comparison of the Solutions

With the 2nd Brookings County - J493-J510 POI 345 kV line, some thermal NUs can be eliminated or reduced, which will save cost of \$59,870,000 if the original NRIS constraint is elected to be mitigated, or save cost of \$31,620,000 if the original NRIS constraint is not elected to be mitigated (Table 4-5).

The 2nd Brookings County - J493-J510 POI 345 kV line is better solution on performance and economics to be selected.

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Figure 4-1: J510 Response and Big Stone 230 kV Voltage with 3PH Fault “J493&510-POI_3ph_BRKNGCO_345”

Table 4-5: Thermal NUs to be Eliminated or Reduced by Addition of 2nd 345 kV Line

Constraint	Owner	Without 2nd 345 kV line		With 2nd 345 kV line		Reduced Cost (\$)
		Mitigation	Cost (\$)	Mitigation	Cost (\$)	
Big Stone-Highway 12 115 kV	OTP GRE	OTP: All of OTP's equipment is sufficient for the 244.1 MVA flow seen in this study. GRE: Complete reconstruction of the line and new equipment at Highway 12. \$3,100,000	\$3,100,000	Not Required	\$0	\$3,100,000
Highway 12-Ortonsville 115 kV	OTP GRE	OTP: All of OTP's equipment is sufficient for the 235.3 MVA flow seen in this study. GRE: Complete reconstruction of the line and new equipment at Highway 12. \$6,900,000	\$6,900,000	Not Required	\$0	\$6,900,000
Big Stone-Big Stone South 230 kV #2	OTP	Line will be required to be fully rebuilt with a larger conductor.	\$1,250,000	Not Required	\$0	\$1,250,000
Big Stone-Blair 230 kV (ERIS)	OTP BEPC	OTP: Conductor clearance mitigation and terminal equipment at Big Stone. OTP's portion \$800,000.	\$800,000	OTP: Some OTP terminal equipment will be required to be replaced at the Big Stone 230 kV sub. OTP's total cost is \$80,000.	\$80,000	\$720,000
Hankinson-Wahpeton 230 kV	OTP	Complete line rebuild and terminal equipment at Hankinson and Wahpeton.	\$21,950,000	Terminal equipment at both Hankinson and Wahpeton substations will be required to be replaced. In addition, conductor clearance mitigations will be required for the line.	\$2,300,000	\$19,650,000
Big Stone-Blair 230 kV (NRIS) ¹	OTP BEPC	OTP: Replace the terminal equipment at Big Stone and completely rebuild the Big Stone–Blair 230 kV line. \$28,250,000	\$28,250,000	Not Required	\$0	\$28,250,000
				Total Saved Cost (\$)		\$59,870,000

Note 1: If the impacted projects do not elect to mitigate this NRIS constraint, the total saved cost by the 2nd line solution will be \$31,620,000.

4.5.1.2 Voltage Collapse for Loss of Big Stone South - J488-J489 POI 345 kV Line

Voltage collapse was identified with the 3PH or SLG faults resulting in tripping of Big Stone South – J488-J489 POI 345 kV line (Table 4-6). The same voltage collapse was also identified in steady state analysis.

Table 4-6: Disturbances Causing Voltage Collapse with Tripping of Big Stone South – J488-J489 POI 345 kV

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Below two potential solutions have been investigated and compared for mitigating the voltage collapse issue. Both solutions can effectively mitigate the identified voltage collapse issue.

Solution 1: Install a (-50, +180) Mvar SVC at Ellendale 230 kV bus, and install a breaker for the fixed line reactor (50 Mvar) at Ellendale 345 kV side (Estimated cost is \$39,140,000).

Solution 2: Build a 2nd Big Stone South - J488&J489 POI 345 kV 25-mile line (Estimated cost is \$44,900,000).

Comparison of the Solutions

The solution of 2nd Big Stone South - J488&J489 POI 345 kV line is better than the solution of SVC in the following aspects:

- Composite Short Circuit Ratio (CSCR) can be increased from 0.953 to 2.679 when one Big Stone South - J488&J489 POI 345 kV line is out of service, which will improve control stability of the nearby wind farms (J488, J489, J436, and J437).
- Several thermal NUs can also be eliminated or reduced, which will save cost of \$20,350,000 (Table 4-7).
- The (-50, +180) Mvar SVC solution is not adequate in resolving the voltage collapse issue when system operational conditions become worse. For example, when nearby six wind farms (J436, J437, J488, J489, Foxtail, Tatanka) are dispatched at their Pmax, at least (-50, +240) Mvar SVC at Ellendale 230 kV and 15 Mvar switched capacitor at Forman 41.6 kV are required to prevent voltage collapse and maintain post-contingency voltages above low limits.

The 2nd Big Stone South - J488&J489 POI 345 kV line is better solution on performance and economics to be selected.

Table 4-7: Thermal NUs to be Eliminated or Reduced by Addition of 2nd 345 kV Line

Constraint	Owner	Without 2nd 345 kV line		With 2nd 345 kV line		Reduced Cost (\$)
		Mitigation	Cost (\$)	Mitigation	Cost (\$)	
Hankinson-Forman 230 kV	OTP	Terminal equipment at the Forman substation will be required to be replaced. In addition, conductor clearance mitigations will be required for the line.	\$2,700,000	Not Required	\$0	\$2,700,000
Oakes-Ellendale 230 kV	OTP MDU	OTP: Full line rebuild and terminal equipment at Oakes. OTP's cost \$20,500,000. MDU: The Ellendale MVP 230 kV bus will be built to 796 MVA. \$0	\$20,500,000	OTP: Terminal equipment at Oakes will be required to be replaced. In addition, conductor clearance mitigation will be required for the line. \$2,850,000 MDU: The Ellendale MVP 230 kV bus will be built to 796 MVA. \$0	\$2,850,000	\$17,650,000
				Total Saved Cost (\$)		\$20,350,000

4.5.1.3 Slow Voltage Recovery at Hickory Creek 345 kV Bus

Per ITCM LPC, voltage at all ITCM transmission system buses should recover to the applicable post-contingency steady-state voltage level (ITCM post-disturbance limits in Table 2-2) within 1.0 second of the clearing of the fault. Under disturbances in Table 4-8, voltage at Hickory Creek 345 kV bus is still below post-contingency steady-state voltage limit (0.93 pu) after 1.0 second with the fault cleared.

Voltage constraint at Hickory Creek 345 kV bus was also identified in the steady state analysis. With addition of the Hickory Creek-Cardinal 345 kV MVP project in the stability model, no stability violations were identified under these disturbances (Table 4-8).

**Table 4-8: Disturbances Causing Slow Voltage Recovery at
Hickory Creek 345 kV Bus**

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4.5.1.4 Out of Step (OOS) Relay at Forbes on M602F Tripping

The OOS relay at Forbes on M602F tripped for 3-phase fault '0890_w_xel_p12_pc3_at_king-eauclaire' (Table 4-9) in both study case and benchmark case. This OOS trip is initiated prior to fault clearing. During the fault OOS relay can operation if the apparent impedance initially remains between blinders long enough for the timer to time out and then moves into the trip area before the fault is cleared (Figure 4-2). M602F OOS relay settings may need to be adjusted when the GNTL 500 kV project goes into service.

The 3-phase fault '0890_w_xel_p12_pc3_at_king-eauclaire' was simulated again by blocking the OOS relay at Forbes on M602F to keep it from operating during the fault. No stability violations were identified.

Table 4-9: Disturbance Causing OOS Relay at Forbes on M602F Tripping

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Figure 4-2: Forbes M602F OOS Relay Apparent Impedance under Disturbance “0890_w_xel_p12_pc3_at_king-eauclaire”

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4.5.1.5 Tripping of Solar Farms by Instantaneous Frequency Protection

Under the disturbances listed in Table 4-8, solar farms J523, J524, and J525 are tripped by their instantaneous high/low frequency protection when a 3PH fault is applied at their POI bus. Frequency excursions that cause protection to operate during the fault are the result of the network solution failing to converge and should be ignored.

These solar farms have full decoupled converters modeled by PSS®E renewable energy converter model REGCAU1. In order to control active and reactive power at each time step, the renewable energy converter model needs a voltage angle reference to compute the complex current injection. Unlike synchronous machines, the solar power converter model does not have inertia to fix a voltage angle reference during the network solution. When voltage at the solar farm POI is driven to zero during simulation of a 3PH fault, the network solution may not converge at buses in the solar farm because the reference frame is lost and the bus voltage angle at the terminals of the renewable energy converter model is unpredictable.

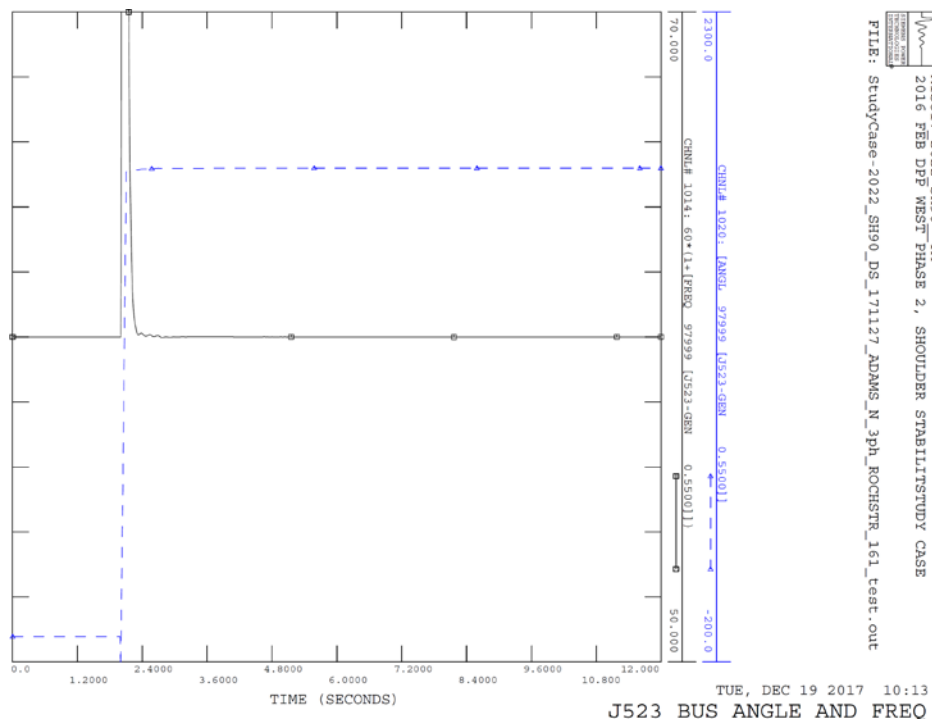
In PSS®E, local bus frequency deviation is calculated from the change in local bus angle. Bus voltage angle changes caused by network convergence failure will yield incorrect calculated local bus frequency as shown in Figure 4-2, Figure 4-4, and Figure 4-5.

Frequency excursions that cause protection to operate during the fault are the result of the network solution failing to converge and should be ignored. With instantaneous frequency protection disabled (Table 4-8), solar farms J523, J524, and J525 do not trip and no stability violations were identified.

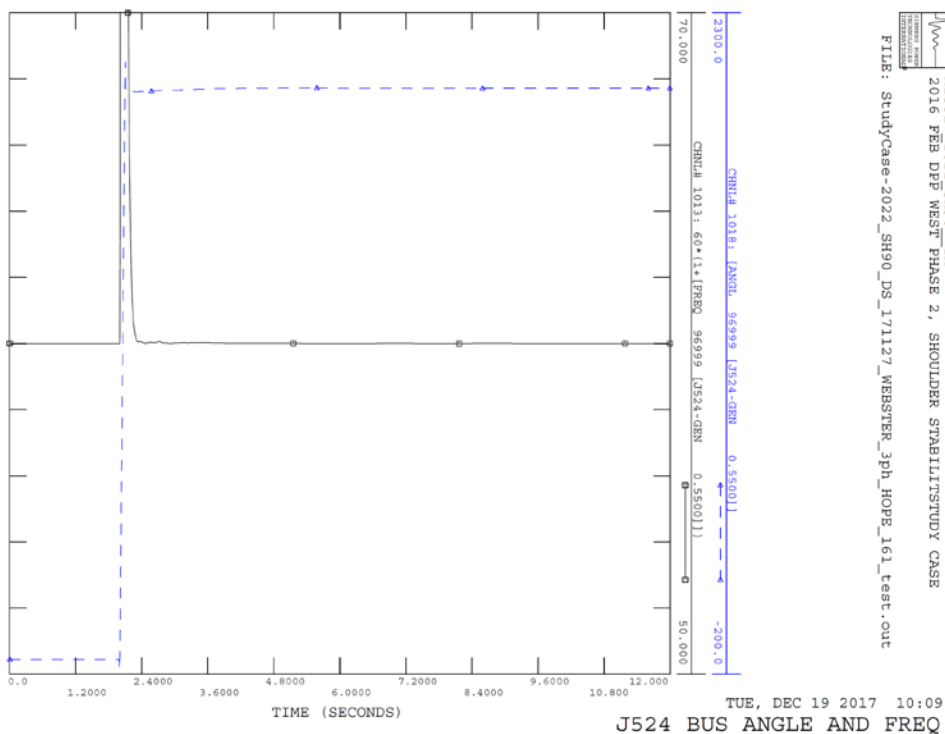
Table 4-10: Disturbances Causing Wind Farm Instantaneous Frequency Protection Tripping

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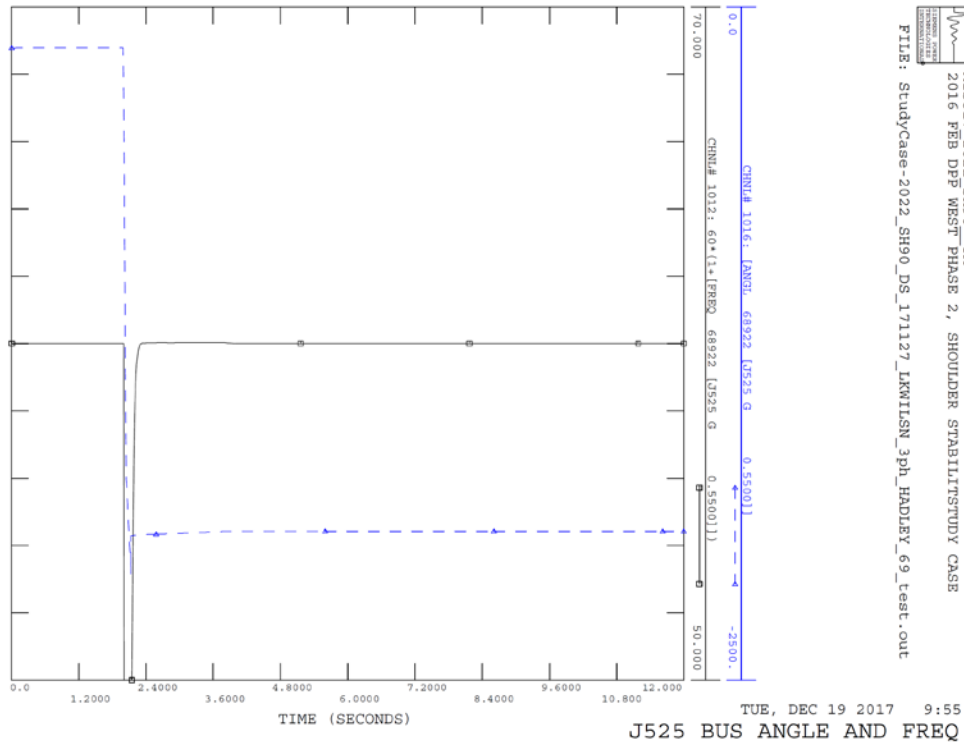
**Figure 4-3: J523 Bus Angle and Bus Frequency for Disturbance
“ADAMS_N_3ph_ROCHSTR_161”**



**Figure 4-4: J524 Bus Angle and Bus Frequency for Disturbance
“WEBSTER_3ph_HOPE_161”**



**Figure 4-5: J525 Bus Angle and Bus Frequency for Disturbance
“LKWILSN_3ph_HADLEY_69”**



4.5.1.6 J525 Solar Farm High / Low Voltage Ride Through

Under the disturbances listed in Table 4-11, J525 solar farm was tripped by its instantaneous high voltage protection (1.2 pu for 0.001 s) if its collector system shunt capacitor (4 Mvar) is switched on, or J525 was tripped by its low voltage protection (0.65 pu for 0.3 s) if its collector system shunt capacitor (4 Mvar) is switched off. The J525 has high / low voltage protection with original settings provided in Table 4-12.

The J525 customer provided updated high voltage protection settings of 1.2 pu for 0.1 s afterwards. With this updated high voltage protection setting, J525 does not trip by the high voltage protection (1.2 pu for 0.1 s) if its collector system shunt capacitor (4 Mvar) is switched on. Mitigation for low voltage ride through requirement under the disturbances listed in Table 4-11 is still required under the condition when the collector system shunt capacitor (4 Mvar) is switched off.

Table 4-11: Disturbances Causing Wind Farm Instantaneous High Voltage Protection Tripping

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Table 4-12: J525 High / Low Voltage Protection Original Settings

V (%)	Delta-V (%)	Relay Time (sec)	Breaker Time (sec)
110	10	1.0	0.0

V (%)	Delta-V (%)	Relay Time (sec)	Breaker Time (sec)
115	15	0.5	0.0
117.5	17.5	0.2	0.0
120	20	0.001	0.0
90	-10	3.0	0.0
75	-25	2.0	0.0
65	-35	0.3	0.0
45	-65	0.15	0.0

4.5.1.7 Tripping of Wind Farm by Low Voltage Protection

Under the disturbances listed in Table 4-13, wind farms were tripped by their low voltage protections. These wind farms have incorrect settings for their low voltage protections. With their settings updated (Table 4-13), no wind farm tripping occurs and no stability violations were identified.

Table 4-13: Disturbances Causing Wind Farm Low Voltage Protection Tripping

CEII Redacted

4.6 Network Upgrades Identified in Stability Analysis

The following Network Upgrades are required for mitigating constraints identified in the stability analysis:

- Build a 2nd Brookings County - J493-J510 POI 345 kV line to resolve voltage collapse and J510 angle instability for loss of Brookings Co - J493-J510 POI 345 kV Line.
- Build a 2nd Big Stone South - J488&J489 POI 345 kV line to resolve voltage collapse for loss of Big Stone South - J488&J489 POI 345 kV line.
- Hickory Creek-Cardinal 345 kV MVP project (MVP project 3127 with in-service date of 12/31/2023).
- Mitigation for J525 to meet low voltage ride through requirement under the condition when the collector system shunt capacitor (4 Mvar) is switched off.

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MWEX Voltage Stability Study

ATC performed steady state voltage stability analysis. Voltage stability analysis is required to determine if the initial conditions of the DPP system models under study are in a stable state as defined by Power-Voltage (PV) curves of the Minnesota Wisconsin Export Interface (MWEX) for loss of the worst contingency.

The voltage stability analysis used 2022 shoulder load cases to compare the Pre-DPP and Post-DPP states.

As shown in Table 5-1, the **Pre-DPP scenario in the 2022 shoulder load (2022SH) case violates ATC Planning Criteria by the nose voltage of the PV curve exceeding 0.95 p.u.** However, Post-DPP scenario meets the criteria and there is no aggravation of the criteria violation and sufficient margin is maintained. Network Upgrades will **NOT** be assigned to the Interconnection Customers, based on the assumptions used in this analysis..

Table 5-1: MWEX Margins to Collapse in the 2022SH Cases

	Real Power Flows (MW)						
	AHD-SLK ¹	MWEX			Margin to Nose ^{2,3}		
Case	N-0 Initial Condition	N-0 I.C.	N-1 I.C.	N-1 Nose	(MW)	(%)	Notes
Pre-DPP	369	975	461	741	280	37.8	Voltage Stable Sufficient Margin ⁴ V(nose)>0.95 p.u. ⁵
Post-DPP	464	1226	603	691	88	12.7	Voltage Stable Sufficient Margin ⁴

Notes:

1. As described in the active MWEX Operating Guide, the AHD-SLK interface is a single element PTDF interface measured at the Minnesota Power 230 kV side of the Arrowhead 230 kV phase shifter.
2. Margin to Nose (MW) is defined as the MWEX N-1 Nose minus the N-1 Initial Condition.
3. Initial Condition flows were measured in the base cases with the worst contingency plus operation of various control systems as needed with all transformer taps, switched shunts, and PARs locked.
4. ATC Planning Criteria requires a 10% voltage stability margin.
5. ATC Planning Criteria requires $V_{nose} < V_{min}$. V_{min} is 0.95 p.u. at the MP Arrowhead 230 kV.

The results of this analysis show that the Post-DPP scenario in the 2022 shoulder load case is voltage stable. The calculated margin to nose is greater than 10%, as required by ATC Planning Criteria. Therefore there will be no Network Upgrade costs assigned to Interconnection Customers for this study.

The MWEX voltage stability study details can be found in Appendix G.

Short Circuit Analysis

6.1 Introduction

Siemens PTI and several transmission owning companies performed short circuit analysis for the DPP 2016 February West Area generation projects.

6.2 J432 Short Circuit Study Performed by Siemens PTI

The J432 short circuit study was performed by Siemens PTI. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J432 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.1.

6.3 J460 Short Circuit Study Performed by Siemens PTI

The J460 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 11577 A (increased by 715.2 A) and SLG fault current is 9751 A (increased by 1399.6 A) at the Hawks Nest 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J460 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.2.

6.4 J475 Short Circuit Study Performed by MEC

The J475 short circuit study was performed by MEC. The study results show that the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican breakers become overdutied as a result of the Interconnection Customer's project.

Study details can be found in Appendix H.3.

6.5 J485 Short Circuit Study Performed by Siemens PTI

The J485 short circuit facilities study was performed by Siemens PTI. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J485 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.4.

6.6 J488 & J489 Short Circuit Study Performed by OTP

The J488 and J489 short circuit study was performed by OTP. With the proposed J488/J489 project additions, the fault currents are estimated to be roughly 8.6 kA at the Twin Brooks 345 kV bus, 11.2 kA at the Big Stone South 345 kV bus, 19.7 kA at the Big Stone South 230 kV bus, and 33.6 kA at the Big Stone South 34.5 kV bus. Based on the results of the study, the fault current ratings of the Transmission Owner's equipment in the area are not exceeded and there are no upgrades required.

Study details can be found in Appendix H.5.

6.7 J493 and J510 Short Circuit Study Performed by OTP

The J493 and J510 short circuit study was performed by OTP.

With the proposed J493 project additions, the fault currents are estimated to be roughly 16 kA at the Big Stone South 230 kV bus and the faults on the 345 kV bus were below these values.

With the proposed J493 and J510 project additions, the fault currents are estimated to be roughly 16.6 kA at the Big Stone South 230 kV bus.

Based on the short circuit analysis performed, the fault current ratings of the Transmission Owner's equipment in the area are not exceeded and there are no upgrades required.

Study details can be found in Appendix H.6.

6.8 J495 Short Circuit Study Performed by ITCM

The J495 short circuit study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J495. Equipment not owned by ITC Midwest was not evaluated for interrupting capability. New circuit breakers at the POI switch station were assumed to have 50 kA interrupting capability.

Study details can be found in Appendix H.7.

6.9 J498 Short Circuit Facilities Study Performed by MEC

The J498 short circuit facilities study was performed by MEC. The study results show that the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican breakers become overdutied as a result of the Interconnection Customer's project.

Study details can be found in Appendix H.8.

6.10 J499 Short Circuit Facilities Study Performed by MEC

The J499 short circuit facilities study was performed by MEC. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the

Transmission Owner's short circuit criteria, interconnection of the J499 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.9.

6.11 J500 Short Circuit Facilities Study Performed by MEC

The J500 short circuit facilities study was performed by MEC. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J500 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.10.

6.12 J504 Short Circuit Facilities Study Performed by ITCM

The J504 short circuit facilities study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J504. Equipment not owned by ITC Midwest was not evaluated for interrupting capability. There are several existing breakers at DAEC 161 kV substation that are expected to be replaced as part of planned work at this location, as well as new breakers being installed with the bus expansion. 40 kA interrupting capability was assumed for the new and replaced breakers.

Study details can be found in Appendix H.11.

6.13 J506 Short Circuit Facilities Study Performed by Siemens PTI

The J506 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 12205.7 A (increased by 806.3 A) and SLG fault current is 9955.5 A (increased by 1698.8 A) at the J506 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J506 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.12.

6.14 J514 Short Circuit Facilities Study Performed by ITCM

The J514 short circuit facilities study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J514. Equipment not owned by ITC Midwest was not evaluated for interrupting capability. Circuit breaker 2680 at Marshalltown 115 kV was previously identified as a constraint for J233, but has since been removed.

Study details can be found in Appendix H.13.

6.15 J523 Short Circuit Facilities Study Performed by ITCM

The J523 short circuit facilities study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J523. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix H.14.

6.16 J524 Short Circuit Facilities Study Performed by Siemens PTI

The J524 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 19095 A (increased by 580.4 A) and SLG fault current is 19487 A (increased by 1244.2 A) at the Webster 161 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J524 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.15.

6.17 J525 Short Circuit Facilities Study Performed by GRE

The J525 short circuit facilities study was performed by GRE. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J525 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix E.2.

6.18 J526 Short Circuit Facilities Study Performed by OTP

The J526 short circuit facilities study was performed by OTP.

With the proposed J526 project additions, the fault currents are estimated to be roughly 11.7 kA at the Big Stone South 345 kV bus, 20.6 kA at the Big Stone South 230 kV bus, and 37.1 kA at the Big Stone South 34.5 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J526 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.16.

6.19 J527 Short Circuit Facilities Study Performed by Siemens PTI

The J527 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 17037 A (increased by 1119.2 A) and SLG fault current is 14696 A (increased by 2339.9 A) at the J527 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J527 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.17.

6.20 J528 Short Circuit Facilities Study Performed by Siemens PTI

The J528 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 16673 A (increased by 868.9 A) and SLG fault current is 14134 A (increased by 1757.2 A) at the J528 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J528 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.18.

6.21 J529 Short Circuit Facilities Study Performed by MEC

The J529 short circuit facilities study was performed by MEC. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J529 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.19.

6.22 J530 Short Circuit Facilities Study Performed by Siemens PTI

The J530 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 13118 A (increased by 1118.4 A) and SLG fault current is 10678 A (increased by 2793.4 A) at the J530 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J530 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.20.

6.23 J534 Short Circuit Facilities Study Performed by Siemens PTI

The J534 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 13335 A (increased by 1119.5 A) and SLG fault current is 11425 A (increased by 2413.0 A) at the J534 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J534 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.21.

6.24 J535 Short Circuit Facilities Study Performed by Siemens PTI

The J535 short circuit facilities study was performed by Siemens PTI. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J535 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.22.

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Deliverability Study

7.1 Project Description

Interconnection requests requesting Network Resource Interconnection Services (NRIS) were considered for deliverability analysis.

7.2 Introduction

Generator interconnection projects have to pass Generator Deliverability Study to be granted Network Resource Interconnection Services (NRIS).

If the generator is determined as not fully deliverable, the customer can choose either to change his project to an Energy Resource (ER) project or proceed with the system upgrades that will make the generator fully deliverable.

Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottlenecked up. The wind generators are tested at 100 % of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

7.3 Study Methodology

MISO Generator Deliverability Study whitepaper describing the algorithm can be found at ["https://www.misoenergy.org/Library/Repository/Study/Generator%20Interconnection/Generator%20Deliverability%20Study%20Methodology.pdf"](https://www.misoenergy.org/Library/Repository/Study/Generator%20Interconnection/Generator%20Deliverability%20Study%20Methodology.pdf).

7.4 Determining the MW restriction

If one facility is overloaded based on the assessed "severe yet credible dispatch" scenario described in the study methodology, and the generator under study is in the "Top 30 DF List" (see white paper for detail), part or all of its output is not deliverable. The restricted MW is calculated as following:

$$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$$

If the result is larger than the maximum output of the generator, 100% of this generator's output is not deliverable.

The generator is also responsible for any NEW base case (pre-shift) overload or NEW "severe yet credible dispatch overload" where the generator is not in the "Top 30 DF List", if the generator's DF is greater than 5%. Please see white paper for detail. The formula above also applies to these situations.

7.5 2022 Deliverability Study Result

7.5.1 J432

J432 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS upgrades and case assumptions)	98 MW (100%)
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7.5.2 J460

J460 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS upgrades and case assumptions)	200 MW (100%)
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7.5.3 J475

J475 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIS upgrades and case assumptions)	84.17 MW (42.09%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraints in ERIS Analysis	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Council Bluffs - Sub 3456 345 kV	120.8	5.59%	No		J475,J499,J500,J527, J528	\$0	\$0 ¹
Lehigh - Webster 345 kV	200.0	6.86%	No		J475,J498,J499,J500, J527,J528,J535	\$3,180	\$100,000

Note 1: Existing MEC emergency rating is 1439 MVA. OPPD equipment is limit. OPPD NU is not required unless it is identified as constraint in affected system study.

7.5.4 J485

J485 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS upgrades and case assumptions)	46.85 MW (100%)
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7.5.5 J493

J493 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIS upgrades and case assumptions)	150 MW (100%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraints in ERIS Analysis	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Big Stone - Blair 230 kV	85.1	41.15%	No	J488,J489,J493,J526	J493,J510,J526	\$5,918,296	\$28,250,000 ¹
Big Stone - Browns Valley 230 kV	121.4	5.77%	Yes	J488,J489,J493,J526	J493,J510,J526		ERIS
Hankinson - Wahpeton 230 kV	150.0	6.34%	Yes	J432,J460,J488,J489,J493,J526	J493,J510,J526		ERIS

Note 1: This is NRIS additional cost. J493, J510, and J526 have impact on this constraint in the NRIS analysis. J493, J510, and J526 will be responsible for this NRIS NU if 2nd J510 POI-Brookings Co 345 kV line NU drops.

7.5.6 J495

J495 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIS upgrades and case assumptions)	200 MW (100%)
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7.5.7 J498

J498 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIS upgrades and case assumptions)	196.3 MW (57.74%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraints in ERIS Analysis	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Lehigh - Webster 345 kV	340.0	44.29%	No		J475,J498,J499,J500,J527,J528,J535	\$34,899	\$100,000

7.5.8 J499

J499 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIS upgrades and case assumptions)	143.08 MW (42.08%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraints in ERIS Analysis	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Council Bluffs - Sub 3456 345 kV	205.4	58.10%	No		J475,J499,J500,J527, J528	\$0	\$0 ¹
Lehigh - Webster 345 kV	340.0	13.47%	No		J475,J498,J499,J500, J527,J528,J535	\$10,614	\$100,000

Note 1: Existing MEC emergency rating is 1439 MVA. OPPD equipment is limit. OPPD NU is not required unless it is identified as constraint in affected system study.

7.5.9 J500

J500 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERI S upgrades and case assumptions)	210.42 MW (42.08%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraints in ERI S Analysis	Projects Associated With ERI S Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Council Bluffs - Sub 3456 345 kV	302.1	17.65%	No		J475,J499,J500,J527, J528	\$0	\$0 ¹
Lehigh - Webster 345 kV	404.9	11.02%	No		J475,J498,J499,J500, J527,J528,J535	\$12,770	\$100,000
J527 POI - Booneville 345 kV	482.6	46.97%	No		J527,J528,J500	\$51,591	\$100,000
J528 POI - Madison 345 kV	500.0	44.86%	No		J527,J528,J500	\$64,744	\$125,000

Note 1: Existing MEC emergency rating is 1439 MVA. OPPD equipment is limit. OPPD NU is not required unless it is identified as constraint in affected system study.

7.5.10 J504

J504 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS upgrades and case assumptions)	50 MW (100%)
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7.5.11 J506

J506 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIS upgrades and case assumptions)	200 MW (100%)
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7.5.12 J510

Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIS upgrades and case assumptions)	266 MW (100%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraints in ERIS Analysis	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Big Stone-Blair 230 kV	150.9	41.15%	No	J488,J489,J493,J526	J493,J510,J526	\$10,495,112	\$28,250,000 ¹
Big Stone - Browns Valley 230 kV	215.3	5.77%	Yes	J488,J489,J493,J526	J493,J510,J526		ERIS
Hankinson - Wahpeton 230 kV	266.0	6.34%	Yes	J432,J460,J488,J489,J493,J526	J493,J510,J526		ERIS

Note 1: This is NRIS additional cost. J493, J510, and J526 have impact on this constraint in the NRIS analysis. J493, J510, and J526 will be responsible for this NRIS NU if 2nd J510 POI-Brookings Co 345 kV line NU drops.

7.5.13 J514

J514 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS upgrades and case assumptions)	30 MW (100%)
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7.5.14 J523

J523 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS upgrades and case assumptions)	50 MW (100%)
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7.5.15 J524

J524 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS upgrades and case assumptions)	100 MW (100%)
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7.5.16 J525

J525 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS upgrades and case assumptions)	33 MW (100%)
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7.5.17 J526

J526 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIS upgrades and case assumptions)	300 MW (100%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraints in ERIS Analysis	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Big Stone - Blair 230 kV	170.2	41.15%	No	J488,J489,J493, J526	J493,J510,J526	\$11,836,592	\$28,250,000 ¹
Big Stone - Browns Valley 230 kV	242.8	6.51%	Yes	J488,J489,J493, J526	J493,J510,J526		ERIS
Hankinson - Wahpeton 230 kV	300.0	6.87%	Yes	J432,J460,J488, J489,J493,J526	J493,J510,J526		ERIS

Note 1: This is NRIS additional cost. J493, J510, and J526 have impact on this constraint in the NRIS analysis. J493, J510, and J526 will be responsible for this NRIS NU if 2nd J510 POI-Brookings Co 345 kV line NU drops.

7.5.18 J527

J527 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIS upgrades and case assumptions)	105.2 MW (42.08%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraints in ERIS Analysis	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Council Bluffs - Sub 3456 345 kV	151.0	17.00%	No		J475,J499,J500,J527, J528	\$0	\$0 ¹
Lehigh - Webster 345 kV	202.5	11.40%	No		J475,J498,J499,J500, J527,J528,J535	\$6,605	\$100,000
J527 POI - Booneville 345 kV	241.3	50.57%	No		J527,J528,J500	\$27,773	\$100,000
J528 POI - Madison 345 kV	250.0	44.86%	No		J527,J528,J500	\$32,372	\$125,000

Note 1: Existing MEC emergency rating is 1439 MVA. OPPD equipment is limit. OPPD NU is not required unless it is identified as constraint in affected system study.

7.5.19 J528

J528 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIS upgrades and case assumptions)	84.16 MW (42.08%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraints in ERIS Analysis	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Council Bluffs - Sub 3456 345 kV	120.8	17.09%	No		J475,J499,J500,J527, J528	\$0	\$0 ¹
Lehigh - Webster 345 kV	162.0	11.28%	No		J475,J498,J499,J500, J527,J528,J535	\$5,228	\$100,000
J527 POI - Booneville 345 kV	193.0	46.97%	No		J527,J528,J500	\$20,636	\$100,000
J528 POI - Madison 345 kV	200.0	48.30%	No		J527,J528,J500	\$27,884	\$125,000

Note 1: Existing MEC emergency rating is 1439 MVA. OPPD equipment is limit. OPPD NU is not required unless it is identified as constraint in affected system study.

7.5.20 J529

J529 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIS upgrades and case assumptions)	250 MW (100%)
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7.5.21 J530

J530 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS upgrades and case assumptions)	250 MW (100%)
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7.5.22 J534

J534 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIS upgrades and case assumptions)	250 MW (100%)
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7.5.23 J535

J535 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIS upgrades and case assumptions)	121.25 MW (57.74%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraints in ERIS Analysis	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Lehigh - Webster 345 kV	210.0	54.87%	No		J475,J498,J499,J500, J527,J528,J535	\$26,704	\$100,000

Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) test for Network Upgrades driven by higher queued interconnection project was performed for this System Impact Study.

Generation projects J500 and J514 are the only two projects to have the SNUs. These Network Upgrades were originally identified in the MISO DPP 2013 August West Area study cycle which are listed below:

- Jasper-Laurel 161 kV (\$600,000, DPP 2013 August)
- J274 POI-Creston 161 kV (\$175,000, DPP 2013 August)

The maximum MW impacts and Shared Network Upgrade (SNU) cost allocations are listed in Table 8-1.

Table 8-1: Maximum MW Impact and SNU Cost Allocations

Network Upgrades	Original Driver Study Cycle	Projects sharing cost	MW Contribution	Total NU Cost (\$)	Cost Responsibility
Jasper-Laurel 161 kV	DPP-2013-AUG	J233	202.31	\$600,000	\$549,631
	DPP-2016-FEB	J514	18.54		\$50,369
J274 POI-Creston 161 kV	DPP-2013-AUG	J274	100.00	\$175,000	\$158,716
	DPP-2016-FEB	J500	10.26		\$16,284

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Section

9

Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

9.1 Cost Assumptions for Network Upgrades

The cost estimate for each network upgrade was provided by the corresponding transmission owning company.

9.2 ERIS Network Upgrades Proposed for DPP West Area Projects

Network upgrades for Energy Resource Interconnection Service (ERIS) were identified in the MISO ERIS analysis and the affected system analysis. The ERIS network upgrades include voltage support network upgrades and thermal network upgrades identified in the MISO steady-state analysis, voltage support network upgrades identified in the MWEX voltage stability analysis, stability and voltage collapse network upgrades identified in the MISO transient stability analysis, short circuit network upgrades identified in the MISO short circuit analysis, and network upgrades identified in the Local Planning Criteria analysis and affected system analysis. The total costs of ERIS network upgrades for the 2022 scenario are summarized in Table 9-1.

Table 9-1: Summary of ERIS Network Upgrades

Category of ERIS Network Upgrades	2022 Cost (\$)
Network Upgrades Identified in MISO Steady-State Analysis	\$96,755,000
Network Upgrades Identified in MWEX Voltage Stability analysis	\$0
Network Upgrades Identified in Stability & Voltage Collapse Analysis	\$76,300,000
Network Upgrades Identified in Short Circuit Analysis	\$0
Additional Network Upgrades Identified in OTP LPC Analysis	\$0
Additional Network Upgrades Identified in GRE LPC Analysis	\$0
Additional Network Upgrades Identified in CIPCO/MPC Affected System	\$600,000
Additional Network Upgrades Identified in PJM Affected System	\$16,993,000
Additional Network Upgrades Identified in SPP Affected System	\$338,500,002
Shared Network Upgrades	\$66,653
Total	\$529,214,655

ERIS network upgrades are listed below.

Table 9-2: Network Upgrades in MISO Steady-State Analysis

Constraint	Owner	Mitigation	Cost (\$)
J530 POI-Hills 345 kV	MEC	MEC: Structure replacements	\$300,000
Hazel Creek 345-230-13.8 kV xfmr #6	XEL	Upgrade Hazel Creek TR9 to 672 MVA	\$4,900,000
Panther-McLeod 230 kV	GRE	Remediate 4 Structures	\$1,000,000
Johnson Jct-Ortonville 115 kV	MRES GRE OTP	OTP: Ortonville terminal equipment requires replacement. OTP's portion of the 0.56 mile line will be required to be fully rebuilt with a larger conductor. \$500,000 GRE: GRE owned switch at Johnson Jct needs to be replaced. \$350K MRES: The MRES owned section (majority of the line) will have to be rebuilt for rating increases. \$16M	\$16,850,000
Johnson Jct-Morris 115 kV	GRE WAPA	GRE: Rebuild 15 miles of 115 kV line from Johnson Junction to Morris. \$8,175,000 WAPA: NU is not required unless it is identified as constraint in affected system study.	\$8,175,000
Big Stone 230-115-13.8 kV xfmr	OTP	Transformer and Terminal equipment will be required to be replaced at the substation.	\$3,250,000
Big Stone-Big Stone South 230 kV #1	OTP	Line will be required to be fully rebuilt with a larger conductor as well as replacement of terminal equipment at Big Stone.	\$1,300,000
Big Stone-Browns Valley 230 kV	OTP MDU	OTP: owns Big Stone and Browns Valley terminal equipment which is sufficient to handle the flow seen in the study. \$0 MDU: Conductor clearance mitigation. \$350,000	\$350,000
Big Stone-Blair 230 kV	OTP BEPC	OTP: Some OTP terminal equipment will be required to be replaced at the Big Stone 230 kV sub. OTP's total cost is \$80,000. BEPC: NU is not required unless it is identified as constraint in affected system study.	\$80,000
Hankinson-Wahpeton 230 kV	OTP	Terminal equipment at both Hankinson and Wahpeton substations will be required to be replaced. In addition, conductor clearance mitigations will be required for the line.	\$2,300,000
Wahpeton-Fergus Falls 230 kV	OTP MRES	Terminal equipment at Wahpeton substation will be required to be replaced. In addition, conductor clearance mitigations will be required for the line.	\$750,000

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Constraint	Owner	Mitigation	Cost (\$)
Oakes-Forman 230 kV	OTP	Terminal equipment at Oakes and Forman substations will be required to be replaced. In addition, conductor clearance mitigations will be required for the line.	\$1,600,000
Oakes-Ellendale 230 kV	OTP MDU	OTP: Terminal equipment at Oakes will be required to be replaced. In addition, conductor clearance mitigation will be required for the line. \$2,850,000 MDU: The Ellendale MVP 230 kV bus will be built to 796 MVA. \$0	\$2,850,000
Marshalltown-Blairstown 115 kV	ITCM	Sag limit at 162F. Raise/replace structures and replace bus conductor at Blairstown. New rating of 120 MVA.	\$2,700,000
Blairstown-Prairie Crk 115 kV	CIPCO ITCM	CIPCO: No Mitigation Needed. \$0 ITCM: Clearance limit at 153F. Raise structures to reach a minimum rating of 114 MVA	\$2,700,000
Hazleton-Washburn 161 kV	ITCM MEC	MEC: Reconductor with structure replacements as appropriate and substation terminal equipment upgrades. \$16.7 M ITCM: Terminal limit is 325 MVA.	\$16,700,000
Boone Jct-Sub T Fort Dodge 161 kV	MEC CIPCO	CIPCO: No Mitigation Needed. \$0 MEC: Reconductor with structure replacements as appropriate and substation terminal equipment upgrades. \$7,000,000	\$7,000,000
Parnell-J438 POI 161 kV	MEC ITCM	MEC: Structure replacements. \$100,000 ITCM: ITCM Limit is 335 MVA at Parnell	\$100,000
Grimes-Sycamore 345 kV #2	MEC	Sycamore substation terminal equipment upgrades.	\$1,200,000
Grimes-Beaver Creek 345 kV	MEC	Structure replacements.	\$600,000
Granger Tap-108th & 54th 161 kV	MEC	Reconductor short section of line and terminal equipment upgrades.	\$100,000
Bondurant-Montezuma 345 kV	MEC	Structure replacements	\$200,000
Webster-Sub T Ft Dodge 161 kV	MEC	Reconductor with structure replacements as appropriate.	\$6,250,000
Voltage constraint identified at Lathrop 161 kV bus	AMRN	25 Mvar switched cap bank at Lathrop 161 kV	\$2,000,000
Voltage constraints in the areas of Hickory Creek, Salem, and Rock Creek	ITCM	Hickory Creek-Cardinal 345 kV project (MVP project 3127); XFMR tap should be locked at neutral position for the two 345-161 kV xfmr's at Salem	\$0
Voltage constraint in the area of Blackhawk	MEC	150 Mvar switched cap bank at Blackhawk 345 kV (\$4 M) and 1x50 Mvar switched cap bank at Midport 161 kV (\$2 M)	\$6,000,000

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Constraint	Owner	Mitigation	Cost (\$)
Voltage constraint in the area of Montezuma, J530 POI	MEC	100 Mvar switched cap bank at Montezuma 345 kV	\$5,000,000
Voltage constraint in the area of J438 POI	MEC	1×25 Mvar switched cap bank at J438 POI 161 kV	\$2,500,000

Table 9-3: Network Upgrades in MWEX Voltage Stability

Constraint	Owner	Mitigation	Cost (\$)
No Constraints			\$0

Table 9-4: Network Upgrades Required for Thermal / Stability / Voltage Collapse

Constraint	Owner	Network Upgrades	Cost (\$)
Hankinson-Forman 230 kV (2nd J488&489-BSS) ¹	OTP	2nd J488&489 POI-BSS 345 kV	\$2,700,000
Oakes-Ellendale 230 kV (2nd J488&489-BSS) ¹	OTP MDU	2nd J488&489 POI-BSS 345 kV	\$17,650,000
Voltage Collapse (2nd J488&489-BSS) ²	J488 J489	2nd J488&489 POI-BSS 345 kV	\$24,550,000
Big Stone-Highway 12 115 kV (2nd J510-Brookings) ³	OTP GRE	2nd J510 POI-Brookings Co 345 kV	\$3,100,000 ⁴
Highway 12-Ortonville 115 kV (2nd J510-Brookings) ³	OTP GRE	2nd J510 POI-Brookings Co 345 kV	\$6,900,000 ⁴
Big Stone-Big Stone South 230 kV #2 (2nd J510-Brookings) ³	OTP	Big Stone-Big Stone South 230 kV #2	\$1,250,000 ⁴
Big Stone-Blair 230 kV (2nd J510-Brookings) ³	OTP BEPC	2nd J510 POI-Brookings Co 345 kV	\$720,000 ⁴
Hankinson-Wahpeton 230 kV (2nd J510-Brookings) ³	OTP	2nd J510 POI-Brookings Co 345 kV	\$19,650,000 ⁴
Transient voltage recovery constraint ⁵	ITCM	Hickory Creek-Cardinal 345 kV project (MVP project 3127); XFMR tap should be locked at neutral position for the two 345-161 kV xfmr's at Salem	\$0

Note 1: 2nd Big Stone South-J488&J489 POI 345 kV line is the NU required for stability and voltage collapse, which will eliminate/reduce the original thermal NUs. This portion of NU cost is allocated based on the thermal NU cost allocation method.

Note 2: The remaining cost of 2nd Big Stone South-J488&J489 POI 345 kV line is allocated based on voltage NU cost allocation method (positive MW impact on branches connecting to J488&J489 POI 345 kV under the most critical contingency Big Stone South - J488&J489 POI 345 kV line).

Note 3: 2nd Brookings County-J510 POI 345 kV line is the NU required for stability and voltage collapse, which will eliminate/reduce the original thermal NUs. This portion of NU cost is allocated based on the thermal NU cost allocation method.

Note 4: The total saved cost of thermal NUs by 2nd Brookings County-J510 POI 345 kV line is \$31,620,000, which is slightly above the estimated cost (\$31,400,000) of 2nd Brookings County-J510 POI 345 kV line. For cost allocation purpose, these thermal NUs cost will be prorated (31.4/31.62) so that the total saved cost will be the same as the 2nd Brookings County-J510 POI line cost.

Note 5: Due to this transient voltage recovery constraint, all projects in DPP 2016 February West Phase 2 study are contingent on the Hickory Creek-Cardinal 345 kV MVP project.

Table 9-5: Additional Network Upgrades in OTP LPC

Constraint	Owner	Mitigation ¹	Comments ¹
Big Stone-Highway 12 115 kV	OTP GRE	OTP: All of OTP's equipment is sufficient for the 244.1 MVA flow seen in this study. GRE: Complete reconstruction of the line and new equipment at Highway 12. \$3,100,000	New NU. Cost is \$3,100,000
Highway 12-Ortonville 115 kV	OTP GRE	OTP: All of OTP's equipment is sufficient for the 235.3 MVA flow seen in this study. GRE: Complete reconstruction of the line and new equipment at Highway 12. \$6,900,000	New NU. Cost is \$6,900,000
Big Stone-Blair 230 kV	OTP BEPC	More NUs are required per OTP LPC. OTP: Conductor clearance mitigation and terminal equipment at Big Stone. OTP's portion \$800,000. BEPC: NU is not required unless it is identified as constraint in affected system study.	NU cost increases by \$720,000
Hankinson-Wahpeton 230 kV	OTP	More NUs are required per OTP LPC. Complete line rebuild and terminal equipment at Hankinson and Wahpeton.	NU cost increases by \$19,650,000
Oakes-Ellendale 230 kV	OTP MDU	More NUs are required per OTP LPC. OTP: Full line rebuild and terminal equipment at Oakes. OTP's cost \$20,500,000. \$20,500,000 MDU: The Ellendale MVP 230 kV bus will be built to 796 MVA. \$0	NU cost increases by \$17,650,000

Note 1: Additional NUs and cost identified in the OTP LPC study will not be required with stability & voltage NUs (Table 9-4)

Table 9-6: GRE Local Planning Criteria Network Upgrades

Constraint	Owner	Mitigation	Total Cost (\$)	J525 Cost (\$)
13 miles of 69 kV lines between Chandler tap and Lyon County	Xcel	rebuilding the existing transmission with new 477 ACSR	\$10,000,000	\$0
Fenton 115-69 kV transformer	Xcel	Install 2nd 115-69 kV transformer at Fenton	\$3,900,000	\$0

Table 9-7: CIPCO Affected System Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
Mqoketa-Wyoming 161 kV	CIPCO	CIPCO: Rebuild 20.3 miles of 161 kV line to T2-477 ACSR conductor. Replace 2 161 kV switches and jumpers. Change 3 CT ratios to minimum 800 amps. \$8,000,000 MEC: Note MEC can install redundant 161-69 kV transformer differential relaying on both 161-69 kV transformers at Hills to eliminate this P5 CON. \$600,000. Cheapest solution.	\$600,000

Table 9-8: PJM Affected System Network Upgrades

Constraint	Owner	Mitigation	Total Cost (\$)	DPP Projects
Twin Branch–Argenta 345 kV	PJM	a. A sag check will be required for the ACSR ~ 954 ~ 45/7 ~ RAIL - Conductor Section 1 to determine if the line section can be operated above its emergency rating of 1409 MVA.	\$208,000	J498, J499, J526, J528
Benton–T-094 345 kV (N5106)	PJM	Reconductor or rebuild (depending on the existing structures) the portions of 345 kV lines between the Benton Harbor and Sagreto 345 kV substations.	\$19,000,000	J412, J515, J498, J499, J500, J526, J527, J529, J530, J534
Benton–T-094 345 kV (N5312.3)	PJM	Replace the Benton H Line Riser and Replace the Segreto Wave trap.	\$1,000,000	J498, J499, J500, J526, J527, J529, J530, J534

Table 9-9: SPP Affected System Network Upgrades

Mitigation	Area	Purpose	ERIS and / or NRIS Affecting Requests	Estimated Cost
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	NPPD / KCPL	Required to mitigate COOPER - ST JOE 345KV CKT 1 NRIS thermal constraints for system intact and contingency conditions	NRIS: J432, J460, J475, J485, J493, J495, J498, J499, J500, J504, J506, J510, J514, J523, J524, J525, J526, J527, J528, J529, J530, J534, J535	\$301,000,000
	NPPD / OPPD / KCPL	Required to mitigate COOPER_S flowgate for system intact conditions	NRIS: J432, J460, J475, J485, J493, J495, J498, J499, J500, J504, J506, J510, J514, J523, J524, J525, J526, J527, J528, J529, J530, J534, J535	
	NPPD / OPPD / KCPL	Required to mitigate various steady state voltage violations observed in NPPD, OPPD, KCPL areas after SPP previously allocated, MISO MVP, and MISO current study upgrades are in service. Please refer to Appendix C-V for the incremental voltage constraints.	NRIS: J432, J460, J475, J485, J493, J495, J498, J499, J500, J504, J506, J510, J514, J523, J524, J525, J526, J527, J528, J529, J530, J534, J535	
Replace terminal equipment	WAPA	Required to mitigate APPLIEDORN 4 - WATERTOWN 230 KV CKT 1 NRIS thermal constraints for contingency conditions.	NRIS: J432, J460, JJ493, J510, J526	\$1,000,000

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Mitigation	Area	Purpose	ERIS and / or NRIS Affecting Requests	Estimated Cost
Rebuild approximately 42 miles of 230kV	WAPA	Required to mitigate BLAIR - GRANITE FALLS 230KV CKT 1 NRIS thermal constraints for contingency conditions.	NRIS: J432, J460, J493, J510, J526	\$33,000,000
Rebuild approximately 3 miles of 230kV	WAPA	Required to mitigate GRANITE FALLS – MN VALLEY TAP 230KV CKT 1 NRIS thermal constraints for contingency conditions.	NRIS: J432, J460, J493, J510, J526	\$2,500,000
Terminal equipment upgrade	WAPA / XEL	Required to mitigate SPLIT ROCK - WHITE 345KV CKT 1 NRIS thermal constraint for contingency conditions to at least 1210 MVA.	NRIS: J432, J460, J485, J493, J510, J523, J525, J526	\$1,000,000

Table 9-10: Network Upgrades in Short Circuit Analysis

Constraint	Owner	Mitigation	Cost (\$)
No Constraints			\$0

Table 9-11: Shared Network Upgrades

Network Upgrades	Original Driver Study Cycle	Projects sharing cost	MW Contribution	Total NU Cost (\$)	Cost Responsibility
Jasper-Laurel 161 kV	DPP-2013-AUG	J233	202.31	\$600,000	\$549,631
	DPP-2016-FEB	J514	18.54		\$50,369
J274 POI-Creston 161 kV	DPP-2013-AUG	J274	100.00	\$175,000	\$158,716
	DPP-2016-FEB	J500	10.26		\$16,284

9.3 Cost Allocation Methodology

The costs of Network Upgrades (NU) for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Post Case. For constraints identified in the shoulder peak scenario, the MW impact is calculated using the shoulder peak post-DPP case. The MW impact on constraints identified in the summer peak scenario is calculated using the summer peak post-DPP case. With all Group Study generation projects dispatched in the Post Case, all thermal and voltage constraints will be identified and a distribution factor from each project on each constraint will be obtained.

Constraints which are mitigated by one or a subset of NU are identified. The MW contribution on these constraints from each generating facility is calculated in the Post Case without any network upgrades. Then the cost of each NU is allocated based on the pro rata share of the

Cost Allocation

MW contribution from each generating facility on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of NU is:

$$\text{Project A cost portion of NU} = \text{Cost of NU} \times \left(\frac{\text{Max}(\text{Proj.A MW contribution on constraint})}{\sum_i \text{Max}(\text{Proj.i MW contribution on constraint})} \right)$$

9.4 Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

The Distribution Factor (DF) from each generating facility is calculated on the constraints identified in the steady-state analysis in the Post Case without any network upgrades. For a voltage support network upgrade required for mitigating voltage constraints identified in the steady-state AC contingency analysis, DFs are calculated under the most critical contingency on all branches (proxy branches for voltage support network upgrade) connecting to the high voltage side of the transformer, where the voltage support network upgrade is located. For a voltage support network upgrade required for mitigating MWEX voltage stability constraints identified in the voltage stability analysis, DFs are calculated under the most critical contingency on all branches (proxy branches) connecting to the high voltage side of the transformer, where the voltage collapse occurs.

For each thermal constraint, the maximum MW contribution (increasing flow) from each generating facility is calculated. MW contribution from one generating facility is set as zero if the MW contribution is less than 1 MW, or the constraint is not categorized as MISO ERIS constraint or affected system constraint for that specific generating facility.

For a voltage support network upgrade, the generators in same local area with positive MW impact will be responsible for mitigating the voltage or voltage stability constraints. No cutoff is applied to the MW contribution from each generating facility.

The calculated DF results and the MW contribution on each constraint are in Appendix I.1 for the 2022 scenario.

Finally, the cost allocation for each NU is calculated based on the MW contribution of each generating facility, as detailed in Appendix I.2 for the 2022 scenario.

Assuming all generating facilities in the DPP 2016 February West Area group advance, a summary of the costs for total NUs (NUs for ERIS, NRIS, and Interconnection Facilities) allocated to each generating facility is listed in Table 9-12.

Table 9-12: Summary of Total NU Costs Allocated to Each Generation Project

Project	Max Output (MW)	Total Cost of NU per Project (\$)	\$/MW	Share %
J432	98	\$119,156	1.63%	\$119,156
J460	200	\$142,998	3.98%	\$142,998
J475	200	\$57,899	1.61%	\$57,899
J485	46.85	\$86,682	0.57%	\$86,682
J488	151.8	\$246,442	5.21%	\$246,442
J489	151.8	\$246,442	5.21%	\$246,442
J493	150	\$205,118	4.29%	\$205,118
J495	200	\$197,255	5.50%	\$197,255
J498	340	\$135,557	6.42%	\$135,557
J499	340	\$128,476	6.09%	\$128,476
J500	500	\$153,757	10.71%	\$153,757
J504	50	\$121,916	0.85%	\$121,916
J506	200	\$129,207	3.60%	\$129,207
J510	284.5	\$134,278	5.32%	\$134,278
J514	65	\$136,875	1.24%	\$136,875
J523	50	\$66,851	0.47%	\$66,851
J524	100	\$136,628	1.90%	\$136,628
J525	33	\$149,164	0.69%	\$149,164
J526	300	\$240,335	10.05%	\$240,335
J527	250	\$151,585	5.28%	\$151,585
J528	200	\$156,536	4.36%	\$156,536
J529	250	\$117,007	4.08%	\$117,007
J530	250	\$87,752	3.06%	\$87,752
J534	250	\$118,362	4.12%	\$118,362
J535	210	\$129,101	3.78%	\$129,101
Total/Average	4870.95	\$143,815	100.00%	\$143,815

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Model Development for Steady-State and Stability Analysis

A.1 DPP 2016 February Generation Projects

Table A-1: DPP 2016 February West Area Projects

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J432	SD	Brookings	XEL	Brookings 345 kV	98	Wind	NRIS
J460	MN	Lincoln	GRE	Brookings-H081 345 kV line	200	Wind	NRIS
J475	IA	Poweshiek	MEC	Montezuma 345 kV	200	Wind	NRIS
J485	MN	Olmsted	RPU	West Side 161 kV	46.85	CT	NRIS
J488	SD	Grant, Roberts	OTP	Big Stone-Ellendale 345 kV	151.8	Wind	ERIS
J489	SD	Grant, Roberts	OTP	Big Stone-Ellendale 345 kV	151.8	Wind	ERIS
J493	MN	Yellow Medicine	OTP	Big Stone-Brookings 345 kV	150	Wind	NRIS
J495	IA	Winnebago	ITCM	Ledyard-Colby 345 kV line	200	Wind	NRIS
J498	IA	Boone, Greene	MEC	Grimes-Lehigh 345 kV	340	Wind	NRIS
J499	IA	Adair, Madison	MEC	Fallow-Grimes 345 kV	340	Wind	NRIS
J500	IA	Adair	MEC	New substation at intersection of Boone-Atchison 345 kV and Rolling Hills-Madison Co 345 kV	500	Wind	NRIS
J504	IA	Linn	ITCM	Bertram-Duane Arnold 161 kV (0.5 mi from Duane Arnold)	50	Solar	NRIS
J506	IA	Plymouth	MEC	Raun-Lakefield Jct 345 kV	200	Wind	NRIS
J510	SD	Deuel	OTP	Scandinavia Township	266 sum / 284.5 win	CT	NRIS
J514	IA	Marshall	ITCM	Marshalltown substation	30 sum / 65 win	CC	NRIS
J523	MN	Mower	ITCM	Adams 161 kV	50	Solar	NRIS
J524	IA	Webster	MEC	Webster 161 kV	100	Solar	NRIS
J525	MN	Murray	XEL	Lake Wilson-Hadley 69 kV	33	Solar	NRIS
J526	SD	Deuel	OTP	Brooking Co-Big Stone South 345 kV	300	Wind	NRIS
J527	IA	Adair	MEC	Booneville-Cooper 345 kV	250	Wind	NRIS
J528	IA	Adair	MEC	Rolling Hills-Madison 345 kV	200	Wind	NRIS
J529	IA	Palo Alto	MEC	Obrien-Kossuth 345 kV	250	Wind	NRIS
J530	IA	Iowa	MEC	Montezuma-Hills 345 kV	250	Wind	NRIS

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J534	IA	Humboldt, Kossuth	MEC	Kossuth–Webster 345 kV	250	Wind	NRIS
J535	IA	Sac	MEC	J411–Lehigh 345 kV	210	Wind	NRIS

Table A-2: Dynamic Modeling for DPP West Area Projects

MISO Project #	Dynamic Modeling	Generator Reactive Power Capability (power factor)
J432	49 Gamesa G9X 2.0 MW (GXX048)	± 0.95
J460	100 Vestas V110 2.0 MW (VS3103)	± 0.95
J475	100 Vestas V110 2.0 MW (VS3103)	± 0.95
J485	5 9.37 MW natural gas fired reciprocating engines	0.8 lagging -0.95 leading
J488	61 GE 2.5 MW WTG (GEWTG0501)	± 0.9
J489	61 GE 2.5 MW WTG (GEWTG0501)	± 0.9
J493	45 V136 GS 3.45MW (GSCOR1)	0.902 lagging 0.918 leading
J495	100 Vestas V110 2.0 MW (VS3103)	± 0.95
J498	170 Vestas V110 2.0 MW Mk 10D (VS3103)	± 0.95
J499	170 Vestas V110 2.0 MW Mk 10D (VS3103)	± 0.95
J500	250 Vestas V110 2.0 MW Mk 10D (VS3103)	± 0.95
J504	14 GE 1500V 4 MW Solar (PVGU1)	± 0.9
J506	100 Vestas V110 2.0 MW (VS3103)	± 0.95
J510	GE CT generator (GENROU)	0.85 lagging 0.95 leading
J514	J233 CC expansion, 2 CTGs, 1 STG (GENROU)	NA
J523	20 SMA SC 2500 ev 2.5 MVA inverters (REGCAU1)	± 0.9
J524	40 SMA SC 2500 ev 2.5 MVA inverters (REGCAU1)	± 0.9
J525	12 FS3000CU15 Solar (REGCAU1)	± 0.9
J526	120 GE 2.5 MW WTG (GEWTG2)	± 0.9
J527	100 GE 2.5 MW WTG (GEWTG2)	± 0.9
J528	80 GE 2.5 MW WTG (GEWTG2)	± 0.9
J529	125 Vestas V110 2.0 MW (VS3103)	0.95 lagging 0.94 leading
J530	100 GE 2.5 MW WTG (GEWTG2)	± 0.9

MISO Project #	Dynamic Modeling	Generator Reactive Power Capability (power factor)
J534	100 GE 2.5 MW WTG (GEWTG2)	± 0.9
J535	84 GE 2.5 MW WTG (GEWTG2)	± 0.9

Table A-3: Collector System and Shunt Compensation Modeling for DPP West Area Non-Synchronous Projects

MISO Project #	Generator Modeling	Collector System Modeling	Shunt Compensation
J432	One 98 MW unit	Req = 0.010510 pu Xeq = 0.069124 pu Beq = 0.018199 pu	30 Mvar capacitor bank on 34.5 kV system
J460	Two 100 MW units	Circuit #1: R=0.008357 pu X=0.0101 B=0.05792 Circuit #2: R=0.019103 X=0.025036 B=0.04728	15 Mvar capacitor bank on each 34.5 kV collector system
J475	One 168 MW unit and one 32 MW unit	Circuit #1: R=0.00727 pu X=0.0116 pu B=0.17079 pu Circuit #2: R=0.03461 pu X=0.05527 pu B=0.05274 pu	3*15 Mvar capacitor bank and 1*5 Mvar reactor bank on 34.5 kV system
J485	Five 9.37 MW units	None	None
J488	One 151.8 MW unit	R=0.01448 pu X=0.02156 pu B=0.0 pu	20 Mvar capacitor bank on 34.5 kV system
J489	One 151.8 MW unit	R=0.010728 pu X=0.01406 pu B=0.0 pu	20 Mvar capacitor bank on 34.5 kV system
J493	One 150 MW unit	Req = 0.008928 pu Xeq = 0.039978 pu Beq = 0.06162 pu	10 Mvar capacitor bank on 34.5 kV system
J495	Two 100 MW units	Two equivalent collector systems, each has: Req = 0.02395 pu Xeq = 0.026499 pu Beq = 0.10306 pu	27 Mvar capacitor bank on each 34.5 kV collector system

MISO Project #	Generator Modeling	Collector System Modeling	Shunt Compensation
J498	Two 170 MW units	None	None
J499	Two 170 MW units	None	None
J500	Two 166 MW units One 168 MW unit	None	None
J504	One 50 MW unit	Req = 0.001848 pu Xeq = 0.001092 pu Beq = 0.00232 pu	None
J506	Two 100 MW units	Two equivalent collector systems, each has: Req = 0.026793 pu Xeq = 0.030726 pu Beq = 0.08777 pu	27.5 Mvar capacitor bank on each 34.5 kV collector system
J510	One 284.5 MW unit	None	None
J514	Two 232 MW units One 257 MW unit	None	None
J523	One 50 MW unit	Req = 0.007079 pu Xeq = 0.006495 pu Beq = 0.00475 pu	12 Mvar capacitor bank on 34.5 kV system
J524	One 100 MW unit	Req = 0.003379 pu Xeq = 0.003045 pu Beq = 0.00907 pu	None
J525	One 33 MW unit	R=0.0045 pu X=0.0055 pu B=0.0046 pu	4 Mvar capacitor bank on 34.5 kV system
J526	Two 150 MW units	Two equivalent collector systems, each has: R=0.0086 pu X=0.0110 pu B=0.1065 pu	10 Mvar capacitor bank on each 34.5 kV collector system
J527	Two 125 MW units	Two equivalent collector systems, each has: R=0.0104 pu X=0.0132 pu B=0.0884 pu	7 Mvar capacitor bank on each 34.5 kV collector system
J528	Two 100 MW units	Two equivalent collector systems, each has: R=0.0130 pu X=0.0166 pu B=0.0703 pu	7 Mvar capacitor bank on each 34.5 kV collector system
J529	Two 125 MW units	None	None

MISO Project #	Generator Modeling	Collector System Modeling	Shunt Compensation
J530	Two 125 MW units	Two equivalent collector systems, each has: R=0.0104 pu X=0.0132 pu B=0.0884 pu	7 Mvar capacitor bank on each 34.5 kV collector system
J534	Two 125 MW units	Two equivalent collector systems, each has: R=0.0104pu X=0.0132pu B=0.0884pu	7 Mvar capacitor bank on each 34.5 kV collector system
J535	One 210 MW unit	R=0.0043pu X=0.0055pu B=0.2131pu	12 Mvar capacitor bank on 34.5 kV bus

Table A-4: DPP 2016 February Central Area Projects

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J351	IN	St. Joseph	NIPS	Stillwell 345 kV	705 sum / 715 win	CC	NRIS
J468	IL	Douglas	AMIL	Sidney-Kansas West 345 kV line	202	Wind	NRIS
J515	IN	Benton, Warren	DEI	Cayuga 345 kV	400	Wind	ERIS

Table A-5: DPP 2016 February Michigan Area Projects

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J533	MI	Gratiot	METC	Slate 345 kV	200	Wind	NRIS
J538	MI	Hillsdale	METC	Moore Rd-Dowling 138 kV	150	Wind	NRIS
J540	MI	Kalkaska	WPSC	Kalkaska County	48.6	CT	NRIS Only

Table A-6: DPP 2016 February ATC Area Projects

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J505	WI	Manitowoc	ATC	Mishicot-Kewaunee 138 kV line	100	Solar	NRIS

A.2 Model Review Comments

Table A-7: Model Review Comments

Company	Python/ Idev File Name	2022 SH	2022 SPK	2022 Stability
ATC	20210917_StoneLake_345kV_STATCOM.idv	x	x	x
ATC	Increase NED xfmr Rate B to 334 MVA.py	x	x	x
ATC	20231231_Z3_SB_EEN_345-138kV.IDV	x	x	x
ATC	20231231_Z3_SB_HCK-EEN_345kV.idv	x	x	x
ATC	20231231_Z3_SB_EEN-CDL_345kV.idv	x	x	x
ATC	20170808_CHC-NLL_Eden2.con	x	x	
BEPC	BEPC_MISO_2016_FEB_West_Phase2_Modelupdate_R1.idv	x	x	x
BEPC	BEPC_MISO_2016_FEB_West_Phase2_Gen_update_SH22-SPTI.idv	x		x
BEPC	BEPC_MISO_2016_FEB_West_Phase2_Gen_update_SP22-SPTI.idv		x	
CIPCO	CIPCO_Gen_SH.idv	x		x
CIPCO	CIPCO_Gen_PK.idv		x	
CIPCO	CIPCO DPP-2016-FEB Phase 2 Additional P6.con	x	x	
DPC	Update DPC Ratings.py	x	x	x
GRE	py-GRE-add-G667-at-RoundLakeTap.py	x	x	x
GRE	py-GRE-add-G549-at-Williams.py	x	x	x
GRE	py-open4X55atRockRiver.py	x	x	x
GRE	GRE_CCR-DKN.CON	x	x	
GRE	Adding H081 Info.idv	x	x	x
ITCM	ITCM DPP 16 Feb Phase 2 Comments.idv	x	x	x
MDU	MDU_Updates-MISO17_2022SUM_TA_SPTI.idv		x	
MDU	MDU_Updates-MISO17_2022SH90_TA_SPTI.idv	x		x
MEC	Adjust MEC Wind output_SH.idv	x		x
MEC	DPP2016FEB Ph2 MEC SH Updates 09.15.2017.py	x		x
MEC	DPP2016FEB Ph2 MEC SUM Updates 09.15.2017.py		x	
MEC	Turn off MEC Reactors.py	x		x
MEC	Adjust J416 SH output.py	x		x
MRES	MRES Summer Model Updates.idv	x	x	x
OTP	1) OTP_Buffalo_disconnect_xfmr2_416.idv	x	x	x
OTP	2) Jamestown345Transformers_10-25-16.idv	x	x	x
OTP	3) BSS XFMR updates.idv	x	x	x
OTP	4) OTP_unlock_BSS_reactors.idv	x	x	x

Company	Python/ Idev File Name	2022 SH	2022 SPK	2022 Stability
OTP	5) BigStone-BigStoneSouth230kV_LineImedanceUpdates_9-13-17.idv	x	x	x
OTP	6) SummerRatingUpdates_9-13-17.idv	x	x	x
OTP	7) BigStoneSouth-Brookings345kV_LineImedanceUpdates_9-13-17.idv	x	x	x
OTP	8) BigStoneSouth-Ellendale345kV_LineImedanceUpdates_9-13-17.idv	x	x	x
SPP	Turn off Ft Calhoun.py	x	x	x
MISO	Adjust J171 Gen output.py	x	x	x
MISO	Update MDU WTG outputs.py	x		x
MISO	Update J320 Pmax.py	x	x	x
MISO	Turn off J391_SH.py	x		x
IC	Address J493 Comments.idv	x	x	x
MISO	SystemModelingShoulderFinalPython.py	x		x
MISO	SystemModelingSummerFinalPython.py		x	
SPTI	Update Area Numbers.idv	x	x	x
GRE	Adjust J171 Gen output.py	x	x	x
MISO	Update MDU WTG outputs.py	x		x
MISO	Update J320 Pmax.py	x	x	x
MISO	Remove Duplicated DPP Projects.idv	x	x	x
MISO	Remove retired gens.idv	x	x	x
IC	J475_SH.idv	x		x
IC	J475_PK.idv		x	
IC	J493&J510_SH.idv	x		x
IC	J493&J510_PK.idv		x	
IC	J514_SH.idv	x		x
BEPC	SH-Turn Off BSS SSL.py	x		x
MISO	RMV_J264.py	x	x	x
MISO	RMV_J298.py	x	x	x
MEC	Cap_ShaulisRd 161.py	x	x	x
MISO	J514_SH.idv	x		x
MISO	J514_PK.idv		x	
MEC	Correct_J475_SH.idv	x		x
MEC	Correct_J475_PK.idv		x	
MISO	Adjust length of J436 J437 POI line.py	x	x	x

Company	Python/ Idev File Name	2022 SH	2022 SPK	2022 Stability
MISO	Turn off BSSE Reactor.py	x	x	x
SPTI	Update DPP Caps.py	x	x	x
OTP	OTPSummerRatingUpdates_10-24-17.idv	x	x	x
MDU	MDU_Updates-Bench_MISO17_2022SH90_NetInt.idv	x		x
MDU	MDU_Updates-MISO17_2022SH90.idv	x		x
MDU	MDU_Updates-MISO17_2022SUM.idv		x	
SPP	RMV_SPP-531449.py	x	x	x
IC	Update J525 GSU.idv	x	x	x
MISO	Correct J442-G736.py	x	x	x
MISO	Add 2015AugDPP changes_171031.py	x	x	x
MEC	MEC Stability Updates.py			x
MDU	MDU-ModelUpdates_StudyCase-2022_SH90_DS_171020.idv			x
MDU	MDU-OtherUpdates_StudyCase-2022_SH90_DS_171020.idv	x	x	x
MISO	Remove the original NUs required for DPP 2015 Aug West group	x	x	x
ITCM	Aug_15_DPP_RestudyMitigation_ITCM.idv	x	x	x
OTP	OTP_Mitigations_11-22-17.idv	x	x	x
MEC	M Ave-New Sharon 69 kV Reconductor.py	x	x	x
MEC	Poweshiek-New Sharon 69 kV Reconductor.py	x	x	x
J493	Add-J493-Cap.py	x	x	x

A.3 MISO Classic as the Study Sink

Table A-8: MISO Classic as the Study Sink

Area #	Area Name	Area #	Area Name
207	HE	600	Xcel
208	DEI	608	MP
210	SIGE	613	SMMPA
216	IPL	615	GRE
217	NIPS	620	OTP
218	METC	627	ALTW
219	ITC	633	MPW
295	WEC	635	MEC
296	MIUP	661	MDU
314	BREC	663	BEPC-MISO
333	CWLD	680	DPC
356	AMMO	694	ALTE
357	AMIL	696	WPS
360	CWLP	697	MGE
361	SIPC	698	UPPC

A.4 PJM Market as PJM Projects Sink

Table A-9: PJM Market as PJM Projects Sink

Area #	Area Name	Area #	Area Name
201	AP	229	PPL
202	ATSI	230	PECO
205	AEP	231	PSE&G
209	DAY	232	BGE
212	DEO&K	233	PEPCO
215	DLCO	234	AE
222	CE	235	DP&L
225	PJM	236	UGI
226	PENELEC	237	RECO
227	METED	320	EKPC
228	JCP&L	345	DVP

A.5 SPP Market as SPP Projects Sink

Table A-10: SPP Market as SPP Projects Sink

Area #	Area Name	Area #	Area Name
515	SWPA	541	KCPL
520	AEPW	542	KACY
523	GRDA	544	EMDE
524	OKGE	545	INDN
525	WFEC	546	SPRM
526	SPS	640	NPPD
527	OMPA	645	OPPD
531	MIDW	650	LES
534	SUNC	652	WAPA
536	WERE	659	BEPC-SPP
540	GMO		

A.6 Contingency Files used in Steady-State Analysis

Table A-11: List of Contingencies used in Steady-State Analysis

Contingency File Name	Description	2022
Automatic single element contingencies	Single element outages at buses 60 kV and above in the study region	x
20170808_CHC-NLL_Eden2.con	Specified category P1, P2 contingencies in ATC, ITCM	x
CIPCO DPP-2016-FEB Phase 2 Additional P6.con	Specified category P6 contingencies in CIPCO	x
CC Bipole Events.con	Specified category P1, P7 contingencies in GRE Coal Creek	x
MEC-DPP2016FEB West Ph2 2022 Cat P1 09.15.2017.con	Specified category P1 contingencies in MEC	x
MEC-DPP2016FEB West Ph2 2022 Cat P2 09.15.2017.con	Specified category P2 contingencies in MEC	x
MEC-DPP2016FEB West Ph2 2022 Cat P5 09.15.2017.con	Specified category P5 contingencies in MEC	x
MEC-DPP2016FEB West Ph2 2022 Cat P7 09.15.2017.con	Specified category P7 contingencies in MEC	x
P1_AMES_MTEP17-2022TA.CON	Specified category P1 contingencies in AMES	x
P2-P7_AMES_MTEP17-2022TA.con	Specified category P2-P7 contingencies in AMES	x
P1_ATC_MTEP17-2022TA.con	Specified category P1 contingencies in ATC	x
P2-P7_ATC_MTEP17-2022TA.con	Specified category P2-P7 contingencies in ATC	x
P1_BEPC_MTEP17-2022TA.con	Specified category P1 contingencies in BEPC	x
P2-P7_BEPC_MTEP17-2022TA.con	Specified category P2-P7 contingencies in BEPC	x
P1_CBPC_MTEP17-2022TA.con	Specified category P1 contingencies in CBPC	x
P2-P7_CBPC_MTEP17-2022TA.con	Specified category P2-P7 contingencies in CBPC	x
P1_CFU_MTEP17-2022TA.con	Specified category P1 contingencies in CFU	x
P2-P7_CFU_MTEP17-2022TA.con	Specified category P2-P7 contingencies in CFU	x
P1_CIPCO_MTEP17-2022TA.con	Specified category P1 contingencies in CIPCO	x
P2-P7_CIPCO_MTEP17-2022TA.con	Specified category P2-P7 contingencies in CIPCO	x
P1_DPC_MTEP17-2022TA.con	Specified category P1 contingencies in DPC	x
P2-P7_DPC_MTEP17-2022TA.con	Specified category P2-P7 contingencies in DPC	x
P1_GRE_MTEP17-2022TA.con	Specified category P1 contingencies in GRE	x
P2-P7_GRE_MTEP17-2022TA.con	Specified category P2-P7 contingencies in GRE	x

Contingency File Name	Description	2022
P1_ITCM_MTEP17-2022TA.con	Specified category P1 contingencies in ITCM	x
P2-P7_ITCM_MTEP17-2022TA.con	Specified category P2-P7 contingencies in ITCM	x
P1_MDU_MTEP17-2022TA.con	Specified category P1 contingencies in MDU	x
P2-P7_MDU_MTEP17-2022TA.con	Specified category P2-P7 contingencies in MDU	x
P1_MP_MTEP17-2022TA.con	Specified category P1 contingencies in MP	x
P2-P7_MP_MTEP17-2022TA.CON	Specified category P2-P7 contingencies in MP	x
P1_MPC_MTEP17-2022TA.con	Specified category P1 contingencies in MPC	x
P2-P7_MPC_MTEP17-2022TA.con	Specified category P2-P7 contingencies in MPC	x
P1_MPW_MTEP17-2022TA.con	Specified category P1 contingencies in MPW	x
P2-P7_MPW_MTEP17-2022TA.CON	Specified category P2-P7 contingencies in MPW	x
P1_MRES_MTEP17-2022TA.con	Specified category P1 contingencies in MRES	x
P2-P7_MRES_MTEP17-2022TA.CON	Specified category P2-P7 contingencies in MRES	x
P1_OTP_MTEP17-2022TA.con	Specified category P1 contingencies in OTP	x
P2-P7_OTP_MTEP17-2022TA.con	Specified category P2-P7 contingencies in OTP	x
P1_RPU_MTEP17-2022TA.con	Specified category P1 contingencies in RPU	x
P2-P7_RPU_MTEP17-2022TA.con	Specified category P2-P7 contingencies in RPU	x
P1_SMPMA_MTEP17-2022TA.con	Specified category P1 contingencies in SMPMA	x
P2-P7_SMPMA_MTEP17-2022TA.con	Specified category P2-P7 contingencies in SMPMA	x
P1_XEL_MTEP17-2022TA.con	Specified category P1 contingencies in XEL	x
P2-P7_XEL_MTEP17-2022TA.con	Specified category P2-P7 contingencies in XEL	x
P1-4_ATC_MTEP17-2022TA.con	Specified category P1-4 contingencies in ATC	x
P1-4_DPC_MTEP17-2022TA.con	Specified category P1-4 contingencies in DPC	x
P1-4_GRE_MTEP17-2022TA.con	Specified category P1-4 contingencies in GRE	x
P1-4_ITCM_MTEP17-2022TA.con	Specified category P1-4 contingencies in ITCM	x
P1-4_MDU_MTEP17-2022TA.con	Specified category P1-4 contingencies in MDU	x
P1-4_MP_MTEP17-2022TA.con	Specified category P1-4 contingencies in MP	x
P1-4_MPC_MTEP17-2022TA.con	Specified category P1-4 contingencies in MPC	x
P1-4_OTP_MTEP17-2022TA.con	Specified category P1-4 contingencies in OTP	x
P1-4_SMPMA_MTEP17-2022TA.con	Specified category P1-4 contingencies in SMPMA	x
P1-4_XEL_MTEP17-2022TA.con	Specified category P1-4 contingencies in XEL	x

Contingency File Name	Description	2022
P1_AMRN_MTEP17-2022TA.con	Specified category P1 contingencies in Ameren	x
P2-P7_AMRN_MTEP17-2022TA.con	Specified category P2-P7 contingencies in Ameren	x
P1_CWLD_MTEP17-2022TA.con	Specified category P1 contingencies in CWLD	x
P2-P7_CWLD_MTEP17-2022TA.con	Specified category P2-P7 contingencies in CWLD	x
P1_CWLP_MTEP17-2022TA.con	Specified category P1 contingencies in CWLP	x
P2-P7_CWLP_MTEP17-2022TA.con	Specified category P2-P7 contingencies in CWLP	x
P1_PPI_MTEP17-2022TA.con	Specified category P1 contingencies in PPI	x
P2-P7_PPI_MTEP17-2022TA.con	Specified category P2-P7 contingencies in PPI	x
P1_SIPC_MTEP17-2022TA.con	Specified category P1 contingencies in SIPC	x
P2-P7_SIPC_MTEP17-2022TA.con	Specified category P2-P7 contingencies in SIPC	x
ComEd_RTEP_Cat_P1.con	Specified category P1 contingencies in ComEd	x
ComEd_RTEP_Cat_P2-P7.con	Specified category P2-P7 contingencies in ComEd	x
P1_AECI_MTEP17-2022TA.con	Specified category P1 contingencies in AECI	x
P2-P7_AECI_MTEP17-2022TA.con	Specified category P2-P7 contingencies in AECI	x
P1_CE_MTEP17-2022TA.con	Specified category P1 contingencies in ComEd	x
P2-P7_CE_MTEP17-2022TA.con	Specified category P2-P7 contingencies in ComEd	x
P1_WAPA_MTEP17-2022TA.con	Specified category P1 contingencies in WAPA	x
P2-P7_WAPA_MTEP17-2022TA.con	Specified category P2-P7 contingencies in WAPA	x
P1-4_AMRN_MTEP17-2022TA.con	Specified category P1-4 contingencies in Ameren	x
P1-4_CWLP_MTEP17-2022TA.con	Specified category P1-4 contingencies in CWLP	x
P1-4_SIPC_MTEP17-2022TA.con	Specified category P1-4 contingencies in SIPC	x
P1-4_WAPA_MTEP17-2022TA.con	Specified category P1-4 contingencies in WAPA	x

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Model Data

B.1 Power Flow Model Data

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B.2 Dynamic Model Data

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B.3 2022 Slider Diagrams

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Appendix

C

Reactive Power Requirement Analysis Results (FERC Order 827)

Table C-1: Reactive Power Requirement Analysis Results

Project #	HV Side Bus #	Lagging Power Factor Results				Lagging Power Factor Results				Turbine Inherent Power Factor
		MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Lagging Power Factor at HV Side	Meet Lagging Power Factor Req.?	MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Leading Power Factor at HV Side	Meet Leading Power Factor Req.?	
J432	61031	96.20	32.90	0.946	Y	95.68	-72.30	-0.798	Y	± 0.95
J460	61042	196.43	74.56	0.935	Y	196.00	-90.26	-0.908	Y	± 0.95
J475	65730	195.06	71.00	0.940	Y	194.70	-108.50	-0.874	Y	± 0.95
J488	60418	147.19	53.73	0.939	Y	144.90	-135.80	-0.730	Y	± 0.9
J489	61418	148.05	55.02	0.937	Y	146.31	-132.36	-0.742	Y	± 0.9
J493	60215	146.81	48.26	0.950	Y	146.24	-109.15	-0.801	Y	0.902 lagging -0.918 leading
J495	61536	192.76	84.72	0.915	Y	192.10	-99.06	-0.889	Y	± 0.95
J504	71089	49.74	15.08	0.957	Y	49.65	-37.37	-0.799	Y	± 0.9
J506	15010	192.32	83.38	0.917	Y	191.54	-102.48	-0.882	Y	± 0.95
J523	97996	48.99	26.23	0.882	Y	48.75	-37.07	-0.796	Y	± 0.9
J524	96996	98.05	39.51	0.928	Y	97.53	-76.36	-0.787	Y	± 0.9
J525	68920	32.58	14.60	0.913	Y	32.46	-24.83	0.794	Y	± 0.9
J526	72032	292.76	119.00	0.926	Y	291.18	-210.46	-0.810	Y	± 0.9
J527	65201	243.98	98.98	0.927	Y	242.86	-169.66	-0.820	Y	± 0.9
J528	65301	195.16	79.84	0.926	Y	194.10	-140.50	-0.810	Y	± 0.9
J530	75731	243.98	98.98	0.927	Y	242.86	-169.66	-0.820	Y	± 0.9
J534	66001	243.98	98.98	0.927	Y	242.86	-169.66	-0.820	Y	± 0.9
J535	66202	205.82	90.16	0.916	Y	204.96	-136.07	-0.833	Y	± 0.9

2022 Contingency Analysis Results

D.1 2022 Summer Shoulder (SH) Constraints

Table D-1: 2022 SH System Intact Thermal Constraints

Table D-2: 2022 SH System Intact Voltage Constraints

Table D-3: 2022 SH Category P1 Thermal Constraints

Table D-4: 2022 SH Category P1 Voltage Constraints

Table D-5: 2022 SH Category P2-P7 Thermal Constraints

Table D-6: 2022 SH Category P2-P7 Voltage Constraints

Table D-7: 2022 SH Non-Converged Contingencies

Table D-8: 2022 SH Non-Converged Contingencies DCCC Results

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D.2 2022 Summer Peak (SPK) Constraints

Table D-9: 2022 SPK System Intact Thermal Constraints

Table D-10: 2022 SPK System Intact Voltage Constraints

Table D-11: 2022 SPK Category P1 Thermal Constraints

Table D-12: 2022 SPK Category P1 Voltage Constraints

Table D-13: 2022 SPK Category P2-P7 Thermal Constraints

Table D-14: 2022 SPK Category P2-P7 Voltage Constraints

Table D-15: 2022 SPK Non-Converged Category P2-P7 Contingencies

Table D-16: 2022 SPK Non-Converged Category P2-P7 Contingencies DCCC Results

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Affected System Contingency Analysis Results

E.1 CIPCO/MPC Companies Affected System Analysis Results

Table E-1: 2022 SH CIPCO/MPC Affected System Analysis Results

Table E-2: 2022 SPK CIPCO/MPC Affected System Analysis Results

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E.2 J525 GRE Local Planning Criteria Analysis

Below is the GRE local planning criteria analysis report provided by GRE.

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E.3 OTP LPC Analysis

E.3.1 OTP LPC Model Updates

Table E-3: Wind Generation Turned Off to Create OTP LPC Light Load No Wind Case

Bus Number	Bus Name	Machine Id	Area Num	Area Name
659194	NDPRAIRWND1W0.6900	1	659	BEPC-SPP
659273	WILTON 2 W0.6900	2	659	BEPC-SPP
659289	DAYCNTYWND W0.6900	1	659	BEPC-SPP
659290	POMONA W0.6900	1	659	BEPC-SPP
659291	HYDE W0.6900	1	659	BEPC-SPP
659294	WILTON 1 W0.6900	1	659	BEPC-SPP
659296	SDPRAIRWND1W0.6900	1	659	BEPC-SPP
659366	WILTON 3 W0.6900	3	659	BEPC-SPP
659411	CAMPBLCNTY W0.6900	1	659	BEPC-SPP
659453	BRADYGEN1 W0.6900	1	659	BEPC-SPP
659457	LINDAHLWNDGW0.6900	1	659	BEPC-SPP
661307	DM WILWW 0.5750	W	661	MDU
661317	CDR HLLW 0.5750	W	661	MDU
661986	FOXTAILW 0.6900	W	661	MDU
661987	G752-2 W 0.6600	W	661	MDU
661989	THNDRSPT G W0.6600	W	661	MDU
661999	TATANKAW 0.5750	W	661	MDU
6060	G736 0.6900	1	620	OTP
600162	J262 COURT W0.6000	W	620	OTP
600163	J263 COURT W0.6000	W	620	OTP
600165	LK AGASSIZ G0.6000	W	620	OTP
615132	G619TAMW 0.6000	W	620	OTP
620112	EDGE GEN 0.5750	1	620	OTP
620115	HRDLKGEN 0.6000	1	620	OTP
620115	HRDLKGEN 0.6000	2	620	OTP
620115	HRDLKGEN 0.6000	3	620	OTP
620115	HRDLKGEN 0.6000	4	620	OTP
620412	L1GEN2 9 0.5750	1	620	OTP
620413	L1GEN1 9 0.5750	1	620	OTP
620451	G474 GRNT CW41.600	W	620	OTP

Bus Number	Bus Name	Machine Id	Area Num	Area Name
657600	L1GEN3 9 0.5750	1	620	OTP
657601	L2GEN 9 0.5750	1	620	OTP
657737	ASH3GEN9 0.6900	1	620	OTP
657745	MPC02100GEN90.6900	1	620	OTP
657964	ASH2GEN9 0.5750	1	620	OTP
657964	ASH2GEN9 0.5750	2	620	OTP
657985	ASH1GEN9 0.5750	1	620	OTP
657985	ASH1GEN9 0.5750	2	620	OTP
562872	G15046_3 0.6900	1	652	WAPA
563302	G15096_3 0.6900	1	652	WAPA
563342	G15098_3 0.6900	1	652	WAPA
652004	G13_001IS_3 0.6900	1	652	WAPA
652007	G14_001IS_3 0.6900	1	652	WAPA
652173	G09_001IS_2 0.6900	1	652	WAPA
652345	G07_023IS_2 0.6900	W	652	WAPA
658019	MPS WIND G 0.6000	W1	652	WAPA
659031	G12_014IS_4 0.6900	1	652	WAPA
659127	G14_004IS_2 0.6900	1	652	WAPA
659141	G14_010IS_2 0.6900	1	652	WAPA
659407	ANTELPHILLSW0.6900	1	652	WAPA
660013	OAKTREE WIND0.7000	1	652	WAPA
660016	B_H WIND 0.7000	1	652	WAPA
50324	J442 G 0.6900	1	620	OTP
50420	J436 G 0.6900	1	620	OTP
50430	J437 G 0.6900	1	620	OTP

E.3.2 OTP LPC Summer Shoulder Study Results

Table E-4: OTP LPC SH System Intact Thermal Constraints

Table E-5: OTP LPC SH System Intact Voltage Constraints

Table E-6: OTP LPC SH Category P1 Thermal Constraints

Table E-7: OTP LPC SH Category P1 Voltage Constraints

Table E-8: OTP LPC SH Category P2-P7 Thermal Constraints

Table E-9: OTP LPC SH Category P2-P7 Voltage Constraints

Table E-10: OTP LPC SH Non-Converged Contingencies

Table E-11: OTP LPC SH Non-Converged Contingencies DCCC Results

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E.3.3 OTP LPC Summer Peak Study Results

Table E-12: OTP LPC SPK System Intact Thermal Constraints

Table E-13: OTP LPC SPK System Intact Voltage Constraints

Table E-14: OTP LPC SPK Category P1 Thermal Constraints

Table E-15: OTP LPC SPK Category P1 Voltage Constraints

Table E-16: OTP LPC SPK Category P2-P7 Thermal Constraints

Table E-17: OTP LPC SPK Category P2-P7 Voltage Constraints

Table E-18: OTP LPC SPK Non-Converged Category P2-P7 Contingencies

Table E-19: OTP LPC SPK Non-Converged Category P2-P7 Contingencies DCCC Results

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E.3.4 OTP LPC Light Load No Wind (LLNW) Study Results

Table E-20: OTP LPC LLNW System Intact Potential High Voltages

Table E-21: OTP LPC LLNW Category P1 Potential High Voltages

Table E-22: OTP LPC LLNW Category P2-P7 Potential High Voltages

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E.4 PJM Affected System Study Results

Below is the PJM affected system study report provided by PJM.

E.4.1 PJM Affected System Steady State Analysis

E.4.2 PJM Affected System J475 Dynamic Stability Analysis

E.4.3 PJM Affected System J530 Dynamic Stability Analysis

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E.5 SPP Affected System Study Results

Below is the SPP affected system study report provided by SPP.

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Appendix

F

Transient Stability Results

F.1 2022 Summer Shoulder Stability Results Summary

Stability simulation was performed in the 2022 summer shoulder case.

If the identified stability related issues (Section 4.5.1) are resolved, all stability simulations will be transiently stable, transient period voltage criteria will be met, oscillations will be well damped. No additional Network Upgrades are required in stability analysis.

Stability study results summary is in Table F-1.

Table F-1: 2022 Summer Shoulder Stability Analysis Results Summary

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F.2 2022 Summer Shoulder Stability Plots

Plots of stability simulations for 2022 summer shoulder study case are in separate files which are listed below:

AppendixF2_2022SH_DPP 2016Feb-West_Phase 2_Plots.zip

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MWEX Voltage Study Details

Below is the MWEX voltage stability study report provided by ATC.

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Short Circuit Analysis

- H.1 J432 Short Circuit Study Performed by Siemens PTI**
- H.2 J460 Short Circuit Study Performed by Siemens PTI**
- H.3 J475 Short Circuit Study Performed by MEC**
- H.4 J485 Short Circuit Study Performed by Siemens PTI**
- H.5 J488 & J489 Short Circuit Study Performed by OTP**
- H.6 J493 and J510 Short Circuit Study Performed by OTP**
- H.7 J495 Short Circuit Study Performed by ITCM**
- H.8 J498 Short Circuit Facilities Study Performed by MEC**
- H.9 J499 Short Circuit Facilities Study Performed by MEC**
- H.10 J500 Short Circuit Facilities Study Performed by MEC**
- H.11 J504 Short Circuit Facilities Study Performed by ITCM**
- H.12 J506 Short Circuit Facilities Study Performed by Siemens PTI**
- H.13 J514 Short Circuit Facilities Study Performed by ITCM**
- H.14 J523 Short Circuit Facilities Study Performed by ITCM**
- H.15 J524 Short Circuit Facilities Study Performed by Siemens PTI**
- H.16 J526 Short Circuit Facilities Study Performed by OTP**
- H.17 J527 Short Circuit Facilities Study Performed by Siemens PTI**
- H.18 J528 Short Circuit Facilities Study Performed by Siemens PTI**

H.19 J529 Short Circuit Facilities Study Performed by MEC

H.20 J530 Short Circuit Facilities Study Performed by Siemens PTI

H.21 J534 Short Circuit Facilities Study Performed by Siemens PTI

H.22 J535 Short Circuit Facilities Study Performed by Siemens PTI

J432 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J432 generating facility (49 Gamesa 2.0 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the Xcel system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "NSP_03202017.GDB"

The J432 short circuit benchmark model was developed as follows:

- Added Big Stone South-Ellendale 345 kV line and Ellendale 345/230/34.5 kV transformer.
- Added J460 project and turned it online.

The J432 short circuit study model was developed starting from the above benchmark case as follows:

- The J432 generating facility with its sequence parameters was added at the POI on Brookings County 345 kV Substation. The associated 345 kV Gentie, 345-34.5-13.2 kV transformer ($Z=9\%$ at 60 MVA, YG-YG-D), 34.5 kV collector system equivalent line and 34.5-0.69 kV pad mount transformer ($Z=11.6\%$ at 115.15 MVA, D-YG) were also added.
- The J432 generator was modeled as a generator with impedances listed below. These values represent the DFIG configuration with the maximum crowbar resistance connected in the rotor.
 - o Positive sequence impedances (i.e. subtransient, transient and synchronous) (Z_1): $0.71474+j0.3344$ pu
 - o Negative sequence impedance (Z_2): $0.04899+j0.1626$ pu
 - o Zero sequence impedance (Z_0): $j999999$ pu (zero sequence currents are blocked by the GSU transformers (Wye-Delta connection, therefore, this value does not have any impact on the short circuit results)

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J432 project) and benchmark case (without the J432 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to "1.00 per unit with multiplier of 1"
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J432 (study case) and without J432 (benchmark case) are listed in Table 1. With the addition of the J432 generating facility, the maximum increase in the 3PH short circuit current is 0.74% (126.1 A), and the maximum increase in the SLG short circuit current is 2.19% (382.1 A), which are both observed at the Brookings County 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

As shown in the Table 1, the changes in fault current at buses around the POI are relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J432 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J432

Substation	Faulted Bus			Benchmark Case without J432 Project		Study Case with J432 Project		Ampere Change with J432 vs Benchmark		% Change with J432 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J432	61031	J432_POI	345	-	-	17514.8	16541.2	-	-	-	-
J432	61032	J432-COL	34.5	-	-	17756.7	11618.6	-	-	-	-
J432	61034	J432_COL	34.5	-	-	8518.4	8546.4	-	-	-	-
J432	61035	J432_TR_TER	13.2	-	-	0.0	21383.1	-	-	-	-
BROOKINGS COUNTY (NS	113200	BOK_345KV_1	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113201	BOK_345KV_2	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113202	BOK_IB_BSS	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113203	BOK_IB_HNL	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113204	BOK_IB_TR09	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113205	BOK_IB_TR10	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113206	BOK_IB_WHI1	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113207	BOK_IB_WHI2	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
WHITE (WAPA)	225600	WHI_345KV_1	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	225601	WHI_345KV_2	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	225602	WHI_IB_BOK1	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	225603	WHI_IB_BOK2	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	225604	WHI_IB_SPK	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	225605	WHI_IB_TR01	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	225606	WHI_IB_WTT	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	300094	WHI_IB_DCR	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
DEER CREEK (WAPA)	300095	DCR_345KV	345	17140.7	16552.6	17476.8	16669.9	336.1	117.3	1.96%	0.71%
J460	61041	J460_POI	345	11512.8	12497.8	11577.8	12538.5	65.0	40.7	0.56%	0.33%
BROOKINGS COUNTY (NS	113208	BOK_115KV_1	115	28120.4	26806.8	28344.7	26889.9	224.3	83.1	0.80%	0.31%
BROOKINGS COUNTY (NS	113209	BOK_115KV_2	115	28120.4	26806.8	28344.7	26889.9	224.3	83.1	0.80%	0.31%
BROOKINGS COUNTY (NS	113210	BOK_IB_TR10	115	28120.4	26806.8	28344.7	26889.9	224.3	83.1	0.80%	0.31%
BROOKINGS COUNTY (NS	113211	BOK_IB_YNK1	115	28120.4	26806.8	28344.7	26889.9	224.3	83.1	0.80%	0.31%
BROOKINGS COUNTY (NS	113212	BOK_IB_YNK2	115	28120.4	26806.8	28344.7	26889.9	224.3	83.1	0.80%	0.31%
J460	61042	J460	345	11150.9	11940.4	11210.8	11977.2	59.9	36.8	0.54%	0.31%
HAWKS NEST (WIND)	300068	HNL_345KV_1	345	9715.8	11548.8	9751.0	11576.7	35.2	27.9	0.36%	0.24%
HAWKS NEST (WIND)	300069	HNL_345KV_2	345	9715.8	11548.8	9751.0	11576.7	35.2	27.9	0.36%	0.24%
HAWKS NEST (WIND)	300070	HNL_BOK	345	9715.8	11548.8	9751.0	11576.7	35.2	27.9	0.36%	0.24%
RED PINE (WIND)	300071	HNL_GEN	345	9658.6	11499.7	9693.3	11527.4	34.7	27.7	0.36%	0.24%
WATERTOWN (WAPA)	221500	WTT_345KV	345	8739.3	9643.6	8760.8	9660.8	21.5	17.2	0.25%	0.18%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J432

Substation	Faulted Bus			Benchmark Case without J432 Project		Study Case with J432 Project		Ampere Change with J432 vs Benchmark		% Change with J432 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
BIG STONE SOUTH (OTP)	109600	BST_345KV	345	8289.1	7472.0	8302.6	7481.3	13.5	9.3	0.16%	0.12%
WHITE (WAPA)	225607	WHI_115KV	115	12163.7	10520.0	12191.8	10532.5	28.1	12.5	0.23%	0.12%
YANKEE (NSPM)	229500	YNK_115KV_1	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229501	YNK_115KV_2	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229502	YNK_IB_BOK1	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229503	YNK_IB_BOK2	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229504	YNK_IB_BRI	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229505	YNK_IB_RB1	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229506	YNK_IB_TR01	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229507	YNK_IB_TR02	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	300448	YNK_REAC1	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
WATERTOWN (WAPA)	221501	WTT_230KV	230	11808.2	13524.8	11825.4	13539.9	17.2	15.1	0.15%	0.11%
LYON COUNTY (NSPM)	168300	LYC_345KV_1	345	10169.7	11844.0	10183.6	11856.6	13.9	12.6	0.14%	0.11%
LYON COUNTY (NSPM)	168301	LYC_345KV_2	345	10169.7	11844.0	10183.6	11856.6	13.9	12.6	0.14%	0.11%
LYON COUNTY (NSPM)	168302	LYC_IB_BOK1	345	10169.7	11844.0	10183.6	11856.6	13.9	12.6	0.14%	0.11%
LYON COUNTY (NSPM)	168303	LYC_IB_CMT2	345	10169.7	11844.0	10183.6	11856.6	13.9	12.6	0.14%	0.11%
LYON COUNTY (NSPM)	168304	LYC_IB_TR09	345	10169.7	11844.0	10183.6	11856.6	13.9	12.6	0.14%	0.11%
BIG STONE SOUTH (OTP)	300906	BSS_230KV	230	16323.8	12559.0	16340.7	12569.7	16.9	10.7	0.10%	0.09%
BIG STONE (OTP)	109601	BST_230KV	230	17057.9	12766.6	17074.9	12777.1	17.0	10.5	0.10%	0.08%
SPLIT ROCK (NSPM)	206100	SPK_345KV_1	345	13532.6	14710.5	13547.9	14721.6	15.3	11.1	0.11%	0.08%
SPLIT ROCK (NSPM)	206101	SPK_345KV_2	345	13532.6	14710.5	13547.9	14721.6	15.3	11.1	0.11%	0.08%
SPLIT ROCK (NSPM)	206102	SPK_IB_NOB	345	13532.6	14710.5	13547.9	14721.6	15.3	11.1	0.11%	0.08%
SPLIT ROCK (NSPM)	206103	SPK_IB_SOC	345	13532.6	14710.5	13547.9	14721.6	15.3	11.1	0.11%	0.08%
SPLIT ROCK (NSPM)	206104	SPK_IB_WHI	345	13532.6	14710.5	13547.9	14721.6	15.3	11.1	0.11%	0.08%

J460 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J460 generating facility (100 Vestas V110 2.0 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the Xcel system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "NSP_03202017.GDB"

The J460 short circuit benchmark model was developed as follows:

- Added Big Stone South-Ellendale 345 kV line and Ellendale 345/230/34.5 kV transformer.
- Added J432 project and turned it online

The J460 short circuit study model was developed starting from the above benchmark case as follows:

- The J460 generating facility with its sequence parameters was added at the POI on Brookings County – Hawks Nest (H081) 345 kV Substation. The associated 345 kV Gentie, two 345-34.5-13.2 kV transformers ($Z=6\%$ at 100 MVA, YG-YG-D), two 34.5 kV collector system equivalent lines and two 34.5-0.690 kV pad mount transformers ($Z+=7.8\%$, $Z0=7.3\%$ at 105 MVA, D-YG) were also added.
- The J460 generation was represented using the conventional generator model with the impedance of the machines set as follows per the data provided in the Vestas manual.
 - Positive saturated sequence impedances (i.e. subtransient, transient and synchronous) ($Z1$): j0.1669 p.u.
 - Negative saturated sequence impedance ($Z2$): j0.1669 p.u. (same as positive sequence impedances)
 - Zero sequence saturated impedance ($Z0$): j0.1669 p.u.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J460 project) and benchmark case (without the J460 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to "1.00 per unit with multiplier of 1"
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J460 (study case) and without J460 (benchmark case) are listed in Table 1. With the addition of the J460 generating facility, the maximum increase in the 3PH short circuit current is 6.58% (715.2 A), and the maximum increase in the SLG short circuit current is 16.76% (1399.6 A), which are both observed at the Hawks Nest (H081) 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 11577 A (increased by 715.2 A) and SLG fault current is 9751 A (increased by 1399.6 A) at the Hawks Nest 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J460 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J460

Substation	Faulted Bus			Benchmark Case without J460 Project		Study Case with J460 Project		Ampere Change with J460 vs Benchmark		% Change with J460 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J460	61041	J460_POI	345	-	-	11577.8	12538.5	-	-	-	-
J460	61042	J460	345	-	-	11210.8	11977.2	-	-	-	-
J460	61043	J460_COL1	34.5	-	-	38909.0	29079.0	-	-	-	-
J460	61046	J460_COL2	34.5	-	-	38436.0	28683.7	-	-	-	-
J460	61047	J460_COL1	34.5	-	-	33279.5	26558.9	-	-	-	-
J460	61048	J460_COL2	34.5	-	-	27091.6	23334.5	-	-	-	-
J460	61049	J460_TER1	13.2	-	-	0.0	53996.6	-	-	-	-
J460	61050	J460_TER2	13.2	-	-	0.0	54574.1	-	-	-	-
HAWKS NEST (WIND)	300068	HNL_345KV_1	345	8351.4	10861.5	9751.0	11576.7	1399.6	715.2	16.76%	6.58%
HAWKS NEST (WIND)	300069	HNL_345KV_2	345	8351.4	10861.5	9751.0	11576.7	1399.6	715.2	16.76%	6.58%
HAWKS NEST (WIND)	300070	HNL_BOK	345	8351.4	10861.5	9751.0	11576.7	1399.6	715.2	16.76%	6.58%
RED PINE (WIND)	300071	HNL_GEN	345	8310.2	10818.4	9693.3	11527.4	1383.1	709.0	16.64%	6.55%
BROOKINGS COUNTY (NS)	113200	BOK_345KV_1	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113205	BOK_IB_TR10	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113206	BOK_IB_WH11	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113201	BOK_345KV_2	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113204	BOK_IB_TR09	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113207	BOK_IB_WH12	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113202	BOK_IB_BSS	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113203	BOK_IB_HNL	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
WHITE (WAPA)	225600	WHI_345KV_1	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	300094	WHI_IB_DCR	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	225601	WHI_345KV_2	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	225604	WHI_IB_SPK	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	225606	WHI_IB_WTT	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	225605	WHI_IB_TR01	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	225603	WHI_IB_BOK2	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	225602	WHI_IB_BOK1	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
J432	61031	J432_POI	345	16705.3	15910.4	17514.8	16541.2	809.5	630.8	4.85%	3.96%
DEER CREEK (WAPA)	300095	DCR_345KV	345	16729.8	16056.2	17476.8	16669.9	747.0	613.7	4.47%	3.82%
LYON COUNTY (NSPM)	168300	LYC_345KV_1	345	9847.1	11563.6	10183.6	11856.6	336.5	293.0	3.42%	2.53%
LYON COUNTY (NSPM)	168301	LYC_345KV_2	345	9847.1	11563.6	10183.6	11856.6	336.5	293.0	3.42%	2.53%
LYON COUNTY (NSPM)	168304	LYC_IB_TR09	345	9847.1	11563.6	10183.6	11856.6	336.5	293.0	3.42%	2.53%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J460

Substation	Faulted Bus			Benchmark Case without J460 Project		Study Case with J460 Project		Ampere Change with J460 vs Benchmark		% Change with J460 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
LYON COUNTY (NSPM)	168302	LYC_IB_BOK1	345	9847.1	11563.6	10183.6	11856.6	336.5	293.0	3.42%	2.53%
LYON COUNTY (NSPM)	168303	LYC_IB_CMT2	345	9847.1	11563.6	10183.6	11856.6	336.5	293.0	3.42%	2.53%
BROOKINGS COUNTY (NS)	113210	BOK_IB_TR10	115	27824.4	26455.5	28344.7	26889.9	520.3	434.4	1.87%	1.64%
BROOKINGS COUNTY (NS)	113211	BOK_IB_YNK1	115	27824.4	26455.5	28344.7	26889.9	520.3	434.4	1.87%	1.64%
BROOKINGS COUNTY (NS)	113208	BOK_115KV_1	115	27824.4	26455.5	28344.7	26889.9	520.3	434.4	1.87%	1.64%
BROOKINGS COUNTY (NS)	113209	BOK_115KV_2	115	27824.4	26455.5	28344.7	26889.9	520.3	434.4	1.87%	1.64%
BROOKINGS COUNTY (NS)	113212	BOK_IB_YNK2	115	27824.4	26455.5	28344.7	26889.9	520.3	434.4	1.87%	1.64%
HAZEL CREEK (NSPM)	148702	HZC_IB_LYC	345	6884.9	7977.3	6974.8	8087.0	89.9	109.7	1.31%	1.38%
HAZEL CREEK (NSPM)	148700	HZC_345KV_1	345	6884.9	7977.3	6974.8	8087.0	89.9	109.7	1.31%	1.38%
CEDAR MOUNTAIN (GRE)	119104	CMT_IB_LYC2	345	8864.5	11514.3	8941.6	11644.0	77.1	129.7	0.87%	1.13%
CEDAR MOUNTAIN (GRE)	119100	CMT_345KV_1	345	8864.5	11514.3	8941.6	11644.0	77.1	129.7	0.87%	1.13%
CEDAR MOUNTAIN (GRE)	119103	CMT_IB_HNA2	345	8864.5	11514.3	8941.6	11644.0	77.1	129.7	0.87%	1.13%
CEDAR MOUNTAIN (GRE)	119101	CMT_345KV_2	345	8864.5	11514.3	8941.6	11644.0	77.1	129.7	0.87%	1.13%
CEDAR MOUNTAIN (GRE)	119105	CMT_IB_TR01	345	8864.5	11514.3	8941.6	11644.0	77.1	129.7	0.87%	1.13%
CEDAR MOUNTAIN (GRE)	119102	CMT_IB_HNA1	345	8864.5	11514.3	8941.6	11644.0	77.1	129.7	0.87%	1.13%
WATERTOWN (WAPA)	221500	WTT_345KV	345	8698.8	9561.6	8760.8	9660.8	62.0	99.2	0.71%	1.04%
LYON COUNTY (NSPM)	168305	LYC_115KV_1	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168306	LYC_115KV_2	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168312	LYC_IB_CLO	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168307	LYC_CAP	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168316	LYC_IB_TR09	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168311	LYC_IB_CAP2	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168310	LYC_IB_CAP1	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168315	LYC_IB_MSH	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168313	LYC_IB_LAY1	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168314	LYC_IB_LAY2	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168308	LYC_CAP1	115	17035.5	16216.2	17208.7	16346.0	173.2	129.8	1.02%	0.80%
LYON COUNTY (NSPM)	168309	LYC_CAP2	115	17035.5	16216.2	17208.7	16346.0	173.2	129.8	1.02%	0.80%
BIG STONE SOUTH (OTP)	109600	BST_345KV	345	8261.4	7429.0	8302.6	7481.3	41.2	52.3	0.50%	0.70%
WATERTOWN (WAPA)	221501	WTT_230KV	230	11771.8	13446.7	11825.4	13539.9	53.6	93.2	0.46%	0.69%
YANKEE (NSPM)	229500	YNK_115KV_1	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	229501	YNK_115KV_2	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	229506	YNK_IB_TR01	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J460

Substation	Faulted Bus			Benchmark Case without J460 Project		Study Case with J460 Project		Ampere Change with J460 vs Benchmark		% Change with J460 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
YANKEE (NSPM)	229507	YNK_IB_TR02	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	229504	YNK_IB_BRI	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	229502	YNK_IB_BOK1	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	300448	YNK_REAC1	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	229505	YNK_IB_RB1	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	229503	YNK_IB_BOK2	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
HAZEL CREEK (NSPM)	148705	HZC_IB_MNV	230	11431.9	11656.2	11497.3	11730.0	65.4	73.8	0.57%	0.63%
HAZEL CREEK (NSPM)	148703	HZC_230KV_1	230	11431.9	11656.2	11497.3	11730.0	65.4	73.8	0.57%	0.63%
WHITE (WAPA)	225607	WHI_115KV	115	12120.3	10468.6	12191.8	10532.5	71.5	63.9	0.59%	0.61%
MINNESOTA VALLEY (NS	176501	MNV_230KV_1	230	13600.1	12962.7	13667.1	13040.9	67.0	78.2	0.49%	0.60%
MINNESOTA VALLEY (NS	176502	MNV_230KV_2	230	13600.1	12962.7	13667.1	13040.9	67.0	78.2	0.49%	0.60%
MINNESOTA VALLEY (NS	176503	MNV_230KV_5	230	13600.1	12962.7	13667.1	13040.9	67.0	78.2	0.49%	0.60%
MINNESOTA VALLEY (NS	176504	MNV_230KV_6	230	13600.1	12962.7	13667.1	13040.9	67.0	78.2	0.49%	0.60%
MINNESOTA VALLEY (NS	176500	MNVT_GRA_PTH	230	13506.2	12925.2	13571.7	13002.5	65.5	77.3	0.48%	0.60%
RED PINE (WIND)	300072	HNL_GEN1_034	34.5	15362.3	15293.5	15653.7	15384.2	291.4	90.7	1.90%	0.59%
RED PINE (WIND)	300073	HNL_GEN2_034	34.5	15362.3	15293.5	15653.7	15384.2	291.4	90.7	1.90%	0.59%
GRANITE FALLS (WAPA)	145100	GRA_230KV_1	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145101	GRA_230KV_2	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145102	GRA_IB_BAR	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145104	GRA_IB_KV8A	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145108	GRA_IB_WMR	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145105	GRA_IB_MNV1	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145107	GRA_IB_MOS	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145103	GRA_IB_KV3A	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145109	GRA_IB_WTT	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145106	GRA_IB_MNV2	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
MARSHALL SW STATION	170600	MSH_115KV	115	9491.8	12041.8	9531.8	12102.6	40.0	60.8	0.42%	0.50%
MARSHALL SW STATION	170603	MSH_IB_SEA	115	9491.8	12041.8	9531.8	12102.6	40.0	60.8	0.42%	0.50%
MARSHALL SW STATION	170601	MSH_IB_LYC	115	9491.8	12041.8	9531.8	12102.6	40.0	60.8	0.42%	0.50%
MARSHALL SW STATION	170602	MSH_IB_NSV	115	9491.8	12041.8	9531.8	12102.6	40.0	60.8	0.42%	0.50%
BIG STONE SOUTH (OTP	300906	BSS_230KV	230	16282.6	12508.0	16340.7	12569.7	58.1	61.7	0.36%	0.49%
BIG STONE (OTP)	109601	BST_230KV	230	17015.1	12715.8	17074.9	12777.1	59.8	61.3	0.35%	0.48%
NORTH SEVENTH STREET	183200	NSV_115KV_1	115	8659.1	11161.9	8690.8	11212.5	31.7	50.6	0.37%	0.45%

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J475 and J555 Short Circuit Study Performed by MEC

The scope of this DPP Optional Study facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J475, a proposed 200 MW wind farm, and J555, a proposed 140 MW expansion, and nearby substations both with and without the Interconnection Customer interconnected. J475 and J555 were assumed to interconnect at the MidAmerican and ITC Midwest jointly-owned Montezuma Substation. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in DPP study cycles of the MISO generation interconnection process, where the system impact studies are not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 13,008 Amps without the Interconnection Customer included and 14,334 Amps with the full 340 MW from the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results, especially the 34.5 kV results that should be calculated by the Interconnection Customer. For example, generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation or system contingencies.

As shown in the table, the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican breakers become overdutied as a result of the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

**Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J475 and J555**

Bus Number	Bus Name	English Name	Base kV	Area Num	Owner	SLG Fault Current Comparison					3 Ph Fault Current Comparison				
						Base SLG w/o wind farms J475/J555	SLG with new J475 wind farm	SLG Difference w/ J475 wind farm vs Base	SLG with J475 & J555 wind farm	SLG Difference w/ J475 & J555 wind farm vs Base	Base 3PH w/o wind farms J475/J555	3PH with new J475 wind farm	3PH Difference w/ J475 wind farm vs Base	3PH with J475 & J555 wind farm	3PH Difference w/ J475 & J555 wind farm vs Base
635730	MNTZUMA3	Montezuma	345	635	MEC	9,020	10,083	1,063	11,078	2,058	13,008	13,705	697	14,334	1,326
		J475 Collector Bus	34.5			19,371	32,795	13,424	32,856	13,485	16,744	27,713	10,969	27,756	11,012
		J555 Collector Bus	34.5			19,371	19,371	-	29,325	9,954	16,744	16,744	-	24,307	7,563
635680	BONDRNT3	Bondurant	345	635	MEC	17,016	17,131	115	17,223	207	19,055	19,188	133	19,299	244
631143	OTTUMWA3	Ottumwa	345	627	ITCM	9,031	9,193	162	9,322	291	10,891	11,055	164	11,194	303
	J530 POI	J530 POI	345	635	MEC	10,293	10,542	249	10,742	449	12,902	13,146	244	13,354	452
635700	SYCAMOR3	Sycamore	345	635	MEC	19,005	19,070	65	19,122	117	20,106	20,193	87	20,265	159
635690	GDMEC	GDMEC	345	635	MEC	18,566	18,638	72	18,697	131	18,624	18,711	87	18,783	159
344000	7ADAIR	Adair/Zachary 345	345	356	AMMO	8,920	8,934	15	8,946	26	12,465	12,494	29	12,517	52
631115	OTTUMWA5	Ottumwa	161	627	ITCM	27,848	27,929	81	27,993	145	23,967	24,035	68	24,091	124
636400	HILLS 3	Hills	345	635	MEC	16,822	16,902	80	16,966	144	19,937	20,054	117	20,151	214
		J530 Collector Bus	34.5			30,409	30,480	71	30,537	128	26,522	26,574	52	26,617	95
635600	GRIMES 3	Grimes	345	635	MEC	18,245	18,292	47	18,330	85	20,460	20,535	75	20,597	137
635701	SYCAMOR5	Sycamore	161	635	MEC	33,003	33,067	64	33,119	116	32,925	33,003	78	33,068	143
635650	SE POLK3	SE Polk	345	635	MEC	17,992	18,057	65	18,110	118	18,356	18,438	82	18,505	149
345435	7MAYWOOD	Maywood	345	356	AMMO	8,222	8,226	4	8,228	6	11,420	11,429	9	11,436	16
344006	5ADAIR3	Adair/Zachary 161	161	356	AMMO	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
631110	WAPELLO5	Wapello	161	627	ITCM	10,727	10,735	8	10,741	14	11,096	11,108	12	11,117	21
631118	LUCAS 5	Lucas Co	161	627	ITCM	3,221	3,222	1	3,222	1	3,339	3,340	1	3,341	2
631116	BRDGPRT5	Bridgeport	161	627	ITCM	15,772	15,794	22	15,811	39	17,111	17,144	33	17,170	59
631134	TRICNTY5	Tri County	161	627	ITCM	15,937	15,960	23	15,978	41	17,358	17,391	33	17,418	60
635880	PLSNT CRNR 5	Pleasant Corner	161	635	MEC	9,837	9,846	9	9,853	15	12,403	12,420	17	12,434	31
636420	TIFFIN 3	Tiffin	345	635	MEC	11,251	11,275	24	11,295	44	14,815	14,864	49	14,905	90
636620	WALCOTT3	Walcott	345	635	MEC	11,076	11,083	7	11,088	12	15,648	15,667	19	15,682	34
636640	LOUISA 3	Louisa	345	635	MEC	19,061	19,085	24	19,105	44	19,077	19,111	34	19,139	62
636645	SUB T 3	Sub T Haskins	345	635	MEC	11,232	11,252	20	11,269	37	15,712	15,759	47	15,797	85
636401	HILLS 5	Hills	161	635	MEC	19,245	19,278	33	19,304	59	19,685	19,723	38	19,755	70
631034	ROSEHLW5	Rose Hollow	161	627	ITCM	5,802	5,803	2	5,804	3	6,551	6,554	3	6,556	5

J485 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J485 generating facility on the adequacy of existing circuit breakers and related equipment in the Project area.

1.2 Short Circuit Model

An ASPEN short circuit database including positive, negative and zero sequence parameters of the project area was provided by MISO. This starting short circuit model is listed below:

- Short Circuit Model: "RPUcApe2014Base_2016.03.01.dxt"

The J485 short circuit study model was developed as follows:

- The J485 gas generating facility with its sequence parameter was added at the West Side 161 kV bus via a 13.8 kV/161 kV transformer ($Z=7.5\%$ at 39 MVA, $X/R=24$, D-YG)
- The J485 gas project was modeled at the West Side 13.8 kV bus with MVA base of 58.38 MVA. All impedance values are set using saturated reactance data.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J485 project) and benchmark case (without the J485 project). The following fault simulation options were used in the short circuit analysis.

- The prefault voltage was set to "from a linear network solution"

Three-phase-to-ground (3PH) and single-phase-to-ground (SLG) faults were simulated on buses within the Study Project area. The results of short circuit currents with J485 (study case) and without J485 (benchmark case) are listed in Table 1. With the addition of the J485 generating facility, the maximum increase in the 3PH short circuit current is 3.89% (643.1 amps), and the maximum increase in the SLG short circuit current is 13.91% (1632.1 amps), which are both observed at the West Side 161 kV substation bus.

1.4 Summary of J485 Short Circuit Analysis

The study results show that the 3PH fault current is 17159 kA (increased by 643.1 A) and SLG fault current is 13367 A (increased by 1632.1 A) at the West Side 161 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J485 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J485

ASPEN Bus Number	ASPEN Bus Name	ASPEN Base kV	LOCATION	AREA	ZONE	Benchmark Case without J485 Project		Study Case with J485 Project		Amps Change with J485 vs Benchmark		% Change with J485 vs Benchmark	
						3PH Fault SC Amps	SLG Fault SC Amps	3PH Fault SC Amps	SLG Fault SC Amps	3PH	SLG	3PH	SLG
16	WS 161	161	WESTSIDE	1	1	16515.8	11734.8	17158.9	13366.9	643.1	1632.1	3.89%	13.91%
98	Q9 T	161	IBM TAP	1	1	17794.7	13096.4	18337.7	14273.9	543.0	1177.5	3.05%	8.99%
61906	ML 161 TAP	161	MAPLE LE	1	1	16233.6	11439.9	16695.2	12412.6	461.6	972.7	2.84%	8.50%
61907	ML 161	161	MAPLE LE	1	1	16233.6	11439.8	16695.1	12412.6	461.5	972.8	2.84%	8.50%
110	IBM 161	161	IBM STAT	1	1	17536.3	12929.2	17993.9	13817.2	457.6	888.0	2.61%	6.87%
13	NH 161	161	NORTHERN	1	1	16699.4	12107.8	17112.2	12889.9	412.8	782.1	2.47%	6.46%
7	CC 161	161	CASCADE	1	1	18754.4	14433.5	19157.9	15116.2	403.5	682.7	2.15%	4.73%
14	CRT 161	161	CROSSTOW	1	1	18179.8	13912.0	18540.8	14496.0	361.0	584.0	1.99%	4.20%
5	SLS 161	161	SILVER L	1	1	18040.5	13804.6	18390.4	14365.0	349.9	560.4	1.94%	4.06%
61948	BYR 161	161	BYRON	1	1	20626.7	16352.2	21002.8	16955.3	376.1	603.1	1.82%	3.69%
8	ZR 161	161	ZUMBRO R	1	1	13178.6	8931.1	13394.3	9261.4	215.7	330.3	1.64%	3.70%
15	CHR 161	161	CHESTER	1	1	18106.0	13270.1	18385.3	13647.1	279.3	377.0	1.54%	2.84%
69547	ROCH 161	161	ROCHESTE	1	1	18074.4	13239.1	18351.8	13612.9	277.4	373.8	1.53%	2.82%
9	BV 161	161	BAMBER V	1	1	13681.0	9389.3	13870.4	9642.4	189.4	253.1	1.38%	2.70%
6	WC 161	161	WILLOW C	1	1	13109.0	8984.5	13272.7	9202.0	163.7	217.5	1.25%	2.42%
100	NR 161	161	NORTH RO	1	1	18282.8	17302.0	18407.0	17482.5	124.2	180.5	0.68%	1.04%
61950	BYR 345	345	BYRON	1	1	16573.2	12891.0	16675.2	13082.7	102.0	191.7	0.62%	1.49%
103	NR 345	345	NORTH RO	1	1	19418.7	15499.6	19513.8	15647.9	95.1	148.3	0.49%	0.96%
63032	PV 345	345	PLEASANT	1	1	16912.7	15416.4	16982.7	15480.9	70.0	64.5	0.41%	0.42%
63033	PV 161	161	PLEASANT	1	1	19807.0	4795.5	19875.0	4831.7	68.0	36.2	0.34%	0.75%
69549	WABACO 161	161	WABACO	1	1	10457.2	7464.1	10484.0	7490.9	26.8	26.8	0.26%	0.36%
60102	ADAMS 345	345	ADAMS	1	1	13874.1	11875.5	13900.4	11890.9	26.3	15.4	0.19%	0.13%

A short circuit analysis was performed to determine the impact of the proposed generation on the adjacent substation fault current levels. The proposed generation, as outlined in this document, was added to the existing Otter Tail Power Company's ASPEN OneLiner system model (BANG13-0503.OLR) and faults were simulated on all busses within one substation from the substation that contained the point of interconnection. As per a recommendation from MISO, the fault current from the J488/J489 wind generation was assumed to be 1 per unit of the full nameplate rating.

Study Results

Based on the results of the study, the fault current ratings of the Otter Tail equipment in the area are not exceeded.

With the proposed J488/J489 project additions, the fault currents are estimated to be roughly 8.6 kA at the Twin Brooks 345 kV bus, 11.2 kA at the Big Stone South 345 kV bus, 19.7 kA at the Big Stone South 230 kV bus, and 33.6 kA at the Big Stone South 34.5 kV bus. There does not appear to be any short circuit related upgrades needed for J488/J489.

A general protective relay coordination review will need to be performed by OTP when the projects approach completion and before energizing the generation source.



Interconnection Facility Study Report

Generation Interconnection
MISO Projects J493 and J510
Prepared for Midcontinent ISO

April 25, 2017

This report contains CEII and Trade Secret information and should not be released without consent from Otter Tail Power Company

2. Short Circuit Study

A short circuit analysis was performed to determine the impact of the proposed generation projects on the adjacent substation fault current levels. The proposed generation, as outlined in this document, was added to the existing Transmission Owner's ASPEN OneLiner system model (BANG13-0503.OLR) and faults were simulated on all busses within one substation from the substation that contained the point of interconnection. As per a recommendation from MISO, the fault current from the J493 wind generation was assumed to be 1 per unit of the full nameplate rating.

Study Results

Based on the results of the study, the fault current ratings of the Transmission Owner's equipment in the area are not exceeded.

With the proposed J493 project additions, the fault currents are estimated to be roughly 16 kA at the Big Stone South 230 kV bus and the faults on the 345 kV bus were below these values. There do not appear to be any short circuit related upgrades needed for J493.

With the proposed J493 and J510 project additions, the fault currents are estimated to be roughly 16.6 kA at the Big Stone South 230 kV bus. There do not appear to be any short circuit related upgrades needed for J493 and J510.

A general protective relay coordination review will need to be performed by Transmission Owner when the projects approach completion and before energizing the generation source.

This report contains CEII and Trade Secret information and should not be released without consent from Otter Tail Power Company



Evaluation of Fault Currents and Interrupting Capabilities for Generation Interconnection Project J495

7-17-2017

Summary

Generator Interconnection Project J495 is a 200 MW wind powered generating facility with proposed interconnection to the 345 kV bus at a new switch station tapping ITC Midwest's Ledyard to Colby 345 kV line. Three phase and single line to ground fault current values were evaluated before and after interconnection of the generating facility using Computer Aided Protection Engineering (CAPE) software by Electrocon International. The base CAPE model maintained and used by ITC Midwest's system protection group was updated to include planned BES future system topology representative of a 2019 system was used in the analysis.

Generating Facility Assumptions

The generating facility was modeled based on available information, and included:

- Two 100 MW equivalent generators
- Two 34.5kV to 690V equivalent transformers
- Two 34.5kV to 345 kV step up transformers
- 1.3 mile 345 kV interconnection line

Figure 1 below shows a one-line diagram of the modeled generating facility and nearby system.

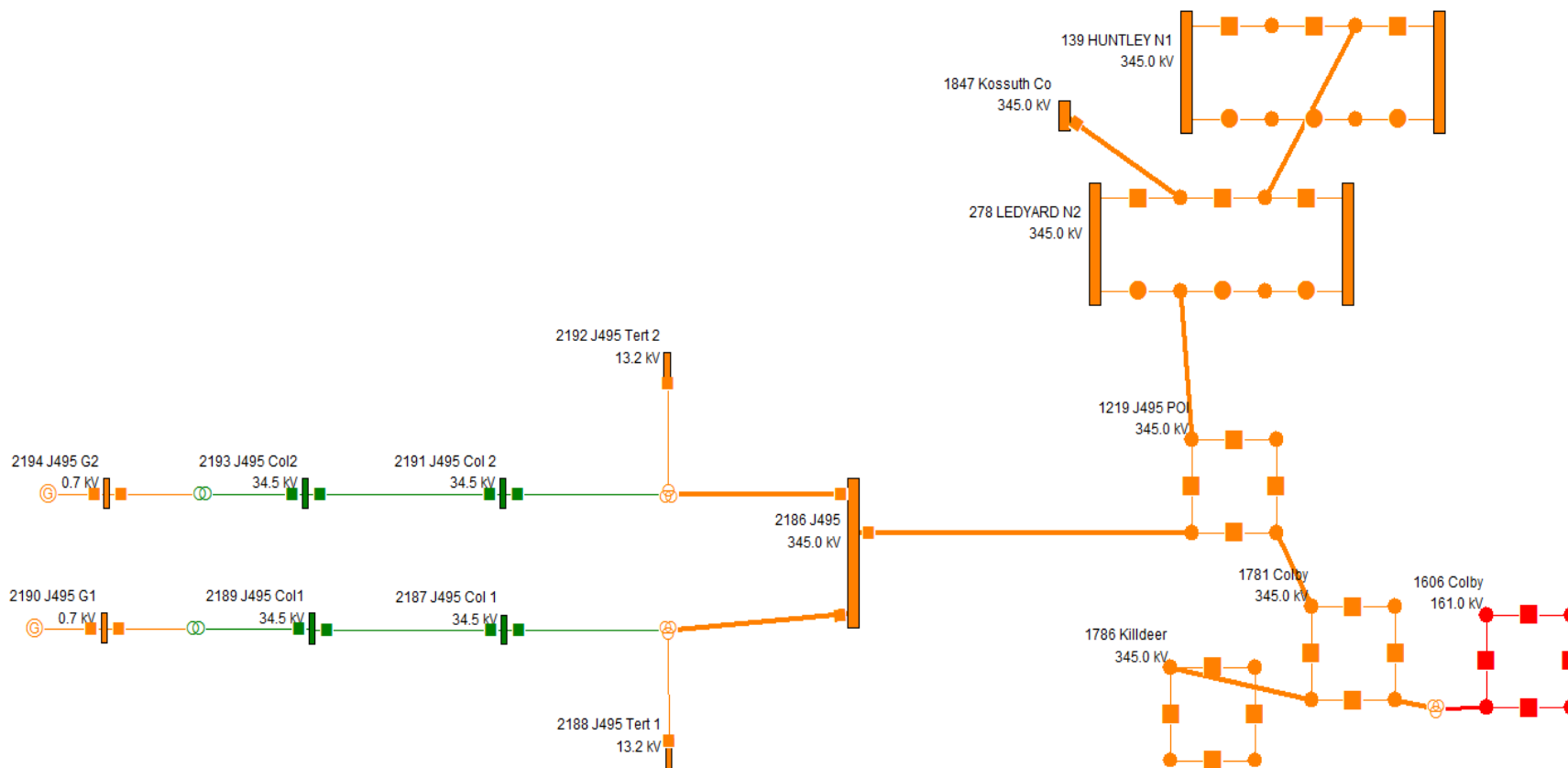


Figure1



Fault Currents

Table 1 below has the three phase and single line to ground fault currents before and after the addition of Project J495.

J495	Ledyard 345	Colby 345	Colby 161	Huntley 345	Killdeer 345	Kossuth
SLG-pre (A)	9121.93	9313.45	20243.8	8496.7	9803.83	10370
3Ph-pre (A)	13062	11044.5	22145.6	11436.2	10822.5	14788.5
SLG-post (A)	10356.4	9530.15	20410.2	8795.42	9901.52	10673.7
3Ph-post (A)	13094.9	11059	22153.7	11449.9	10833	14804.2
Contribution SLG (A)	1234.47	216.7	166.4	298.72	97.69	303.7
Contribution 3Ph (A)	32.9	14.5	8.1	13.7	10.5	15.7

Conclusion

Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J495. Equipment not owned by ITC Midwest was not evaluated for interrupting capability. New circuit breakers at the POI switch station were assumed to have 50kA interrupting capability.

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J498 Short Circuit Study Performed by MEC

The scope of this DPP Optional Study facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J498, a proposed 340 MW wind farm, and nearby substations both with and without the Interconnection Customer interconnected. J498 was assumed to interconnect off the Lehigh-Grimes 345 kV line. MidAmerican is the operating agent of this jointly-owned line. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in a DPP study cycle of the MISO generation interconnection process where the system impact study is not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 11,055 Amps without the Interconnection Customer included and 12,498 Amps with the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results. For example, generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation or system contingencies.

As shown in the table, the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J498

Bus Number	Bus Name	English Name	Base kV	Owner	SLG Fault Current Comparison			3 Ph Fault Current Comparison		
					Base SLG w/o new wind farm	SLG with new wind farm	SLG Difference w/ wind farm vs Base	Base 3PH w/o new wind farm	3PH with new wind farm	3PH Difference w/ wind farm vs Base
12083	J498_TAP	J498 POI	345	MEC	7,799	9,576	1,776	11,055	12,498	1,443
11798	J498_COLHI	J498 Collector Bus	34.5		NA	31,416	NA	NA	28,289	NA
636010	LEHIGH 3	Lehigh	345	MEC	9,962	10,784	822	12,680	13,501	821
635600	GRIMES 3	Grimes	345	MEC	18,987	19,347	360	21,429	21,860	431
636000	WEBSTER3	Webster	345	MEC	10,304	10,683	379	12,492	12,946	454
6201	J411 POI	Ida Co	345	MEC	7,133	7,197	65	9,527	9,621	94
635700	SYCAMOR3	Sycamore	345	MEC	19,630	19,888	259	20,933	21,270	337
12084	J499_TAP	J499 POI	345	MEC	9,223	9,253	31	11,409	11,462	53
635630	BOONVIL3	Booneville	345	MEC	15,390	15,501	111	18,010	18,186	176
635601	GRIMES 5	Grimes	161	MEC	23,008	23,148	140	24,022	24,182	160
635369	KOSSUTH 3	Kossuth Co	345	MEC	12,312	12,389	77	14,979	15,093	113
636001	WEBSTER5	Webster	161	MEC	17,538	17,777	239	18,015	18,222	207
635200	RAUN 3	Raun	345	MEC	27,097	27,157	60	25,447	25,509	61
635680	BONDRNT3	Bondurant	345	MEC	17,504	17,629	125	19,708	19,907	200
635701	SYCAMOR5	Sycamore	161	MEC	33,629	33,864	235	33,694	33,974	281
635590	FALLOW 3	Fallow Ave	345	MEC	8,708	8,725	17	10,215	10,244	29
12085	J500_TAP	J500 POI	345	MEC	13,423	13,454	31	17,020	17,083	63
635635	MADISON3	Madison Co	345	MEC	13,218	13,279	61	16,259	16,374	115
635631	BOONVIL5	Booneville	161	MEC	18,692	18,754	63	21,728	21,825	97
635602	142ST 5	142nd St	161	MEC	17,682	17,762	80	21,324	21,447	123
635614	100 54 5	100th St & 54th Ave	161	MEC	20,682	20,778	97	25,122	25,280	158
635609	141 44 5	Hwys 141 & 44	161	MEC	15,237	15,286	49	19,093	19,182	89

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J499 Short Circuit Study Performed by MEC

The scope of this DPP Optional Study facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J499, a proposed 340 MW wind farm, and nearby substations both with and without the Interconnection Customer interconnected. J499 was assumed to interconnect off the Fallow-Grimes 345 kV line. MidAmerican is the operating agent for this jointly-owned line. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in a DPP study cycle of the MISO generation interconnection process where the system impact study is not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 10,017 Amps without the Interconnection Customer included and 11,462 Amps with the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results. For example, generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation or system contingencies.

As shown in the table, the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J499

Bus Number	Bus Name	English Name	Base kV	Owner	SLG Fault Current Comparison			3 Ph Fault Current Comparison		
					Base SLG w/o new wind farm	SLG with new wind farm	SLG Difference w/ wind farm vs Base	Base 3PH w/o new wind farm	3PH with new wind farm	3PH Difference w/ wind farm vs Base
12084	J499_TAP	J499 POI	345	MEC	7,553	9,253	1,700	10,017	11,462	1,445
11802	J499_COLHI	J499 Collector Bus	34.5		NA	31,310	NA	NA	28,180	NA
635590	FALLOW 3	Fallow Ave	345	MEC	7,984	8,725	741	9,476	10,244	768
635600	GRIMES 3	Grimes	345	MEC	18,785	19,347	562	21,164	21,860	695
635589	ECLIPSE3	Eclipse	345	MEC	6,642	7,086	443	7,914	8,428	514
635014	POTTCO 3	Overland Trail	345	MEC	24,626	24,800	173	24,686	24,909	222
635700	SYCAMOR3	Sycamore	345	MEC	19,481	19,888	408	20,727	21,270	543
12083	J498_TAP	J498 POI	345	MEC	9,543	9,576	33	12,434	12,498	64
635630	BOONVIL3	Booneville	345	MEC	15,317	15,501	185	17,889	18,186	297
635601	GRIMES 5	Grimes	161	MEC	22,931	23,148	217	23,928	24,182	255
635000	CBLUFFS3	Council Bluffs	345	MEC	31,629	31,829	199	28,966	29,192	226
635015	POTTCO 5	Overland Trail	161	MEC	25,444	25,496	52	26,439	26,516	77
635680	BONDRNT3	Bondurant	345	MEC	17,430	17,629	199	19,583	19,907	324
635701	SYCAMOR5	Sycamore	161	MEC	33,503	33,864	361	33,534	33,974	440
636010	LEHIGH 3	Lehigh	345	MEC	10,764	10,784	20	13,464	13,501	37
12085	J500_TAP	J500 POI	345	MEC	13,393	13,454	61	16,957	17,083	126
635635	MADISON3	Madison Co	345	MEC	13,176	13,279	102	16,177	16,374	197
635631	BOONVIL5	Booneville	161	MEC	18,650	18,754	105	21,664	21,825	161
635602	142ST 5	142nd St	161	MEC	17,637	17,762	124	21,252	21,447	196
635614	100 54 5	100th St & 54th Ave	161	MEC	20,629	20,778	149	25,031	25,280	249
635609	141 44 5	Hwys 141 & 44	161	MEC	15,211	15,286	76	19,043	19,182	139

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J500 Short Circuit Study Performed by MEC

The scope of this DPP Optional Study facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J500, a proposed 500 MW wind farm, and nearby substations both with and without the Interconnection Customer interconnected. J500 was assumed to interconnect off the Booneville-Atchison County and Rolling Hills-Madison County 345 kV lines. MidAmerican is the sole owner or operating agent for these line facilities. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in a DPP study cycle of the MISO generation interconnection process where the system impact study is not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 14,936 Amps without the Interconnection Customer included and 17,083 Amps with the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results. For example, generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation or system contingencies.

As shown in the table, the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J500

Bus Number	Bus Name	English Name	Base kV	Owner	SLG Fault Current Comparison			3 Ph Fault Current Comparison		
					Base SLG w/o new wind farm	SLG with new wind farm	SLG Difference w/ wind farm vs Base	Base 3PH w/o new wind farm	3PH with new wind farm	3PH Difference w/ wind farm vs Base
12085	J500_TAP	J500 POI	345	MEC	10,841	13,454	2,613	14,936	17,083	2,147
11806	J500_COLHI	J500 Collector Bus	34.5		NA	31,689	NA	NA	28,412	NA
635100	RLHILLS3	Rolling Hills	345	MEC	9,295	10,160	865	12,196	12,955	759
635017	ATCHSN 3	Atchison Co	345	MEC	11,181	11,313	132	13,827	14,022	195
635630	BOONVIL3	Booneville	345	MEC	14,857	15,501	645	17,368	18,186	818
635635	MADISON3	Madison Co	345	MEC	12,711	13,279	568	15,633	16,374	741
635013	PNYCRK 3	Pony Creek	345	MEC	27,296	27,490	194	26,333	26,579	246
640139	COOPER 3	Cooper	345	NPPD	22,334	22,450	116	21,758	21,912	154
635600	GRIMES 3	Grimes	345	MEC	18,989	19,347	358	21,312	21,860	548
635631	BOONVIL5	Booneville	161	MEC	18,471	18,754	284	21,493	21,825	332
635640	NORWLK3	Norwalk	345	MEC	13,834	14,137	304	15,932	16,365	433
635016	STHLND 3	Southland	345	MEC	27,266	27,459	193	26,317	26,562	245
635000	CBLUFFS3	Council Bluffs	345	MEC	31,608	31,829	221	28,936	29,192	256
645458	S3458 3	Sub 3458 (Neb City)	345	OPPD	17,438	17,478	40	18,418	18,483	65
640277	MOORE 3	Moore	345	NPPD	11,222	11,231	10	12,317	12,334	17
541199	ST JOE 3	St Joseph	345	GMO	12,113	12,120	8	14,541	14,557	16
300039	7FAIRPT	Fairport	345	AECI	8,838	8,844	6	10,509	10,522	13
640140	COOPER 5	Cooper	161	NPPD	11,210	11,222	12	9,931	9,945	13
635700	SYCAMOR3	Sycamore	345	MEC	19,566	19,888	323	20,790	21,270	480
12083	J498_TAP	J498 POI	345	MEC	9,552	9,576	24	12,447	12,498	51
12084	J499_TAP	J499 POI	345	MEC	9,211	9,253	43	11,377	11,462	85
635601	GRIMES 5	Grimes	161	MEC	22,956	23,148	192	23,917	24,182	266
635621	SUGRCRK5	Sugar Creek	161	MEC	14,764	14,883	119	19,689	19,914	225
635632	EARLHAM5	Earlham	161	MEC	6,936	6,958	22	8,482	8,516	35
635641	NORWLK5	Norwalk	161	MEC	21,027	21,255	228	22,740	23,041	301
635650	SE POLK3	SE Polk	345	MEC	18,175	18,463	288	18,592	18,985	393



Evaluation of Fault Currents and Interrupting Capabilities for Generation Interconnection Project J504

7-17-2017

Summary

Generator Interconnection Project J504 is a 50 MW solar powered generating facility with proposed interconnection to the 161 kV bus at ITC Midwest's Duane Arnold Energy Center (DAEC) Substation. Three phase and single line to ground fault current values were evaluated before and after interconnection of the generating facility using Computer Aided Protection Engineering (CAPE) software by Electrocon International. The base CAPE model maintained and used by ITC Midwest's system protection group was updated to include planned BES future system topology representative of a 2019 system was used in the analysis.

Generating Facility Assumptions

The generating facility was modeled based on available information, and included:

- A 50 MW equivalent current source
- 34.5kV to 600V equivalent transformer
- 34.5kV to 161 kV step up transformer

Figure 1 below shows a one-line diagram of the modeled generating facility and nearby system.

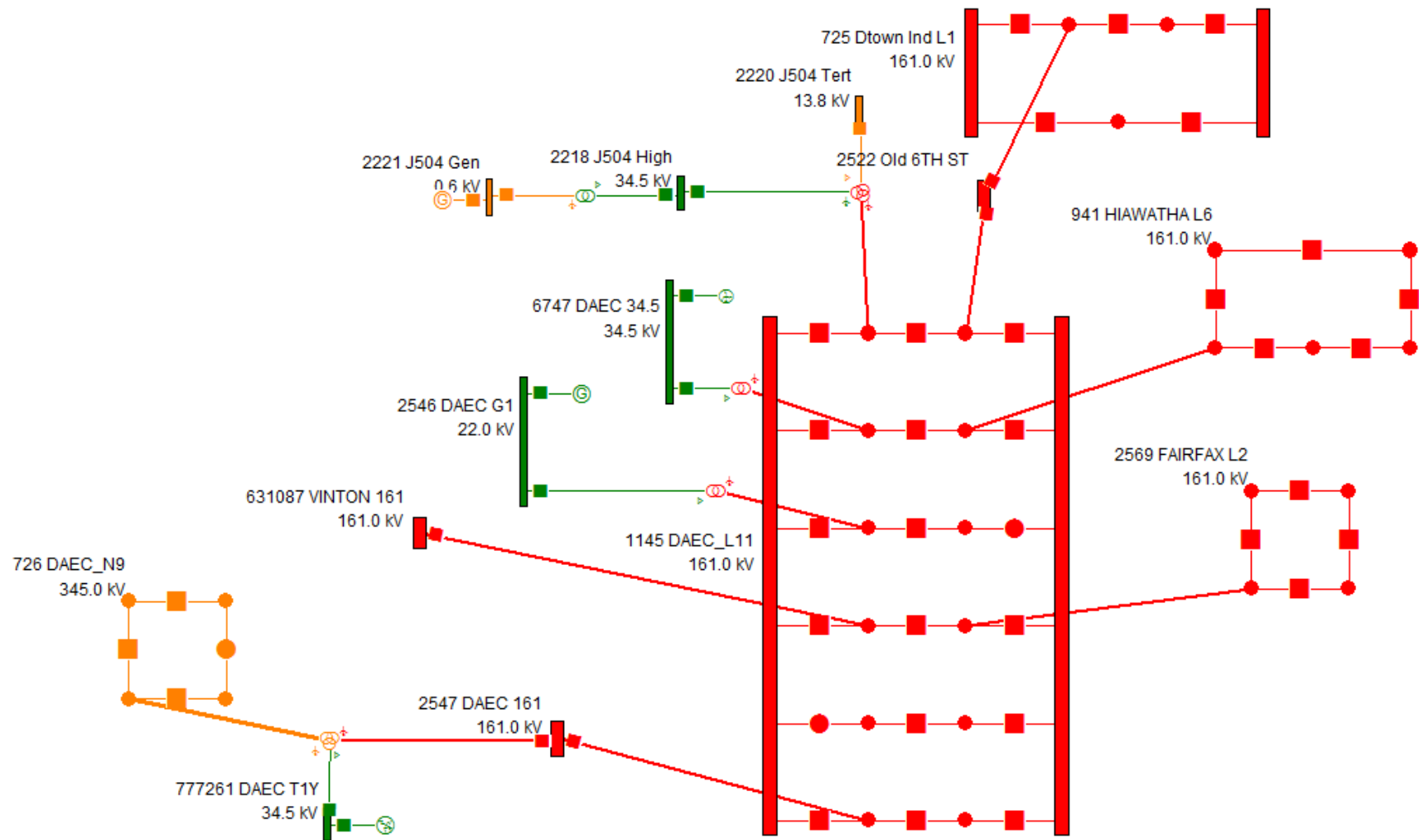


Figure1

Fault Currents

Table 1 below has the three phase and single line to ground fault currents before and after the addition of Project J504.

J504	DAEC 161	DAEC 345	Hiawatha 161	Fairfax 161	DTI 161	Vinton 161
SLG-pre (A)	29067.2	11730.1	18425.9	11886.1	13286	8448.33
3Ph-pre (A)	25349.7	13413.1	14772.8	12818.1	14390.9	10401.1
SLG-post (A)	29391.4	11771.3	18450.6	11898.1	13301.5	8454.19
3Ph-post (A)	25461.9	13428.7	14795.8	12834.3	14410.7	10408.3
Contribution SLG (A)	324.2	41.2	24.7	12	15.5	5.86
Contribution 3Ph (A)	112.2	15.6	23	16.2	19.8	7.2

Conclusion

Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J504. Equipment not owned by ITC Midwest was not evaluated for interrupting capability. There are several existing breakers at DAEC 161 kV substation that are expected to be replaced as part of planned work at this location, as well as new breakers being installed with the bus expansion. 40kA interrupting capability was assumed for the new and replaced breakers.

J506 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J506 generating facility (100 Vestas V110 2.0 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J506 short circuit study model was developed as follows:

- Added J524, J527, J528, J530, J534, J535 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J506 generating facility with its sequence parameters at approximately 35 miles from Raun on Raun – Lakefield Jct 345 kV line. The associated 345 kV interconnection generation tie line (3 miles), two 345-34.5-13.2 kV transformers ($Z_+=9\%$, $Z_0=8.01\%$ at 75 MVA, YG-YG-D), two 34.5 kV collector system equivalent lines and two 34.5-0.69 kV pad mount transformers ($Z_+=9.79\%$, $Z_0=9.07\%$ at 105 MVA, D-YG) were also added.
- The J506 generation was represented using the conventional generator model with the impedance of the machines set as follows per the data provided in the Vestas manual.
 - Positive saturated sequence impedances (i.e. subtransient, transient and synchronous) (Z_1): j0.1669 p.u.
 - Negative saturated sequence impedance (Z_2): j0.1669 p.u. (same as positive sequence impedances)
 - Zero sequence saturated impedance (Z_0): j0.1669 p.u.

The J506 short circuit benchmark model was developed as follows:

- Turned off J506 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J506 project) and benchmark case (without the J506 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to "1.00 per unit with multiplier of 1"
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J506 (study case) and without J506 (benchmark case) are listed in Table 1. With the addition of the J506 generating facility, the maximum increase in the 3PH short circuit current is 7.07% (806.3 A), and the maximum increase in the SLG short circuit current is 20.57% (1698.8 A), which are both observed at the J506 POI 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 12205.7 A (increased by 806.3 A) and SLG fault current is 9955.5 A (increased by 1698.8 A) at the J506 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J506 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J506

Substation	Faulted Bus			Benchmark Case without J506 Project		Study Case with J506 Project		Ampere Change with J506 vs Benchmark		% Change with J506 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J506	16030	J506-COL2	34.5	-	-	16639.9	16313.0	-	-	-	-
J506	16020	J506_COL2	34.5	-	-	23877.6	18046.6	-	-	-	-
J506	16011	J506_TR2_TER	13.2	-	-	0.0	31964.5	-	-	-	-
J506	15030	J506-COL1	34.5	-	-	16639.9	16313.0	-	-	-	-
J506	15020	J506_COL1	34.5	-	-	23877.6	18046.6	-	-	-	-
J506	15011	J506_TR1_TER	13.2	-	-	0.0	31964.5	-	-	-	-
J506	15010	J506	345	-	-	9166.4	11157.3	-	-	-	-
J506	65400	J506_POI	345	8256.7	11399.4	9955.5	12205.7	1698.8	806.3	20.57%	7.07%
SHD HIGHLAND WIND FA	14839	SHD GEN BUS1	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	10839	SHD-345	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	473	SHD GEN BUS3	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	472	SHD GEN BUS2	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	471	SHD-345	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	470	SHD-345	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	469	SHD-345	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	468	SHD-345	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
ROCK ISLAND CLEAN LI	10838	RICL-345	345	10711.0	13314.3	11019.8	13577.2	308.8	262.9	2.88%	1.97%
MVP SBC OBRIEN COUNT	500001	OBRIEN-345	345	11415.5	14032.0	11674.8	14268.0	259.3	236.0	2.27%	1.68%
SBC OBRIEN COLLECTOR	11830	COLECTOR-GEN	345	11415.5	14032.0	11674.8	14268.0	259.3	236.0	2.27%	1.68%
SBC OBRIEN COLLECTOR	10830	COLECTOR-345	345	11415.5	14032.0	11674.8	14268.0	259.3	236.0	2.27%	1.68%
RAU RAUN	10840	RAU-345-A	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	485	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	484	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	483	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	482	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	481	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	479	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	478	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	477	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	476	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	475	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	474	RAU-345-B	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
NLN NEAL NORTH	11542	NLN 3 345	345	26055.5	24691.2	26396.1	25004.5	340.6	313.3	1.31%	1.27%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J506

Substation	Faulted Bus			Benchmark Case without J506 Project		Study Case with J506 Project		Ampere Change with J506 vs Benchmark		% Change with J506 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
NL4 NEAL SOUTH	11207	NEAL 4 345	345	25102.2	23337.7	25384.8	23606.8	282.6	269.1	1.13%	1.15%
UPLAND PRARIE WIND F	500036	J455-345	345	9346.1	12092.4	9468.7	12228.8	122.6	136.4	1.31%	1.13%
RED ROCK NORTH STAR	500030	RED ROCK TAP	345	9632.6	11451.1	9689.0	11518.8	56.4	67.7	0.59%	0.59%



Evaluation of Fault Currents and Interrupting Capabilities for Generation Interconnection Project J514

7-19-2017

Summary

Generator Interconnection Project J514 is an increase of 30MW summer and 60 MW winter to an existing generating facility interconnected to ITC Midwest's Marshalltown 161 kV substation. Three phase and single line to ground fault current values were evaluated and compared to the previous values from the original J233 interconnection study using Computer Aided Protection Engineering (CAPE) software by Electrocon International. The base CAPE model maintained and used by ITC Midwest's system protection group was updated to include planned BES future system topology representative of a 2019 system was used in the analysis.

Generating Facility Assumptions

The generating facility was modeled based on available information, and included:

- Two 262 MVA generators (combustion turbines)
- One 306 MVA generator (steam turbine)
- Three 18kV to 161 kV step up transformers

Sutherland Steam units 1 and 3 have approved Attachment Y retirements, and were taken out of service in the analysis of J514 which changes the fault currents as well.

Figure 1 below shows a one-line diagram of the modeled generating facility and nearby system.

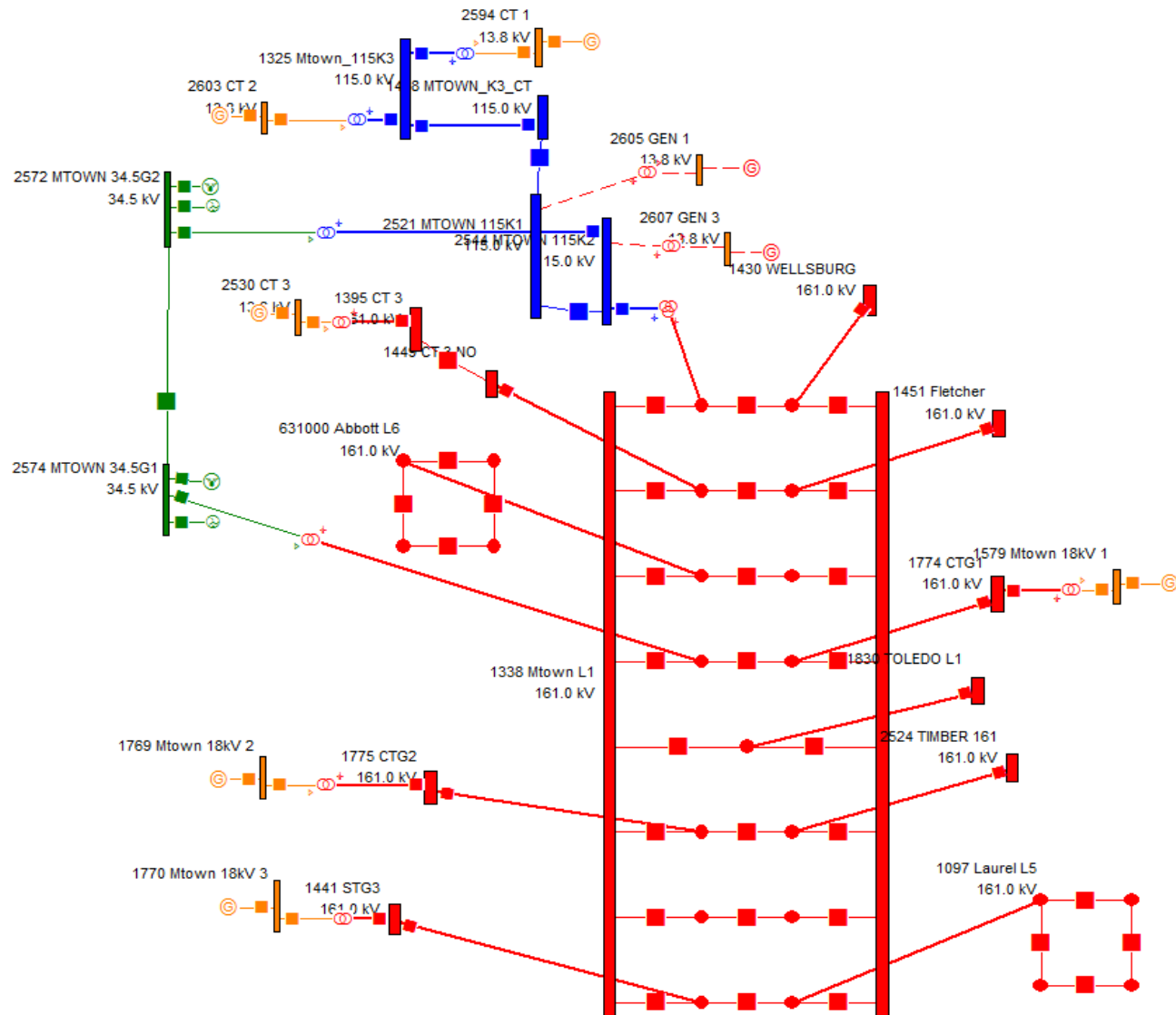


Figure1

Fault Currents

Table 1 below has the three phase and single line to ground fault currents comparing the results from the J233 study and J514.

J514	Marshalltown 161	Marshalltown 115	Marshalltown 34.5	Wellsburg 161	Abbott 161	Toledo 161	Timber Creek 161	Laurel 161
SLG-J233 (A)	28504.7	19667.1	3372.16		7223.32		11473	9971.34
3Ph-J233 (A)	24795.4	16976.8	15773.3		10640.5		13556.6	12807.7
SLG-J514 (A)	28730.9	16514.7	3381.4	7462.9	9278.54	6641.68	12510.3	9955.3
3Ph-J514 (A)	24202.4	14536.9	15466.2	8312.51	10941.6	7754.12	13534.4	12145.9
Difference SLG (A)	226.2	-3152.4	9.24	N/A	2055.22	N/A	1037.3	-16.04
Difference 3Ph (A)	-593	-2439.9	-307.1	N/A	301.1	N/A	-22.2	-661.8

Conclusion

Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J514. Equipment not owned by ITC Midwest was not evaluated for interrupting capability. Circuit breaker 2680 at Marshalltown 115 kV was previously identified as a constraint for J233, but has since been removed.



Evaluation of Fault Currents and Interrupting Capabilities for Generation Interconnection Project J523

7-17-2017

Summary

Generator Interconnection Project J523 is a 50 MW solar powered generating facility with proposed interconnection to the 161 kV bus at ITC Midwest's Adams Substation. Three phase and single line to ground fault current values were evaluated before and after interconnection of the generating facility using Computer Aided Protection Engineering (CAPE) software by Electrocon International. The base CAPE model maintained and used by ITC Midwest's system protection group was updated to include planned BES future system topology representative of a 2019 system was used in the analysis.

Generating Facility Assumptions

The generating facility was modeled based on available information, and included:

- A 50 MW equivalent current source
- 34.5kV to 600V equivalent transformer
- 34.5kV to 161 kV step up transformer
- 0.8 mile 161 kV interconnection line

Figure 1 below shows a one-line diagram of the modeled generating facility and nearby system.

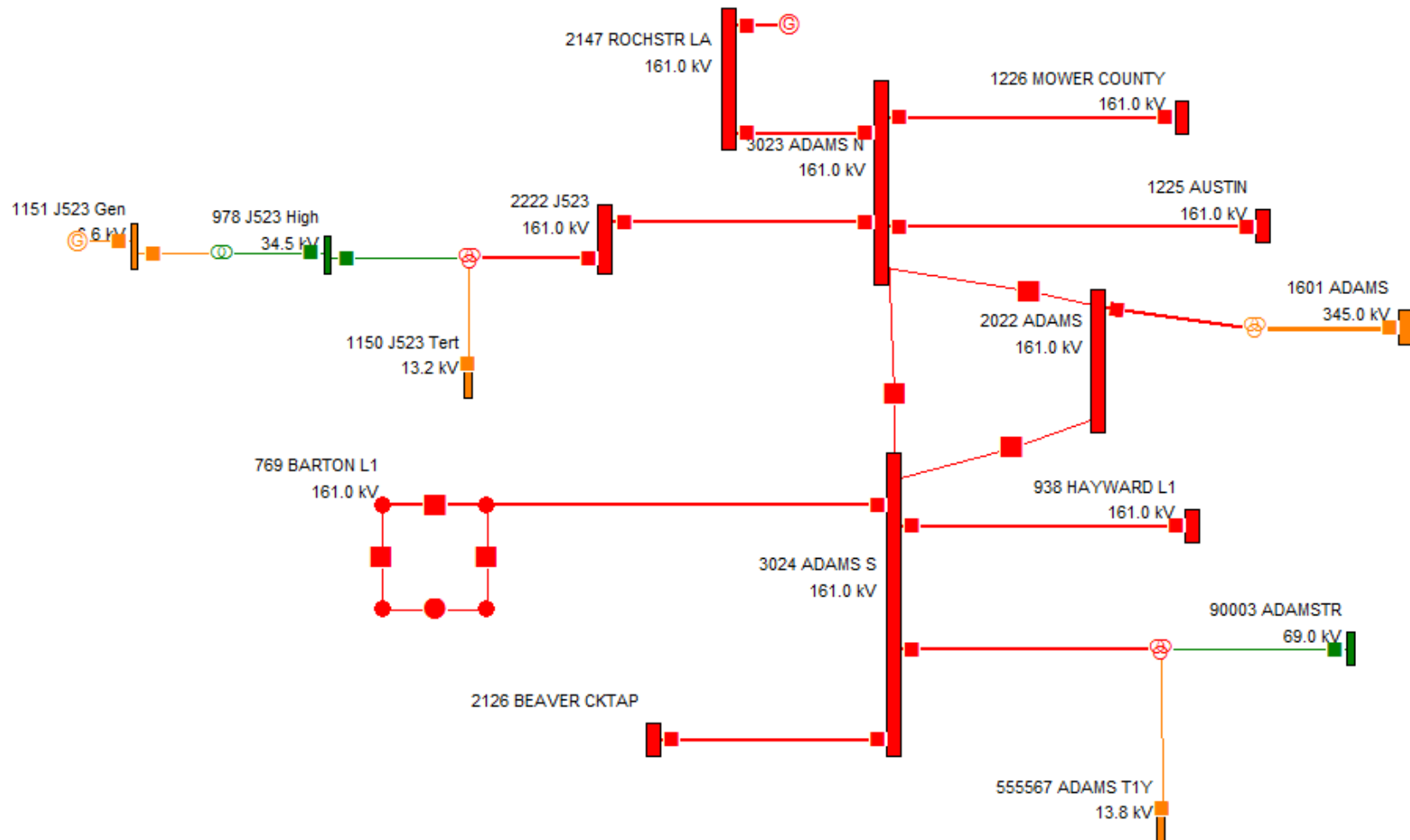


Figure1

Fault Currents

Table 1 below has the three phase and single line to ground fault currents before and after the addition of Project J523.

J523	Adams 161	Adams 345	Adams 69	Mower Co 161	Rochester 161	Austin 161	Barton 161	Beaver Creek 161	Hayward 161
SLG-pre (A)	17100.7	9853.96	12922.1	8904.79	5240.59	10452.2	9788.32	6865.9	11568.2
3Ph-pre (A)	17353.5	10595.8	12845	9170.35	4482.69	12145.4	11117	8544.83	14138.6
SLG-post (A)	17790.5	9918.36	13011.4	8971.37	5244.7	10486.1	9803.9	6901.44	11584.3
3Ph-post (A)	17464	10607.2	12860.9	9197.78	4486.18	12161.8	11126	8559.99	14151.5
Contribution SLG (A)	689.8	64.4	89.3	66.58	4.11	33.9	15.58	35.54	16.1
Contribution 3Ph (A)	110.5	11.4	15.9	27.43	3.49	16.4	9	15.16	12.9

Conclusion

Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J523. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

J524 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J524 Solar PV generating facility (40 SMA SC 2.5 MVA inverters) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J524 short circuit study model was developed as follows:

- Added J506, J527, J528, J530, J534, J535 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J524 generating facility with its sequence parameters at Webster 161 kV substation. The associated 161 kV interconnection generation tie line, 161-34.5-13.2 kV transformer ($Z_+ = 9.0\%$, $Z_0 = 8.0\%$ at 75 MVA, YG-YG-D), 34.5 kV collector system equivalent line and 34.5-0.550 kV pad mount transformer ($Z_+ = 6\%$, $Z_0 = 5.4\%$ at 112 MVA, D-YG) were also added.
- Per CAPE's manual, solar generation project should be modeled the same way as Type 4 wind generation. The J524 solar project was modeled as Type 4 controlled power and current-limited generator. It is assumed that the project has the maximum short circuit contribution of 1.5 p.u. This corresponds to a maximum fault current limit of 598 amps at 161 kV side of the GSU transformer.

The J524 short circuit benchmark model was developed as follows:

- Turned off J524 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J524 project) and benchmark case (without the J524 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to "1.00 per unit with multiplier of 1"
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J524 (study case) and without J524 (benchmark case) are

listed in Table 1. With the addition of the J524 generating facility, the maximum increase in the 3PH short circuit current is 3.63% (1028.1 A) observed at Webster 13.8 kV bus. The maximum increase in the SLG short circuit current is 6.82% (1244.2 A) observed at the Webster 161 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 19095 A (increased by 580.4 A) and SLG fault current is 19487 A (increased by 1244.2 A) at the Webster 161 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J524 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J524

Substation	Faulted Bus			Benchmark Case without J524 Project		Study Case with J524 Project		Ampere Change with J524 vs Benchmark		% Change with J524 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J524	97000	J524_TR_TER	13.2	-	-	0.0	28842.8	-	-	-	-
J524	96998	J524-COL1	34.5	-	-	19577.9	14160.9	-	-	-	-
J524	96997	J524_COL1	34.5	-	-	20175.1	14679.4	-	-	-	-
J524	96996	J524	161	-	-	17620.2	17752.8	-	-	-	-
WEB WEBSTER	10663	WEB-13.8-9T1	13.8	0.0	28298.2	0.0	29326.3	0.0	1028.1	0.00%	3.63%
WEB WEBSTER	10652	WEB-161	161	18242.8	18515.0	19487.0	19095.4	1244.2	580.4	6.82%	3.13%
WEB WEBSTER	10642	WEB-69	69	9937.1	9570.4	10246.7	9837.1	309.6	266.7	3.12%	2.79%
WEB WEBSTER	10645	WEB-12.4-8T1	12.5	0.0	13579.2	0.0	13949.8	0.0	370.6	0.00%	2.73%
CBPC-HOPE	10600	CB HOPE 161	161	7023.7	8906.4	7234.1	9138.7	210.4	232.3	3.00%	2.61%
CBPC-VINCENT JCT	10362	VNCNT JCT 69	69	3659.4	5108.1	3754.8	5238.2	95.4	130.1	2.61%	2.55%
CBPC-SNELL	10364	SNELL 69	69	3485.2	4818.2	3573.7	4938.3	88.5	120.1	2.54%	2.49%
FDD SUB D FD	401100	FDD-13.8	13.8	2033.2	2833.5	2033.2	2903.9	0.0	70.4	0.00%	2.48%
FDT SUB T FD	4607	FDT-8T6-13.2	13.2	9677.2	9467.4	9917.5	9702.6	240.3	235.2	2.48%	2.48%
FDT SUB T FD	2101	FDT-8T2-13.2	13.2	8180.0	8272.4	8180.0	8477.9	0.0	205.5	0.00%	2.48%
FDT SUB T FD	1100	FDT-8T1-13.2	13.2	8026.7	8039.2	8026.7	8238.9	0.0	199.7	0.00%	2.48%
FDH HARVEST AVE	629	FDH-8T2-13.2	13.2	7765.5	7608.8	7765.5	7797.3	0.0	188.5	0.00%	2.48%
FDH HARVEST AVE	628	FDH-8T1-13.2	13.2	7765.5	7608.8	7765.5	7797.3	0.0	188.5	0.00%	2.48%
FTL TATE AND LYLE	6262	FTL-8T2-13.2	13.2	10892.3	10500.3	11162.1	10760.4	269.8	260.1	2.48%	2.48%
FTL TATE AND LYLE	6261	FTL-8T1-13.2	13.2	10892.3	10500.3	11162.1	10760.4	269.8	260.1	2.48%	2.48%
FDT SUB T FD	3610	FDT-8T5-13.8	13.8	0.0	7317.9	0.0	7498.9	0.0	181.0	0.00%	2.47%
FDT SUB T FD	610	FDT-69-BUS#1	69	6394.9	6300.0	6556.7	6455.8	161.8	155.8	2.53%	2.47%
FDQ SUB Q FD	2520	FDQ-6T2-13.2	13.2	9335.5	8543.8	9566.0	8754.8	230.5	211.0	2.47%	2.47%
FDQ SUB Q FD	620	FDQ-69-B2	69	5144.9	6305.1	5274.2	6460.8	129.3	155.7	2.51%	2.47%
FDQ SUB Q FD	448	FDQ-69-B1	69	5144.9	6305.1	5274.2	6460.8	129.3	155.7	2.51%	2.47%
FDQ SUB Q FD	1620	FDQ-6T1-13.2	13.2	9517.3	8695.5	9752.3	8910.2	235.0	214.7	2.47%	2.47%
FDQ SUB Q FD	2529	FDQ-FD-FOODS	13.2	2787.1	3775.1	2787.1	3868.3	0.0	93.2	0.00%	2.47%
FDN SUB N FD	402520	FDN-13.8	13.8	2540.2	3431.2	2540.2	3515.9	0.0	84.7	0.00%	2.47%
FHA HAYES	617	FHA-69	69	10652.1	9552.7	10920.9	9788.5	268.8	235.8	2.52%	2.47%
FHA HAYES	613	FHA-13.8-8T1	13.8	0.0	17504.2	0.0	17936.2	0.0	432.0	0.00%	2.47%
FHA HAYES	619	FHA-6T1-13.2	13.2	6516.1	6252.2	6516.1	6406.5	0.0	154.3	0.00%	2.47%
FDG SUB G FD	1618	FDG-6T1-13.2	13.2	5377.1	5145.8	5377.1	5272.7	0.0	126.9	0.00%	2.47%
FDB SUB B BARNUM FD	630	FDB-69	69	4444.3	5738.7	4554.9	5880.2	110.6	141.5	2.49%	2.47%
FDT SUB T FD	454	FDT-161	161	9717.5	11130.1	9999.0	11404.4	281.5	274.3	2.90%	2.46%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J524

Substation	Faulted Bus			Benchmark Case without J524 Project		Study Case with J524 Project		Ampere Change with J524 vs Benchmark		% Change with J524 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
FDT SUB T FD	453	FDT-161	161	9717.5	11130.1	9999.0	11404.4	281.5	274.3	2.90%	2.46%
FDT SUB T FD	452	FDT-161	161	9717.5	11130.1	9999.0	11404.4	281.5	274.3	2.90%	2.46%
FDT SUB T FD	451	FDT-161	161	9717.5	11130.1	9999.0	11404.4	281.5	274.3	2.90%	2.46%
FDT SUB T FD	450	FDT-161	161	9717.5	11130.1	9999.0	11404.4	281.5	274.3	2.90%	2.46%
FDT SUB T FD	100	FDT-161	161	9717.5	11130.1	9999.0	11404.4	281.5	274.3	2.90%	2.46%
FDM SUB M FD	634	FDM-69	69	1923.9	3059.7	1971.5	3135.1	47.6	75.4	2.47%	2.46%
FDK SUB K FD	1615	FDK-6T1-13.2	13.2	9490.9	8883.6	9724.8	9102.5	233.9	218.9	2.46%	2.46%
FDG SUB G FD	618	FDG-69	69	6235.7	7467.3	6392.6	7651.3	156.9	184.0	2.52%	2.46%
FDK SUB K FD	615	FDK-69	69	8008.0	8632.3	8210.5	8844.9	202.5	212.6	2.53%	2.46%
FDT SUB T FD	611	FDT-69-BUS#2	69	8821.0	7517.1	9040.7	7702.1	219.7	185.0	2.49%	2.46%
FDT SUB T FD	455	FDT-69-TIE	69	8821.0	7517.1	9040.7	7702.1	219.7	185.0	2.49%	2.46%
FDT SUB T FD	1610	FDT-8T3-13.8	13.8	0.0	13494.4	0.0	13826.5	0.0	332.1	0.00%	2.46%
FTL TATE AND LYLE	626	FTL-161-B1	161	8793.0	10958.0	9040.3	11227.6	247.3	269.6	2.81%	2.46%
FTL TATE AND LYLE	449	FTL-161-B2	161	8793.0	10958.0	9040.3	11227.6	247.3	269.6	2.81%	2.46%
FDH HARVEST AVE	627	FDH-161	161	8866.0	11023.1	9115.5	11294.3	249.5	271.2	2.81%	2.46%
FDT SUB T FD	2510	FDT-8T4-13.8	13.8	0.0	13349.1	0.0	13677.5	0.0	328.4	0.00%	2.46%
FHA HAYES	616	FHA-161	161	9277.0	11381.1	9539.0	11660.8	262.0	279.7	2.82%	2.46%
FDK-FDT-XFR TAP POIN	612	FDK-FDT-XFR	69	7788.8	7755.9	7983.3	7946.5	194.5	190.6	2.50%	2.46%
XFR FRONTIER ETHANOL	641	XFR-69	69	1962.1	2737.5	2009.5	2803.3	47.4	65.8	2.42%	2.40%
CBPC-SNELL SW STR	10365	SNSLL SW 69	69	3517.3	4552.4	3602.1	4660.9	84.8	108.5	2.41%	2.38%
FDV SUB V FD TAP	10370	SUB V FD TAP	69	3204.4	4040.2	3279.7	4134.5	75.3	94.3	2.35%	2.33%
WRI WRIGHT	10653	WRI-161	161	8459.7	10040.3	8683.2	10274.3	223.5	234.0	2.64%	2.33%
CBPC-DUNCOMBE	10373	DUNCOMBE-69	69	2958.9	3824.5	3027.3	3912.2	68.4	87.7	2.31%	2.29%
CBPC-HOPE	10361	CB HOPE 69	69	9993.2	12653.3	10239.2	12942.3	246.0	289.0	2.46%	2.28%
HUM HUMBOLDT CENTRAL	11369	HUM-6T1-12.4	12.5	8088.4	7554.9	8088.4	7726.4	0.0	171.5	0.00%	2.27%
HUM HUMBOLDT CENTRAL	10369	HUM-69	69	4391.0	6691.8	4493.5	6843.3	102.5	151.5	2.33%	2.26%
HUMBOLDT CENTRAL TAP	10368	HUM-69-TAP	69	5496.9	7969.2	5626.2	8149.6	129.3	180.4	2.35%	2.26%
HUE HUMBOLDT EAST	11371	HUE-6T1-12.4	12.5	4996.5	4802.4	4996.5	4911.0	0.0	108.6	0.00%	2.26%
GMC GILMORE CITY	10322	GMC-69	69	1156.2	1846.3	1156.4	1888.0	0.2	41.7	0.02%	2.26%
BOD BODE	10326	BOD-69	69	1047.1	1700.2	1047.3	1738.6	0.2	38.4	0.02%	2.26%
CBPC-SWEAZY JCT	10366	SWEAZY J 69	69	3534.5	3928.2	3614.7	4016.9	80.2	88.7	2.27%	2.26%
HUE HUMBOLDT EAST	10371	HUE-69	69	4866.7	7192.7	4980.2	7355.0	113.5	162.3	2.33%	2.26%
RUTLAND TAP	10324	RUT-69-TAP	69	4728.5	7021.8	4838.7	7180.2	110.2	158.4	2.33%	2.26%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J524

Substation	Faulted Bus			Benchmark Case without J524 Project		Study Case with J524 Project		Ampere Change with J524 vs Benchmark		% Change with J524 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
RUTLAND JCT	10323	RUT-69-JCT	69	1951.2	3098.9	1995.9	3168.8	44.7	69.9	2.29%	2.26%
CBPC-BODE	10325	CB BODE 69	69	1258.4	2035.0	1258.7	2080.9	0.3	45.9	0.02%	2.26%
HUM HUMBOLDT CENTRAL	19369	HUM-TIE	69	3075.6	4760.1	3146.5	4867.4	70.9	107.3	2.31%	2.25%
CBPC-HOPE	10363	CB HOPE MP	99	40344.1	61437.5	41854.5	62819.5	1510.4	1382.0	3.74%	2.25%
WRI WRIGHT	10610	WRI-69	69	10152.2	9079.0	10384.8	9282.9	232.6	203.9	2.29%	2.25%
XFR FRONTIER ETHANOL	642	XFR-6T1-13.8	13.8	5782.8	4972.5	5782.8	5083.9	0.0	111.4	0.00%	2.24%
CBPC-PASSWATER	11360	PASSWATER-13	13.2	0.0	7630.5	0.0	7801.0	0.0	170.5	0.00%	2.23%
CBPC-PASSWATER	10360	PASSWATER-69	69	4012.3	4076.2	4102.1	4167.0	89.8	90.8	2.24%	2.23%
THO THOR	11332	THO-6T1-12.4	12.5	4423.3	4198.2	4423.3	4290.9	0.0	92.7	0.00%	2.21%
THO THOR	10332	THO-69	69	3006.0	4671.2	3073.4	4774.0	67.4	102.8	2.24%	2.20%
CBPC-WEBSTER CITY MU	10609	CB WCMU 69	69	3905.4	5206.5	3991.8	5320.8	86.4	114.3	2.21%	2.20%
FGF GOLDFIELD	11343	FGF-6T1-12.4	12.5	6290.3	5823.9	6290.3	5949.6	0.0	125.7	0.00%	2.16%
CBPC-WEBSTER	10608	CB WBSTR J69	69	4421.2	5190.7	4517.0	5302.4	95.8	111.7	2.17%	2.15%
FGF GOLDFIELD	10343	FGF-69	69	2891.7	4488.8	2954.7	4585.3	63.0	96.5	2.18%	2.15%
EAG EAGLE GROVE NORT	18372	EAG TO AGP	12.5	4810.0	5114.4	4810.0	5222.4	0.0	108.0	0.00%	2.11%
EAG EAGLE GROVE NORT	12372	EAG-6T2-12.4	12.5	7165.8	6669.5	7165.8	6810.3	0.0	140.8	0.00%	2.11%
EAG EAGLE GROVE NORT	19372	EAG TO CLR	12.5	6868.3	6414.7	6868.3	6550.1	0.0	135.4	0.00%	2.11%
EAG EAGLE GROVE NORT	11372	EAG-6T1-12.4	12.5	6869.4	6415.7	6869.4	6551.1	0.0	135.4	0.00%	2.11%
EAG EAGLE GROVE NORT	10372	EAG-69	69	3817.0	5544.9	3898.2	5661.5	81.2	116.6	2.13%	2.10%
CLR CLARION	10329	CLR-69	69	2036.3	3158.1	2036.5	3215.4	0.2	57.3	0.01%	1.81%
CBPC-LAKE CORNELIA T	10328	CORNELIA TAP	69	1959.1	3041.5	1959.2	3095.4	0.1	53.9	0.01%	1.77%
CBPC-LAKE CORNELIA	10341	LKE CORNELIA	69	1497.7	2375.9	1497.8	2418.0	0.1	42.1	0.01%	1.77%
GMC GILMORE CITY	10327	GMC-69-TIE	69	2272.0	3548.2	2272.4	3611.0	0.4	62.8	0.02%	1.77%
CBPC-GILMORE JUNCTIO	10318	CB GILMJ 69	69	2479.6	3864.7	2524.1	3933.1	44.5	68.4	1.79%	1.77%
CBPC-JEWEL	10619	JEWEL-69	69	3473.1	4304.5	3534.6	4380.5	61.5	76.0	1.77%	1.77%
CBPC-RENWICK TAP	10273	CB-RENWICK69	69	906.2	1476.0	906.3	1501.8	0.1	25.8	0.01%	1.75%
LVM MAPCO	13289	LVM MAPCO 69	69	1827.0	2950.4	1827.2	3001.9	0.2	51.5	0.01%	1.75%
LUV LUVERNE	10289	LUV-69	69	1827.0	2950.4	1827.2	3001.9	0.2	51.5	0.01%	1.75%
IPS TAP LIVERMORE /	10288	IPS TAP 69	69	2488.4	4000.5	2532.4	4070.3	44.0	69.8	1.77%	1.74%
CBPC-GARDEN CITY JCT	10612	GARDEN-69	69	3546.9	4555.6	3605.8	4630.9	58.9	75.3	1.66%	1.65%
CBPC-WILLIAMS	10614	WILLIAMS-69	69	3367.4	4309.1	3422.5	4379.6	55.1	70.5	1.64%	1.64%
CBPC-BOONDOCKS	10618	BOONDOCKS-69	69	3499.1	4496.5	3555.6	4569.1	56.5	72.6	1.61%	1.61%
CBPC-SHER HUMB TAP	636	SHER HUM TAP	69	1911.2	2516.1	1911.3	2556.1	0.1	40.0	0.01%	1.59%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J524

Substation	Faulted Bus			Benchmark Case without J524 Project		Study Case with J524 Project		Ampere Change with J524 vs Benchmark		% Change with J524 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
CBPC-SHER HUMB TAP	635	SUB M TAP 69	69	1911.2	2516.1	1911.3	2556.1	0.1	40.0	0.01%	1.59%
FDM SUB M FD	633	FDM-69-CBPC	69	1581.5	2167.6	1581.6	2202.0	0.1	34.4	0.01%	1.59%
CBPC-BLAIRSBURG	10615	BLAIRSRBG-69	69	4913.2	5931.3	4991.1	6025.2	77.9	93.9	1.59%	1.58%
FWL WALL LAKE WIND F	18625	FWL-13.8-8T3	13.8	0.0	15624.6	0.0	15869.3	0.0	244.7	0.00%	1.57%
FWL WALL LAKE WIND F	17625	FWL-69	69	9590.8	8876.4	9741.5	9015.2	150.7	138.8	1.57%	1.56%
FWL WALL LAKE WIND F	10625	FWL-161	161	8688.3	9811.2	8827.2	9964.6	138.9	153.4	1.60%	1.56%
CBPC-WILLEMSEN	10333	WILMSEN-69	69	4568.6	6076.6	4638.1	6168.7	69.5	92.1	1.52%	1.52%
CBPC-TWIN LAKES	10198	TWN LKS 69	69	1525.2	1902.3	1525.3	1931.0	0.1	28.7	0.01%	1.51%
CBPC-TWIN LAKES	10191	TWN LKS TAP	69	1965.9	2479.0	1966.0	2516.4	0.1	37.4	0.01%	1.51%
CBPC-GALBRAITH	10291	CB GALBR 69	69	2720.6	4306.7	2721.0	4369.8	0.4	63.1	0.01%	1.47%
CBPC-ALGONA SOUTH	10391	ALGONA SOUTH	69	2670.5	4205.3	2670.8	4264.0	0.3	58.7	0.01%	1.40%
CBPC-ALGONA SOUTH	10390	ALGONA SOUTH	69	2670.5	4205.3	2670.8	4264.0	0.3	58.7	0.01%	1.40%
WEB WEBSTER	10662	WEB-345	345	11464.8	13570.3	11813.0	13759.6	348.2	189.3	3.04%	1.39%
GAMESA POCAHONTAS PR	10649	GAMESA-161	161	7585.7	8951.0	7689.9	9072.6	104.2	121.6	1.37%	1.36%
PMY POMEROY 161KV	10648	PMY-161	161	7585.7	8951.0	7689.9	9072.6	104.2	121.6	1.37%	1.36%
ITC MIDWEST-COULTER	10309	COULTER TAP	69	1975.0	3011.2	1975.1	3051.5	0.1	40.3	0.01%	1.34%
CBPC-BURT	11128	CB BURT 161	161	4864.2	6360.1	4935.1	6445.0	70.9	84.9	1.46%	1.33%
CBPC-POC-POCAHONTAS	10320	POC-69	69	2764.2	3930.8	2764.3	3982.9	0.1	52.1	0.00%	1.33%
WIL WILLIAMS PIPELIN	10287	WIL-69	69	1624.5	2491.0	1624.5	2522.9	0.0	31.9	0.00%	1.28%
LATIMER TAP	10302	LAT-69-TAP	69	2091.6	3169.2	2091.6	3209.7	0.0	40.5	0.00%	1.28%
CBPC-ALGONA EAST	10395	ALGONA EAST	69	3294.4	4580.4	3336.6	4638.8	42.2	58.4	1.28%	1.27%
CBPC-ALGONA NORTH	10397	ALGONA NORTH	69	3768.9	4770.5	3816.6	4830.4	47.7	59.9	1.27%	1.26%
IPL-KOSSUTH	10398	IPL-KOSSUTH	69	3491.9	4509.1	3536.0	4565.7	44.1	56.6	1.26%	1.26%
LAT LATIMER	10301	LAT-69	69	2209.4	3328.5	2209.4	3369.5	0.0	41.0	0.00%	1.23%
CBPC-BURT	10248	CB BURT 69	69	6334.3	6064.2	6413.8	6138.7	79.5	74.5	1.26%	1.23%
CBPC-BURT	11127	CB BURT MP	99	4189.7	4151.6	4241.6	4202.0	51.9	50.4	1.24%	1.21%
CBPC-BOONE VALLEY JU	10263	CB-BV-JCT-69	69	1995.7	3123.9	1995.8	3161.3	0.1	37.4	0.01%	1.20%
CBPC-EAST	10197	CBPC EAST 69	69	1664.3	1875.2	1664.3	1897.1	0.0	21.9	0.00%	1.17%
CBPC-EAST	10192	CBPC EASTTAP	69	2472.2	2791.1	2472.2	2823.6	0.0	32.5	0.00%	1.16%
FVR FARNHAMVILLE	10190	FVR-69	69	1563.8	1781.6	1563.8	1802.3	0.0	20.7	0.00%	1.16%
PMY POMEROY 161KV	10651	PMY-69-A27	69	5020.2	4726.7	5078.4	4781.4	58.2	54.7	1.16%	1.16%
PMY POMEROY 161KV	10650	PMY-69	69	5020.2	4726.7	5078.4	4781.4	58.2	54.7	1.16%	1.16%
FPC POCAHONTAS WIND	10430	FPC-161	161	7940.4	9119.6	8032.3	9224.6	91.9	105.0	1.16%	1.15%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J524

Substation	Faulted Bus			Benchmark Case without J524 Project		Study Case with J524 Project		Ampere Change with J524 vs Benchmark		% Change with J524 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
CBPC-RINARD	10196	RINARD 69	69	2536.4	2861.3	2536.4	2893.8	0.0	32.5	0.00%	1.14%
PMY POMEROY 161KV	10654	PMY-34.5-AB	34.5	785.4	2604.7	785.4	2634.2	0.0	29.5	0.00%	1.13%
PMY POMEROY 161KV	10644	PMY-34.5	34.5	785.4	2604.7	785.4	2634.2	0.0	29.5	0.00%	1.13%
CBPC-WHITTEMORE JUNC	10274	WHITTEMORE J	69	3393.4	5207.1	3393.6	5261.7	0.2	54.6	0.01%	1.05%
FWL WALL LAKE WIND F	12625	FWL-34.5-2	34.5	15068.5	15726.4	15226.4	15890.8	157.9	164.4	1.05%	1.05%
FWL WALL LAKE WIND F	11625	FWL-34.5-1	34.5	14896.4	15653.0	15051.9	15816.1	155.5	163.1	1.04%	1.04%
CBPC-DOVER	10164	CB-DOVER-69	69	2461.9	3760.8	2461.9	3799.1	0.0	38.3	0.00%	1.02%
FWL WALL LAKE WIND F	14625	FWL-13.8-8T2	13.8	0.0	20321.1	0.0	20526.8	0.0	205.7	0.00%	1.01%
LGR LUNDGREN WIND EN	10777	LGR-345	345	11487.0	14089.3	11674.5	14231.8	187.5	142.5	1.63%	1.01%
LEH LEHIGH	10775	LEH-345	345	11518.2	14128.7	11706.5	14271.5	188.3	142.8	1.63%	1.01%
LEH LEHIGH	188	LEH-345	345	11518.2	14128.7	11706.5	14271.5	188.3	142.8	1.63%	1.01%
LEH LEHIGH	187	LEH-345	345	11518.2	14128.7	11706.5	14271.5	188.3	142.8	1.63%	1.01%
LEH LEHIGH	186	LEH-345	345	11518.2	14128.7	11706.5	14271.5	188.3	142.8	1.63%	1.01%
FWL WALL LAKE WIND F	13625	FWL-13.8-8T1	13.8	0.0	20260.2	0.0	20464.9	0.0	204.7	0.00%	1.01%

A short circuit analysis was performed to determine the impact of the proposed generation on the adjacent substation fault current levels. The proposed generation, as outlined in this document, was added to the existing Otter Tail Power Company's ASPEN OneLiner system model (BANG13-0503.OLR) and faults were simulated on all busses within one substation from the substation that contained the point of interconnection. As per a recommendation from MISO, the fault current from the J526 wind generation was assumed to be 1 per unit of the full nameplate rating.

Study Results

Based on the results of the study, the fault current ratings of the Otter Tail equipment in the area are not exceeded.

With the proposed J526 project additions, the fault currents are estimated to be roughly 11.7 kA at the Big Stone South 345 kV bus, 20.6 kA at the Big Stone South 230 kV bus, and 37.1 kA at the Big Stone South 34.5 kV bus. There does not appear to be any short circuit related upgrades needed for J526.

A general protective relay coordination review will need to be performed by OTP when the projects approach completion and before energizing the generation source.

J527 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J527 generating facility (100 GE 2.5 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J527 short circuit study model was developed as follows:

- Added J506, J524, J528, J530, J534, J535 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J527 generating facility with its sequence parameters at approximately 33 miles from Booneville on Booneville - Cooper 345 kV line. The associated 345 kV interconnection generation tie line, two 345-34.5-13.8 kV transformers ($Z+=9\%$, $Z0=6.7\%$ at 90 MVA, YG-YG-D), two 34.5 kV collector system equivalent lines and two 34.5-0.690 kV pad mount transformers ($Z=5.75\%$ at 140 MVA, D-YG) were also added.
- Modeled the J527 generator facility as a generator with impedances listed below. These values are based on the conservative assumption that the converters "crowbar" (disconnect to protect the power electronics within) for close-in faults. In this case the turbine rotor is short-circuited like that of a squirrel cage induction generator. The behavior can be approximated to $X' = 0.2$, contributing a maximum of 5 per unit fault current.
 - Positive sequence impedances (i.e. subtransient, transient and synchronous) ($Z1$): j0.2 p.u.
 - Negative sequence impedance ($Z2$): j0.2 p.u. (same as positive sequence impedances)
 - Zero sequence impedance ($Z0$): j0.2 p.u.

The J527 short circuit benchmark model was developed as follows:

- Turned off J527 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J527 project) and benchmark case (without the J527 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to “1.00 per unit with multiplier of 1”
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J527 (study case) and without J527 (benchmark case) are listed in Table 1. With the addition of the J527 generating facility, the maximum increase in the 3PH short circuit current is 7.03% (1119.2 A), and the maximum increase in the SLG short circuit current is 18.94% (2339.9 A), which are both observed at the J527 POI 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 17037 A (increased by 1119.2 A) and SLG fault current is 14696 A (increased by 2339.9 A) at the J527 POI 345 kV bus. Based on the Transmission Owner’s short circuit criteria, interconnection of the J527 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J527

Substation	Faulted Bus			Benchmark Case without J527 Project		Study Case with J527 Project		Ampere Change with J527 vs Benchmark		% Change with J527 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J527	65209	J527_TER2	13.8	-	-	0.0	39002.8	-	-	-	-
J527	65208	J527_TER1	13.8	-	-	0.0	39002.8	-	-	-	-
J527	65207	J527_COL2	34.5	-	-	27952.6	22515.4	-	-	-	-
J527	65206	J527_COL1	34.5	-	-	27952.6	22515.4	-	-	-	-
J527	65204	J527 COL2	34.5	-	-	31599.5	23596.0	-	-	-	-
J527	65202	J527 COL	34.5	-	-	31599.5	23596.0	-	-	-	-
J527	65201	J527	345	-	-	14510.2	16812.7	-	-	-	-
J527	65200	J527_POI	345	12356.5	15917.7	14696.4	17036.9	2339.9	1119.2	18.94%	7.03%
ORIENT WIND PROJECT	20921	J500-345-BT	345	14798.1	17781.6	16501.8	18769.8	1703.7	988.2	11.51%	5.56%
ORIENT WIND PROJECT	20911	J500-345	345	14798.1	17781.6	16501.8	18769.8	1703.7	988.2	11.51%	5.56%
J528	65300	J528_POI	345	13148.1	15972.6	14134.3	16673.3	986.2	700.7	7.50%	4.39%
J528	65301	J528	345	12987.8	15770.3	13941.5	16451.9	953.7	681.6	7.34%	4.32%
CRH ROLLING HILLS WI	21916	CRH-345-BT	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
CRH ROLLING HILLS WI	20916	CRH-345	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
CRH ROLLING HILLS WI	426	CRH-345-B1	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
CRH ROLLING HILLS WI	425	CRH-345	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
CRH ROLLING HILLS WI	424	CRH-345-B3	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
CRH ROLLING HILLS WI	423	CRH-345	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
CRH ROLLING HILLS WI	422	CRH-345-B2	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
DEB BOONEVILLE	20905	DEB-345	345	15855.6	18515.1	16268.6	18933.1	413.0	418.0	2.60%	2.26%
DEB BOONEVILLE	16	DEB-345	345	15855.6	18515.1	16268.6	18933.1	413.0	418.0	2.60%	2.26%
DEB BOONEVILLE	15	DEB-345	345	15855.6	18515.1	16268.6	18933.1	413.0	418.0	2.60%	2.26%
DEB BOONEVILLE	14	DEB-345	345	15855.6	18515.1	16268.6	18933.1	413.0	418.0	2.60%	2.26%
DEB BOONEVILLE	13	DEB-345	345	15855.6	18515.1	16268.6	18933.1	413.0	418.0	2.60%	2.26%
DEB BOONEVILLE	10	DEB-345	345	15855.6	18515.1	16268.6	18933.1	413.0	418.0	2.60%	2.26%
DED MADISON COUNTY	20904	MADISON 345	345	13668.6	16692.6	13964.1	17034.9	295.5	342.3	2.16%	2.05%
DED MADISON COUNTY	212	MADISON 345	345	13668.6	16692.6	13964.1	17034.9	295.5	342.3	2.16%	2.05%
DED MADISON COUNTY	211	MADISON 345	345	13668.6	16692.6	13964.1	17034.9	295.5	342.3	2.16%	2.05%
DED MADISON COUNTY	210	MADISON 345	345	13668.6	16692.6	13964.1	17034.9	295.5	342.3	2.16%	2.05%
DGN NORWALK	20955	DGN-345	345	14316.1	16542.0	14467.6	16737.6	151.5	195.6	1.06%	1.18%
DGG GRIMES	20920	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DGG GRIMES	54	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DGG GRIMES	53	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J527

Substation	Faulted Bus			Benchmark Case without J527 Project		Study Case with J527 Project		Ampere Change with J527 vs Benchmark		% Change with J527 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
DGG GRIMES	52	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DGG GRIMES	51	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DGG GRIMES	50	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DGG GRIMES	49	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DGG GRIMES	48	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DAG SYCAMORE EC	20900	DAG-345	345	20091.6	21496.7	20259.4	21722.2	167.8	225.5	0.84%	1.05%
DAG SYCAMORE EC	185	DAG-345	345	20091.6	21496.7	20259.4	21722.2	167.8	225.5	0.84%	1.05%
DAG SYCAMORE EC	184	DAG-345	345	20091.6	21496.7	20259.4	21722.2	167.8	225.5	0.84%	1.05%
DAG SYCAMORE EC	183	DAG-345	345	20091.6	21496.7	20259.4	21722.2	167.8	225.5	0.84%	1.05%
DAG SYCAMORE EC	182	DAG-345	345	20091.6	21496.7	20259.4	21722.2	167.8	225.5	0.84%	1.05%
DAG SYCAMORE EC	181	DAG-345	345	20091.6	21496.7	20259.4	21722.2	167.8	225.5	0.84%	1.05%
DAC SE POLK	20907	DAC-345	345	18605.3	19127.6	18747.6	19305.9	142.3	178.3	0.76%	0.93%
DGC GREATER DES MOIN	20893	GDMEC CT TR2	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	20892	GDMEC CT TR1	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	20891	GDMEC ST TR1	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	20890	GDMEC 345	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	82	GDMEC 345	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	81	GDMEC 345	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	80	GDMEC 345	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	79	GDMEC 345	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	78	GDMEC 345	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DAH BONDURANT	20901	DAH-345	345	17701.5	20025.2	17815.0	20198.8	113.5	173.6	0.64%	0.87%
DEB BOONEVILLE	20831	DEB-161	161	18929.4	21979.6	19097.2	22134.6	167.8	155.0	0.89%	0.71%
DEB BOONEVILLE	21	DEB-161	161	18929.4	21979.6	19097.2	22134.6	167.8	155.0	0.89%	0.71%
DEB BOONEVILLE	20	DEB-161	161	18929.4	21979.6	19097.2	22134.6	167.8	155.0	0.89%	0.71%
DEB BOONEVILLE	19	DEB-161	161	18929.4	21979.6	19097.2	22134.6	167.8	155.0	0.89%	0.71%
DEB BOONEVILLE	18	DEB-161	161	18929.4	21979.6	19097.2	22134.6	167.8	155.0	0.89%	0.71%
DEB BOONEVILLE	17	DEB-161	161	18929.4	21979.6	19097.2	22134.6	167.8	155.0	0.89%	0.71%
DAG SYCAMORE EC	20800	DAG-161-B2	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	504	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	503	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	502	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	501	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J527

Substation	Faulted Bus			Benchmark Case without J527 Project		Study Case with J527 Project		Ampere Change with J527 vs Benchmark		% Change with J527 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
DAG SYCAMORE EC	500	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	499	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	498	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	497	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	496	DAG-161-B1	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%

J528 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J528 generating facility (80 GE 2.5 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J528 short circuit study model was developed as follows:

- Added J506, J524, J527, J530, J534, J535 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J528 generating facility with its sequence parameters at approximately 32 miles from Madison on Madison – Rolling Hills 345 kV line. The associated 345 kV interconnection generation tie line, two 345-34.5-13.8 kV transformers ($Z+=9\%$, $Z0=6.7\%$ at 66 MVA, YG-YG-D), two 34.5 kV collector system equivalent lines and two 34.5-0.690 kV pad mount transformers ($Z=5.75\%$ at 112 MVA, D-YG) were also added.
- Modeled the J528 generator facility as a generator with impedances listed below. These values are based on the conservative assumption that the converters "crowbar" (disconnect to protect the power electronics within) for close-in faults. In this case the turbine rotor is short-circuited like that of a squirrel cage induction generator. The behavior can be approximated to $X' = 0.2$, contributing a maximum of 5 per unit fault current.
 - Positive sequence impedances (i.e. subtransient, transient and synchronous) ($Z1$): j0.2 p.u.
 - Negative sequence impedance ($Z2$): j0.2 p.u. (same as positive sequence impedances)
 - Zero sequence impedance ($Z0$): j0.2 p.u.

The J528 short circuit benchmark model was developed as follows:

- Turned off J528 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J528 project) and benchmark case (without the J528 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to “1.00 per unit with multiplier of 1”
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J528 (study case) and without J528 (benchmark case) are listed in Table 1. With the addition of the J528 generating facility, the maximum increase in the 3PH short circuit current is 5.50% (868.9 A), and the maximum increase in the SLG short circuit current is 14.20% (1757.2 A), which are both observed at the J528 POI 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 16673 A (increased by 868.9 A) and SLG fault current is 14134 A (increased by 1757.2 A) at the J528 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J528 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J528

Substation	Faulted Bus			Benchmark Case without J528 Project		Study Case with J528 Project		Ampere Change with J528 vs Benchmark		% Change with J528 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J528	65309	J528_TER2	13.8	-	-	0.0	29537.6	-	-	-	-
J528	65308	J528_TER1	13.8	-	-	0.0	29537.6	-	-	-	-
J528	65307	J528_COL2	34.5	-	-	21600.0	17422.8	-	-	-	-
J528	65306	J528_COL1	34.5	-	-	21600.0	17422.8	-	-	-	-
J528	65304	J528 COL2	34.5	-	-	24201.3	18134.2	-	-	-	-
J528	65302	J528 COL1	34.5	-	-	24201.3	18134.2	-	-	-	-
J528	65301	J528	345	-	-	13941.5	16451.9	-	-	-	-
J528	65300	J528_POI	345	12377.1	15804.4	14134.3	16673.3	1757.2	868.9	14.20%	5.50%
ORIENT WIND PROJECT	20921	J500-345-BT	345	15219.2	18001.0	16501.8	18769.8	1282.6	768.8	8.43%	4.27%
ORIENT WIND PROJECT	20911	J500-345	345	15219.2	18001.0	16501.8	18769.8	1282.6	768.8	8.43%	4.27%
J527	65200	J527_POI	345	13924.2	16471.8	14696.4	17036.9	772.2	565.1	5.55%	3.43%
J527	65201	J527	345	13763.9	16263.3	14510.2	16812.7	746.3	549.4	5.42%	3.38%
DED MADISON COUNTY	20904	MADISON 345	345	13666.7	16732.3	13964.1	17034.9	297.4	302.6	2.18%	1.81%
DED MADISON COUNTY	212	MADISON 345	345	13666.7	16732.3	13964.1	17034.9	297.4	302.6	2.18%	1.81%
DED MADISON COUNTY	211	MADISON 345	345	13666.7	16732.3	13964.1	17034.9	297.4	302.6	2.18%	1.81%
DED MADISON COUNTY	210	MADISON 345	345	13666.7	16732.3	13964.1	17034.9	297.4	302.6	2.18%	1.81%
CRH ROLLING HILLS WI	21916	CRH-345-BT	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
CRH ROLLING HILLS WI	20916	CRH-345	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
CRH ROLLING HILLS WI	426	CRH-345-B1	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
CRH ROLLING HILLS WI	425	CRH-345	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
CRH ROLLING HILLS WI	424	CRH-345-B3	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
CRH ROLLING HILLS WI	423	CRH-345	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
CRH ROLLING HILLS WI	422	CRH-345-B2	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
DEB BOONEVILLE	20905	DEB-345	345	16007.2	18634.0	16268.6	18933.1	261.4	299.1	1.63%	1.61%
DEB BOONEVILLE	16	DEB-345	345	16007.2	18634.0	16268.6	18933.1	261.4	299.1	1.63%	1.61%
DEB BOONEVILLE	15	DEB-345	345	16007.2	18634.0	16268.6	18933.1	261.4	299.1	1.63%	1.61%
DEB BOONEVILLE	14	DEB-345	345	16007.2	18634.0	16268.6	18933.1	261.4	299.1	1.63%	1.61%
DEB BOONEVILLE	13	DEB-345	345	16007.2	18634.0	16268.6	18933.1	261.4	299.1	1.63%	1.61%
DEB BOONEVILLE	10	DEB-345	345	16007.2	18634.0	16268.6	18933.1	261.4	299.1	1.63%	1.61%
DGN NORWALK	20955	DGN-345	345	14330.9	16572.9	14467.6	16737.6	136.7	164.7	0.95%	0.99%
DGG GRIMES	20920	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DGG GRIMES	54	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DGG GRIMES	53	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J528

Substation	Faulted Bus			Benchmark Case without J528 Project		Study Case with J528 Project		Ampere Change with J528 vs Benchmark		% Change with J528 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
DGG GRIMES	52	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DGG GRIMES	51	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DGG GRIMES	50	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DGG GRIMES	49	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DGG GRIMES	48	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DAG SYCAMORE EC	20900	DAG-345	345	20136.0	21551.9	20259.4	21722.2	123.4	170.3	0.61%	0.79%
DAG SYCAMORE EC	185	DAG-345	345	20136.0	21551.9	20259.4	21722.2	123.4	170.3	0.61%	0.79%
DAG SYCAMORE EC	184	DAG-345	345	20136.0	21551.9	20259.4	21722.2	123.4	170.3	0.61%	0.79%
DAG SYCAMORE EC	183	DAG-345	345	20136.0	21551.9	20259.4	21722.2	123.4	170.3	0.61%	0.79%
DAG SYCAMORE EC	182	DAG-345	345	20136.0	21551.9	20259.4	21722.2	123.4	170.3	0.61%	0.79%
DAG SYCAMORE EC	181	DAG-345	345	20136.0	21551.9	20259.4	21722.2	123.4	170.3	0.61%	0.79%
DAC SE POLK	20907	DAC-345	345	18629.9	19161.5	18747.6	19305.9	117.7	144.4	0.63%	0.75%
DGC GREATER DES MOIN	20893	GDMEC CT TR2	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	20892	GDMEC CT TR1	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	20891	GDMEC ST TR1	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	20890	GDMEC 345	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	82	GDMEC 345	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	81	GDMEC 345	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	80	GDMEC 345	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	79	GDMEC 345	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	78	GDMEC 345	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DAH BONDURANT	20901	DAH-345	345	17726.3	20063.0	17815.0	20198.8	88.7	135.8	0.50%	0.68%
DEB BOONEVILLE	20831	DEB-161	161	18987.4	22019.4	19097.2	22134.6	109.8	115.2	0.58%	0.52%
DEB BOONEVILLE	21	DEB-161	161	18987.4	22019.4	19097.2	22134.6	109.8	115.2	0.58%	0.52%
DEB BOONEVILLE	20	DEB-161	161	18987.4	22019.4	19097.2	22134.6	109.8	115.2	0.58%	0.52%
DEB BOONEVILLE	19	DEB-161	161	18987.4	22019.4	19097.2	22134.6	109.8	115.2	0.58%	0.52%
DEB BOONEVILLE	18	DEB-161	161	18987.4	22019.4	19097.2	22134.6	109.8	115.2	0.58%	0.52%
DEB BOONEVILLE	17	DEB-161	161	18987.4	22019.4	19097.2	22134.6	109.8	115.2	0.58%	0.52%

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J529 Short Circuit Study Performed by MEC

The scope of this DPP Optional Study facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J529, a proposed 250 MW wind farm along with J590, a proposed 90 MW expansion at the same interconnection point, and nearby substations both with and without the Interconnection Customer interconnected. J529/J590 was assumed to interconnect off the Obrien County-Kossuth County 345 kV line. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in a DPP study cycle of the MISO generation interconnection process where the system impact study is not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 10,000 Amps without the Interconnection Customer included and 11,705 Amps with the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results. For example, generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation or system contingencies.

As shown in the table, the changes in fault current at buses more than a couple buses away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J529/J590

Bus Number	Bus Name	English Name	Base kV	Owner	SLG Fault Current Comparison			3 Ph Fault Current Comparison		
					Base SLG w/o new wind farm	SLG with new wind farm	SLG Difference w/ wind farm vs Base	Base 3PH w/o new wind farm	3PH with new wind farm	3PH Difference w/ wind farm vs Base
		J529J590 POI	345		7,660	9,703	2,043	10,000	11,705	1,705
		J529J590 Collector	34.5		23,730	37,957	14,227	20,331	32,060	11,730
55368	J455 POI	J455 POI	345	MEC	9,827	10,320	493	11,388	11,993	606
635369	KOSSUTH 3	Kossuth Co	345	MEC	8,502	9,587	1,085	11,804	12,852	1,048
635368	OBRIEN 3	O'Brien Co	345	MEC	11,489	11,804	315	13,740	14,192	452
631197	LEDYARD3	Ledyard	345	ITCM	8,258	8,705	448	11,552	12,112	560
636000	WEBSTER3	Webster	345	MEC	9,825	10,033	208	11,901	12,150	249
631138	LAKEFLD3	ITCM Lakefield	345	ITCM	15,041	15,234	192	16,469	16,736	267
635400	HIGHLND 3	Highland	345	MEC	10,112	10,283	171	12,330	12,588	258
631193	HUNTLEY3	Huntley	345	ITCM	7,823	8,019	196	10,432	10,725	293
631198	COLBY3	Colby	345	ITCM	7,974	8,039	65	9,386	9,484	98
636010	LEHIGH 3	Lehigh	345	MEC	9,802	9,910	109	12,385	12,541	156
636001	WEBSTER5	Webster	161	MEC	17,205	17,352	147	17,701	17,831	130
601034	NOBLES 3	Nobles Co	345	XCEL	5,670	5,687	16	8,543	8,587	43
601029	LKFLDXL3	Xcel Lakefield	345	XCEL	20,018	20,173	156	14,337	14,438	101
631041	LAKEFLD5	ITCM Lakefield	161	ITCM	17,185	17,286	101	18,941	19,081	140
635200	RAUN 3	Raun	345	MEC	27,244	27,345	101	25,743	25,839	95

J530 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J530 generating facility (100 GE 2.5 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J530 short circuit study model was developed as follows:

- Added J506, J524, J527, J528, J534, J535 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J530 generating facility with its sequence parameters at approximately 26 miles from Montezuma on Montezuma – Hills 345 kV line. The associated 345 kV interconnection generation tie line, two 345-34.5-13.8 kV transformers ($Z+=9\%$, $Z0=6.7\%$ at 90 MVA, YG-YG-D), two 34.5 kV collector system equivalent lines and two 34.5-0.69 kV pad mount transformers ($Z=5.75\%$ at 140 MVA, D-YG) were also added.
- Modeled the J530 generator facility as a generator with impedances listed below. These values are based on the conservative assumption that the converters "crowbar" (disconnect to protect the power electronics within) for close-in faults. In this case the turbine rotor is short-circuited like that of a squirrel cage induction generator. The behavior can be approximated to $X' = 0.2$, contributing a maximum of 5 per unit fault current.
 - Positive sequence impedances (i.e. subtransient, transient and synchronous) ($Z1$): j0.2 p.u.
 - Negative sequence impedance ($Z2$): j0.2 p.u. (same as positive sequence impedances)
 - Zero sequence impedance ($Z0$): j0.2 p.u.

The J530 short circuit benchmark model was developed as follows:

- Turned off J530 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J530 project) and benchmark case (without the J530 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to “1.00 per unit with multiplier of 1”
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J530 (study case) and without J530 (benchmark case) are listed in Table 1. With the addition of the J530 generating facility, the maximum increase in the 3PH short circuit current is 9.32% (1118.4 A), and the maximum increase in the SLG short circuit current is 35.43% (2793.4 A), which are both observed at the J530 POI 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 13118 A (increased by 1118.4 A) and SLG fault current is 10678 A (increased by 2793.4 A) at the J530 POI 345 kV bus. Based on the Transmission Owner’s short circuit criteria, interconnection of the J530 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J530

Substation	Faulted Bus			Benchmark Case without J530 Project		Study Case with J530 Project		Ampere Change with J530 vs Benchmark		% Change with J530 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J530	75739	J530_TER2	13.8	-	-	0.0	38565.2	-	-	-	-
J530	75738	J530_TER1	13.8	-	-	0.0	38565.2	-	-	-	-
J530	75737	J530_COL2	34.5	-	-	27565.4	22179.3	-	-	-	-
J530	75736	J530_COL1	34.5	-	-	27565.4	22179.3	-	-	-	-
J530	75734	J530 COL2	34.5	-	-	31050.5	23176.8	-	-	-	-
J530	75732	J530 COL1	34.5	-	-	31050.5	23176.8	-	-	-	-
J530	75731	J530	345	-	-	10605.1	12990.5	-	-	-	-
IOWA COUNTY WIND FAR	30143	J530-345	345	7884.2	11999.6	10677.6	13118.0	2793.4	1118.4	35.43%	9.32%
DBM MONTEZUMA	30150	DBM-345	345	9487.7	13350.5	10189.5	13789.9	701.8	439.4	7.40%	3.29%
DBM MONTEZUMA	3	DBM-345	345	9487.7	13350.5	10189.5	13789.9	701.8	439.4	7.40%	3.29%
DBM MONTEZUMA	2	DBM-345	345	9487.7	13350.5	10189.5	13789.9	701.8	439.4	7.40%	3.29%
DBM MONTEZUMA	1	DBM-345	345	9487.7	13350.5	10189.5	13789.9	701.8	439.4	7.40%	3.29%
IHL HILLS	30120	IHL-345	345	16284.1	19520.5	17023.0	20052.7	738.9	532.2	4.54%	2.73%
IHL HILLS	302	IHL-345	345	16284.1	19520.5	17023.0	20052.7	738.9	532.2	4.54%	2.73%
IHL HILLS	301	IHL-345	345	16284.1	19520.5	17023.0	20052.7	738.9	532.2	4.54%	2.73%
IHL HILLS	300	IHL-345	345	16284.1	19520.5	17023.0	20052.7	738.9	532.2	4.54%	2.73%
IHL HILLS	299	IHL-345	345	16284.1	19520.5	17023.0	20052.7	738.9	532.2	4.54%	2.73%
IHL HILLS	298	IHL-345	345	16284.1	19520.5	17023.0	20052.7	738.9	532.2	4.54%	2.73%
NORTH ENGLISH WIND F	30152	J555-345	345	7080.8	10350.4	7466.1	10612.6	385.3	262.2	5.44%	2.53%
ENGLISH FARMS WIND F	31151	J475-GEN-BT	345	7134.4	9894.0	7467.6	10123.3	333.2	229.3	4.67%	2.32%
ENGLISH FARMS WIND F	30151	J475-345	345	7134.4	9894.0	7467.6	10123.3	333.2	229.3	4.67%	2.32%
ICK SUB K IOWA CITY	30125	ICK 345	345	11112.0	14647.8	11315.5	14870.4	203.5	222.6	1.83%	1.52%
ICT SUB T HASKINS	30126	ICT-345	345	11124.8	15559.9	11284.2	15762.5	159.4	202.6	1.43%	1.30%
Q92 SUB 92	30118	SUB 92 345	345	12960.6	17700.9	13131.1	17926.8	170.5	225.9	1.32%	1.28%
OGS	602660	OGS 345	345	9121.7	10976.8	9226.6	11082.8	104.9	106.0	1.15%	0.97%
IHL HILLS	30282	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	309	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	308	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	307	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	306	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	305	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	304	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	303	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J530

Substation	Faulted Bus			Benchmark Case without J530 Project		Study Case with J530 Project		Ampere Change with J530 vs Benchmark		% Change with J530 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
Q93 SUB 93	30128	Q93-345	345	19011.0	18963.7	19136.1	19116.7	125.1	153.0	0.66%	0.81%
Q93 SUB 93	30127	Q93-345RATBT	345	19011.0	18963.7	19136.1	19116.7	125.1	153.0	0.66%	0.81%
Q93 SUB 93	317	Q93-345	345	19011.0	18963.7	19136.1	19116.7	125.1	153.0	0.66%	0.81%
Q93 SUB 93	316	Q93-345	345	19011.0	18963.7	19136.1	19116.7	125.1	153.0	0.66%	0.81%
Q93 SUB 93	315	Q93-345	345	19011.0	18963.7	19136.1	19116.7	125.1	153.0	0.66%	0.81%
Q93 SUB 93	314	Q93-345	345	19011.0	18963.7	19136.1	19116.7	125.1	153.0	0.66%	0.81%
IEN HYDROCARBON SUB	30288	IEN-161	161	11971.3	14732.5	12065.3	14825.5	94.0	93.0	0.79%	0.63%
ICP SUB P CORALVILLE	30296	ICP-161	161	11283.3	13577.8	11353.5	13654.8	70.2	77.0	0.62%	0.57%
ICP SUB P CORALVILLE	313	ICP-161	161	11283.3	13577.8	11353.5	13654.8	70.2	77.0	0.62%	0.57%
ICP SUB P CORALVILLE	312	ICP-161	161	11283.3	13577.8	11353.5	13654.8	70.2	77.0	0.62%	0.57%
ICP SUB P CORALVILLE	311	ICP-161	161	11283.3	13577.8	11353.5	13654.8	70.2	77.0	0.62%	0.57%
ICJ SUB J IOWA CITY	30290	ICJ-161	161	10866.8	13294.4	10934.5	13368.6	67.7	74.2	0.62%	0.56%
ICJ SUB J IOWA CITY	332	ICJ-161-B2	161	10866.8	13294.4	10934.5	13368.6	67.7	74.2	0.62%	0.56%
ICU SUB U IOWA CITY	30294	ICU-161-B1	161	10425.2	13260.2	10485.8	13333.7	60.6	73.5	0.58%	0.55%
ICU SUB U IOWA CITY	297	ICU-161-B2	161	10425.2	13260.2	10485.8	13333.7	60.6	73.5	0.58%	0.55%
ICR CORAL RIDGE SUBS	30298	ICR-161	161	10158.2	12727.6	10216.0	12796.1	57.8	68.5	0.57%	0.54%
QWL WALCOTT	30117	QWL-345	345	11081.5	15598.6	11126.0	15682.5	44.5	83.9	0.40%	0.54%
QWL WALCOTT	293	QWL-345	345	11081.5	15598.6	11126.0	15682.5	44.5	83.9	0.40%	0.54%
QWL WALCOTT	292	QWL-345	345	11081.5	15598.6	11126.0	15682.5	44.5	83.9	0.40%	0.54%
QWL WALCOTT	291	QWL-345	345	11081.5	15598.6	11126.0	15682.5	44.5	83.9	0.40%	0.54%
ICK SUB K IOWA CITY	30303	ICK 161	161	10535.8	12157.9	10599.9	12222.4	64.1	64.5	0.61%	0.53%
ICE SUB E IOWA CITY	30302	ICE-161-B2	161	9279.8	12250.4	9335.8	12314.6	56.0	64.2	0.60%	0.52%
ICE SUB E IOWA CITY	331	ICE-161-B1	161	9279.8	12250.4	9335.8	12314.6	56.0	64.2	0.60%	0.52%
ICY SUB Y IOWA CITY	30301	ICY-161	161	8893.4	11843.9	8943.4	11903.5	50.0	59.6	0.56%	0.50%
ICG NORTHGATE IOWA C	30300	ICG-161	161	8990.9	11938.3	9039.1	11998.3	48.2	60.0	0.54%	0.50%

J534 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J534 generating facility (100 GE 2.5 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J534 short circuit study model was developed as follows:

- Added J506, J524, J527, J528, J530, J535 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J534 generating facility with its sequence parameters at approximately 17 miles from Kossuth on Kossuth – Webster 345 kV line. The associated 345 kV interconnection generation tie line, two 345-34.5-13.8 kV transformers ($Z+=9\%$, $Z0=6.7\%$ at 90 MVA, YG-YG-D), two 34.5 kV collector system equivalent lines and two 34.5-0.69 kV pad mount transformers ($Z=5.75\%$ at 140 MVA, D-YG) were also added.
- Modeled the J534 generator facility as a generator with impedances listed below. These values are based on the conservative assumption that the converters "crowbar" (disconnect to protect the power electronics within) for close-in faults. In this case the turbine rotor is short-circuited like that of a squirrel cage induction generator. The behavior can be approximated to $X' = 0.2$, contributing a maximum of 5 per unit fault current.
 - Positive sequence impedances (i.e. subtransient, transient and synchronous) ($Z1$): j0.2 p.u.
 - Negative sequence impedance ($Z2$): j0.2 p.u. (same as positive sequence impedances)
 - Zero sequence impedance ($Z0$): j0.2 p.u.

The J534 short circuit benchmark model was developed as follows:

- Turned off J534 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J534 project) and benchmark case (without the J534 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to “1.00 per unit with multiplier of 1”
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J534 (study case) and without J534 (benchmark case) are listed in Table 1. With the addition of the J534 generating facility, the maximum increase in the 3PH short circuit current is 9.16% (1119.5 A), and the maximum increase in the SLG short circuit current is 26.78% (2413.0 A), which are both observed at the J534 POI 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 13335 A (increased by 1119.5 A) and SLG fault current is 11425 A (increased by 2413.0 A) at the J534 POI 345 kV bus. Based on the Transmission Owner’s short circuit criteria, interconnection of the J534 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J534

Substation	Faulted Bus			Benchmark Case without J534 Project		Study Case with J534 Project		Ampere Change with J534 vs Benchmark		% Change with J534 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J534	66009	J534_TER2	13.8	-	-	0.0	38787.0	-	-	-	-
J534	66008	J534_TER1	13.8	-	-	0.0	38787.0	-	-	-	-
J534	66007	J534_COL2	34.5	-	-	27709.5	22309.0	-	-	-	-
J534	66006	J534_COL1	34.5	-	-	27709.5	22309.0	-	-	-	-
J534	66004	J534 COL2	34.5	-	-	31209.3	23318.2	-	-	-	-
J534	66002	J534 COL1	34.5	-	-	31209.3	23318.2	-	-	-	-
J534	66001	J534	345	-	-	11331.0	13204.1	-	-	-	-
J534	66000	J534_POI	345	9011.9	12215.5	11424.9	13335.0	2413.0	1119.5	26.78%	9.16%
MVP SCS KOSSUTH COUN	500103	KOSSUTH CO	345	12236.9	15105.6	13296.1	15802.7	1059.2	697.1	8.66%	4.61%
PLUM CREEK WIND FARM	10834	J501-345-BT	345	12236.9	15105.6	13296.1	15802.7	1059.2	697.1	8.66%	4.61%
WEB WEBSTER	10662	WEB-345	345	11283.7	13342.4	11813.0	13759.6	529.3	417.2	4.69%	3.13%
MVP SCS LEDYARD	500104	LEDYARD 345	345	9597.2	13156.1	9877.2	13444.6	280.0	288.5	2.92%	2.19%
PALO ALTO CNTY WIND	500132	J529-345-BT	345	10494.1	12549.1	10777.8	12822.8	283.7	273.7	2.70%	2.18%
PALO ALTO CNTY WIND	500032	J529-345	345	10494.1	12549.1	10777.8	12822.8	283.7	273.7	2.70%	2.18%
LEH LEHIGH	10775	LEH-345	345	11483.7	14032.2	11706.5	14271.5	222.8	239.3	1.94%	1.71%
LEH LEHIGH	188	LEH-345	345	11483.7	14032.2	11706.5	14271.5	222.8	239.3	1.94%	1.71%
LEH LEHIGH	187	LEH-345	345	11483.7	14032.2	11706.5	14271.5	222.8	239.3	1.94%	1.71%
LEH LEHIGH	186	LEH-345	345	11483.7	14032.2	11706.5	14271.5	222.8	239.3	1.94%	1.71%
LGR LUNDGREN WIND EN	10777	LGR-345	345	11453.1	13993.8	11674.5	14231.8	221.4	238.0	1.93%	1.70%
RED ROCK NORTH STAR	500030	RED ROCK TAP	345	9585.3	11390.5	9689.0	11518.8	103.7	128.3	1.08%	1.13%
MVP SCS HUNTLEY	500105	HUNTLEY 345	345	8367.7	11202.8	8463.1	11327.8	95.4	125.0	1.14%	1.12%
WEB WEBSTER	10652	WEB-161	161	19220.4	18918.6	19487.0	19095.4	266.6	176.8	1.39%	0.93%
J524	96996	J524	161	17407.6	17600.3	17620.2	17752.8	212.6	152.5	1.22%	0.87%
BEAVER CREEK WIND PR	11778	J498-345-BT	345	9868.2	12803.8	9943.1	12909.4	74.9	105.6	0.76%	0.82%
BEAVER CREEK WIND PR	10778	J498-345	345	9868.2	12803.8	9943.1	12909.4	74.9	105.6	0.76%	0.82%
RED ROCK WIND FARM J	500044	RR-9T3-345	345	7998.9	8914.3	8059.3	8987.6	60.4	73.3	0.76%	0.82%
RED ROCK WIND FARM J	500043	RR-9T2-345	345	7998.9	8914.3	8059.3	8987.6	60.4	73.3	0.76%	0.82%
RED ROCK WIND FARM J	500042	RR-9T1-345	345	7998.9	8914.3	8059.3	8987.6	60.4	73.3	0.76%	0.82%
RED ROCK WIND FARM J	500041	RROCK-GEN-BT	345	7998.9	8914.3	8059.3	8987.6	60.4	73.3	0.76%	0.82%
RED ROCK WIND FARM J	500040	RED ROCK-345	345	7998.9	8914.3	8059.3	8987.6	60.4	73.3	0.76%	0.82%
UPLAND PRARIE WIND F	500036	J455-345	345	9417.1	12147.5	9468.7	12228.8	51.6	81.3	0.55%	0.67%
NORTH STAR WIND FARM	500050	N STAR-345	345	5860.4	7247.0	5892.3	7295.0	31.9	48.0	0.54%	0.66%
MVP SBC OBRIEN COUNT	500001	OBRIEN-345	345	11622.8	14191.7	11674.8	14268.0	52.0	76.3	0.45%	0.54%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J534

Substation	Faulted Bus			Benchmark Case without J534 Project		Study Case with J534 Project		Ampere Change with J534 vs Benchmark		% Change with J534 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
SBC OBRIEN COLLECTOR	11830	COLECTOR-GEN	345	11622.8	14191.7	11674.8	14268.0	52.0	76.3	0.45%	0.54%
SBC OBRIEN COLLECTOR	10830	COLECTOR-345	345	11622.8	14191.7	11674.8	14268.0	52.0	76.3	0.45%	0.54%

J535 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J535 generating facility (84 GE 2.5 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J535 short circuit study model was developed as follows:

- Added J506, J524, J527, J528, J530, J534 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J535 generating facility with its sequence parameters at approximately 44 miles from Lehigh on Lehigh – J411 345 kV line. The associated 345 kV interconnection generation tie line, 345-34.5-13.8 kV transformer ($Z_+ = 9\%$, $Z_0 = 6.7\%$ at 150 MVA, YG-YG-D), 34.5 kV collector system equivalent line and 34.5-0.69 kV pad mount transformer ($Z = 5.75\%$ at 235.2 MVA, D-YG) were also added.
- Modeled the J535 generator facility as a generator with impedances listed below. These values are based on the conservative assumption that the converters "crowbar" (disconnect to protect the power electronics within) for close-in faults. In this case the turbine rotor is short-circuited like that of a squirrel cage induction generator. The behavior can be approximated to $X' = 0.2$, contributing a maximum of 5 per unit fault current.
 - Positive sequence impedances (i.e. subtransient, transient and synchronous) (Z_1): j0.2 p.u.
 - Negative sequence impedance (Z_2): j0.2 p.u. (same as positive sequence impedances)
 - Zero sequence impedance (Z_0): j0.2 p.u.

The J535 short circuit benchmark model was developed as follows:

- Turned off J535 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J535 project) and benchmark case (without the J535 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to “1.00 per unit with multiplier of 1”
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J535 (study case) and without J535 (benchmark case) are listed in Table 1. With the addition of the J535 generating facility, the maximum increase in the 3PH short circuit current is 10.66% (950.0 A), and the maximum increase in the SLG short circuit current is 32.09% (2040.0 A), which are both observed at the J535 POI 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 9858 A (increased by 950.0 A) and SLG fault current is 8398 A (increased by 2040.0 A) at the J535 POI 345 kV bus. Based on the Transmission Owner’s short circuit criteria, interconnection of the J535 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J535

Substation	Faulted Bus			Benchmark Case without J535 Project		Study Case with J535 Project		Ampere Change with J535 vs Benchmark		% Change with J535 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J535	66206	J535_TER	13.8	-	-	0.0	61330.5	-	-	-	-
J535	66205	J535_COL	34.5	-	-	44527.2	35055.8	-	-	-	-
J535	66203	J535 COL	34.5	-	-	47736.3	35775.1	-	-	-	-
J535	66202	J535	345	-	-	8352.8	9788.2	-	-	-	-
J535	66201	J535_POI	345	6358.0	8908.1	8398.0	9858.1	2040.0	950.0	32.09%	10.66%
FIC IDA CO SUB IDA G	11776	GEN BUS-345	345	7273.9	9733.8	8223.5	10299.9	949.6	566.1	13.05%	5.82%
FIC IDA CO SUB IDA G	10776	IDA CNTY-345	345	7273.9	9733.8	8223.5	10299.9	949.6	566.1	13.05%	5.82%
IDA CO WEST WIND FAR	10779	J412-345	345	7218.9	10014.2	7899.1	10479.1	680.2	464.9	9.42%	4.64%
LEH LEHIGH	10775	LEH-345	345	11256.0	13929.3	11706.5	14271.5	450.5	342.2	4.00%	2.46%
LEH LEHIGH	188	LEH-345	345	11256.0	13929.3	11706.5	14271.5	450.5	342.2	4.00%	2.46%
LEH LEHIGH	187	LEH-345	345	11256.0	13929.3	11706.5	14271.5	450.5	342.2	4.00%	2.46%
LEH LEHIGH	186	LEH-345	345	11256.0	13929.3	11706.5	14271.5	450.5	342.2	4.00%	2.46%
LGR LUNDGREN WIND EN	10777	LGR-345	345	11226.9	13891.5	11674.5	14231.8	447.6	340.3	3.99%	2.45%
WEB WEBSTER	10662	WEB-345	345	11624.2	13575.3	11813.0	13759.6	188.8	184.3	1.62%	1.36%
BEAVER CREEK WIND PR	11778	J498-345-BT	345	9798.7	12757.5	9943.1	12909.4	144.4	151.9	1.47%	1.19%
BEAVER CREEK WIND PR	10778	J498-345	345	9798.7	12757.5	9943.1	12909.4	144.4	151.9	1.47%	1.19%
RAU RAUN	10840	RAU-345-A	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	485	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	484	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	483	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	482	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	481	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	479	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	478	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	477	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	476	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	475	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	474	RAU-345-B	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
NLN NEAL NORTH	11542	NLN 3 345	345	26199.5	24808.5	26396.1	25004.5	196.6	196.0	0.75%	0.79%
NL4 NEAL SOUTH	11207	NEAL 4 345	345	25215.9	23438.7	25384.8	23606.8	168.9	168.1	0.67%	0.72%
J534	66000	J534_POI	345	11374.6	13268.4	11424.9	13335.0	50.3	66.6	0.44%	0.50%
J534	66001	J534	345	11281.7	13138.9	11331.0	13204.1	49.3	65.2	0.44%	0.50%

2022 Cost Allocation Results

I.1 Distribution Factor (DF) and MW Contribution Results for Cost Allocation in 2022

Table I-1: Distribution Factor and MW Contribution on Constraints for Thermal NU Cost Allocation in 2022

Table I-2: Distribution Factor and MW Contribution on Voltage / Stability Constraints for NU Cost Allocation in 2022

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I.2 Cost Allocation Details

Table I-1: Network Upgrades Cost Allocation in 2022

Table I-3: Network Upgrades Cost Allocation in 2022

	Monitored Element	English Name	Cost	J432	J460	J475	J485	J488	J489	J493	J495	J498	J499	J500	J504	J506	J510	J514	J523	J524	J525	J526	J527	J528	J529	J530	J534	J535	Upgrade For	
75739	J530 POI	345 636409 HILLS 3	345 1						\$115,097																				\$H MISO	
001080	SPLIT RNC3	345 652557 WHITE 3	345 1																										\$H MISO	
001031	BROOKINGS	345 652557 WHITE 3	345 2																										\$H MISO	
	Hazel Creek 345-230-13.8 kV xfr #6	Hazlet Creek 345-230-13.8 kV xfr #6			\$4,990,800		\$2,843,511															\$2,056,489							\$H MISO	
615529	GRE-PANTHER4	230 619940 HUC-MCLEOD 4	230 1		\$1,000,000																	\$1,000,000							\$H MISO	
G16082	GRE-JOHNNCT7	115 628216 ORTOWHL7 115	1		\$16,850,000				\$3,263,129	\$3,263,129	\$3,433,161											\$6,890,648							\$H MISO	
G16082	GRE-JOHNNCT7	115 625355 MORRIS 7	115 1		\$0				\$1,362,305	\$1,362,305	\$1,005,999											\$3,343,640							\$H MISO	
	Big Stone 230-115-13.8 kV xfr	Big Stone 230-115-13.8 kV xfr			\$3,250,000				\$632,318	\$632,318	\$661,755											\$0							\$H MISO	
G20314	BIGSTON4	230 628322 BSSOUTH4 230	1		\$1,300,000				\$249,611	\$249,611	\$266,921											\$0							\$H MISO	
G20314	BIGSTON4	230 620325 BROWNSV4 230	1		\$350,000				\$55,925	\$55,925	\$64,824							\$43,678				\$0							\$H MISO and/or NRIS	
G20314	BIGSTON4	230 625903 BLAIR 4	230 1		\$80,000				\$15,566	\$15,566	\$16,288							\$0				\$32,570							\$H MISO	
G20327	HANKSON4	230 628329 WAHPETN4 230	1		\$2,300,000	\$74,552	\$151,083		\$399,253	\$399,253	\$361,874							\$207,562				\$0							\$H MISO and/or NRIS	
G20329	WAHPETN4	230 658109 FERGSFL4 230	1		\$750,000					\$0	\$249,158							\$0				\$500,842							\$H MISO	
G20362	OAKES 4	230 629063 FCMCLAN 4	230 1		\$1,000,000				\$800,000	\$800,000	\$0							\$0				\$0							\$H MISO	
G20362	OAKES 4	230 661098 ELLENDRLVP4 230	1		\$2,950,000			\$1,425,000	\$1,425,000	\$0	\$0							\$0				\$0							\$H MISO	
G51094	N-TOWN 7	115 631013 BLKSTWN7	115 1		\$2,700,000				\$150	\$150	\$0							\$2,700,000				\$0							\$H MISO	
G31013	BLKSTWN7	115 631017 PRAR CK 7	115 1		\$2,700,000				\$0	\$0	\$0							\$2,700,000				\$0							\$H MISO	
G31047	LIME CK L1 5 161	631169 LIME CK L2 5 161 2	161 2		\$0				\$0	\$0	\$0							\$0				\$0							\$H MISO	
G31051	HAZLETN L2 5 161	636218 WASHGRN 161	161		\$16,700,000				\$0	\$0	\$0			\$16,700,000				\$0				\$0							\$H MISO	
G31079	BNE JCT5	161 636028 SUB T F0 5 161	1		\$7,000,000				\$0	\$0	\$0		\$7,000,000					\$0				\$0							\$H MISO	
G31083	TRAEAR 5	161 631086 DYSART 5 161	1		\$0				\$0	\$0	\$0							\$0				\$0							\$H MISO	
G31085	PARNEL 5	161 51113 J438 POI 161	1		\$100,000				\$0	\$0	\$0							\$0				\$0							\$H MISO	
G31088	ARNOLD 5	161 631094 FAIRFAX 5 161	1		\$0				\$0	\$0	\$0							\$0				\$0							\$H MISO	
G31088	WYOMING 161	161 631099 WYOMING 161	1		\$600,000				\$0	\$0	\$0				\$78,072			\$56,176				\$0							\$H MISO	
G31113	POWESHKS	161 631117 REASNORS 161	1		\$0				\$0	\$0	\$0				\$0			\$0				\$0							\$H MISO	
G31142	ARNOLD 3	345 631148 MORGAN VLLY 3	345 1		\$0				\$0	\$0	\$0							\$0				\$0							\$H MISO	
G31148	MORGAN VLLY 3	345 636420 TIFFIN 3	345 1		\$0				\$0	\$0	\$0							\$0				\$0							\$H MISO	
G35500	RAUN 345	345 635451 SVAL 345	1		\$0				\$0	\$0	\$0							\$0				\$0							\$H MISO	
G35600	GRIMES 3	345 635700 SYCAMORE 345	345 2		\$1,200,000				\$0	\$0	\$0		\$519,779	\$680,221				\$0				\$0							\$H MISO	
G35600	GRIMES 3	345 636003 BVR CRK 3	345 1		\$600,000				\$0	\$0	\$0			\$318,015				\$0				\$21,171							\$H MISO	
G35605	GRNGRTPS	161 635614 100 54 5 161	1		\$100,000				\$0	\$0	\$0		\$44,481	\$55,519				\$0				\$0							\$H MISO	
G35680	BONDURNTS	345 635730 MHTZJMS 345	1		\$200,000				\$0	\$0	\$0		\$54,095	\$75,810				\$0				\$38,504							\$H MISO	
G36001	WEBSTER	161 630620 SUB T F0 5 161	1		\$6,250,000				\$0	\$0	\$0			\$0				\$0		\$0,250,000		\$0							\$H MISO	
G52593	BLAIR 4	230 652558 GRANITFA 4	230 1		\$0				\$0	\$0	\$0							\$0				\$0							\$H MISO	
	13.3 miles of 69 kV Lines between Chandler tap and Lyon County	13.3 miles of 69 kV Lines between Chandler tap and Lyon County			\$10,000,000				\$0	\$0	\$0							\$0				\$0							\$H MISO	
	Fenton 115-69 kV transformer	Fenton 115-69 kV transformer			\$3,900,000				\$0	\$0	\$0							\$0				\$0							\$H MISO	
	Twin Branch-Argenta 345 kV	Twin Branch-Argenta 345 kV			\$200,000				\$0	\$0	\$0							\$0				\$0							\$H MISO	
Benton-T-694 345 kV (N5106)		Benton-T-694 345 kV (N5106)			\$10,000,000				\$0	\$0	\$0		\$2,250,000	\$2,220,000	\$2,584,000			\$0			\$0								\$H MISO	
Benton-T-694 345 kV (N5312.3)		Benton-T-694 345 kV (N5312.3)			\$1,000,000				\$0	\$0	\$0		\$143,000	\$141,000	\$164,000			\$0			\$0								\$H MISO	
Jasper-Laural 161 kV		Jasper-Laural 161 kV			\$600,000				\$0	\$0	\$0							\$50,300				\$0							\$H MISO	
J274 POI-Creston 161 kV		J274 POI-Creston 161 kV			\$175,000				\$0	\$0	\$0				\$16,284			\$0				\$0								\$H MISO
Council Bluffs-Sub 3456 345 kV		Council Bluffs-Sub 3456 345 kV			\$0				\$0	\$0	\$0							\$0				\$0							\$H MISO	
Lehigh-Webster 345 kV		Lehigh-Webster 345 kV			\$100,000			\$3,180			\$0		\$34,899	\$10,614	\$12,778			\$0				\$0							\$H MISO	
Webster 345-161-13.8 kV xfr		Webster 345-161-13.8 kV xfr			\$0			\$0		\$0	\$0							\$0				\$0							\$H MISO	
J527 POI-Bonneville 345 kV		J527 POI-Bonneville 345 kV			\$100,000			\$0		\$0	\$0			\$51,501				\$0				\$27,773							\$H MISO	
J528 POI-Madison 345 kV		J528 POI-Madison 345 kV			\$125,000			\$0		\$0	\$0			\$64,744				\$0				\$0							\$H MISO	
G20314	BIGSTON4	230 652503 BLAIR 4	230 1 (NRIS)		\$0				\$0	\$0	\$0							\$0				\$0							\$H MISO	
G20314	BIGSTON4	230 628322 BSSOUTH4 230	2 (2nd J510-Brookings)		\$0				\$0	\$0	\$0							\$0				\$0							\$H MISO	
G20314	BIGSTON4	230 620325 BROWNSV4 230	1 (NRIS)		\$0				\$0	\$0	\$0							\$0				\$0							\$H MISO	
G20327	HANKSON4	230 628329 WAHPETN4 230	1 (NRIS)		\$0				\$0	\$0	\$0							\$0				\$0							\$H MISO	
G20327	HANKSON4	230 628329 WAHPETN4 230	1 (2nd J488&489-BSS)		\$2,700,000				\$1,350,000	\$1,350,000	\$0							\$0				\$0							\$H MISO	
G20362	OAKES 4	230 661098 ELLENDRLVP4 230	1 (2nd J488&489-BSS)		\$17,050,000				\$0	\$0	\$0							\$0				\$0							\$H MISO	
	Voltage Collapse (2nd J488&489-BSS)	Voltage Collapse (2nd J488&489-BSS)			\$24,550,000				\$12,275,000	\$12,275,000	\$0							\$0				\$0							\$H MISO	
G20214	BIGSTON7	115 620215 HWY12 7	115 1 (2nd J510-Brookings)		\$3,078,431				\$598,119	\$598,119	\$627,398							\$0				\$0							\$H MISO	
G20215	HWY12 7	115 620216 ORTOWHL7 115	1 (2nd J510-Brookings)		\$6,851,992				\$0	\$0	\$1,331,296			\$1,331,296	\$1,396,467			\$0				\$0							\$H MISO	
G20215	HWY12 7	115 620216 ORTOWHL7 115	1 (2nd J510-Brookings)		\$0				\$0	\$0	\$0							\$0				\$2,792,933							\$H MISO	
G20314	BIGSTON4	230 628322 BSSOUTH4 230	2 (2nd J510-Brookings)		\$1,241,303				\$238,340	\$238,340	\$254,869							\$0				\$509,753							\$H MISO	
G20314	BIGSTON4	230 652503 BLAIR 4	230 1 (2nd J510-Brookings)		\$714,991				\$139,124	\$139,124	\$154,576							\$0				\$291,167							\$H MISO	
G20327	HANKSON4	230 628329 WAHPETN4 230	1 (2nd J510-Brookings)		\$19,513,283	\$632,506	\$1,287,905		\$3,310,917	\$3,310,917	\$3,069,920							\$1,760,965				\$0							\$H MISO	
G20327	HANKSON4	230 628329 WAHPETN4 230	1 (2nd J510-Brookings)		\$0				\$0	\$0	\$0							\$0				\$0							\$H MISO	
	Voltage Collapse & Instability (2nd J510-Brookings)	Voltage Collapse & Instability (2nd J510-Brookings)			\$0				\$0	\$0	\$0							\$0				\$0							\$H MISO	
25 Mvar switched cap bank at Latchrop 161 kV (344923)		25 Mvar switched cap bank at Latchrop 161 kV (344923)			\$2,000,000	\$36,468	\$72,936	\$52,476	\$0	\$54,435	\$55,238	\$69,695	\$179,945	\$201,022	\$329,705	\$4,285	\$94,805	\$113,933	\$6,419	\$22,171	\$6,380	\$110,993	\$162,762	\$190,440	\$182,575	\$40,009	\$103,499	\$183,288	Reactive Power NU	
25 Mvar switched cap bank at J438 POI 161 kV (51113)		25 Mvar switched cap bank at J438 POI 161 kV (51113)			\$2,500,000	\$55,305	\$22,532	\$0	\$0	\$36,468	\$44,143	\$35,71																		

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